FIT PM12 twelve-channel test campaign

FIT hardware acceptance bench

November 14, 2026

1 Selective HDMI stimulation: 14 November 2026

Resident image `c7531465`, MCU1803. Four PM data outputs were each pulsed160 times independently: no SYS restart in32 completed trials. Eight idle controls were also stable. SPI signatures were read with data outputs held static zero. No PLL changes, output-enable release, direction reversal or FPGA reload was performed.

SPI divider	Approximate SCK	Restarts / completed frames
60	1.002MHz	8/16
600	100.197kHz	3/9
6000	10.020kHz	4/9
65535	917.34Hz	5/9

One additional fast frame was interrupted by a read timeout; its raw trace is retained and excluded from the completed denominator. A bounded repeat completed. All returned signatures were correct. Two long-frame series each show5 restarts in9 reads. A busy sample preceding a blocked counter read does not prove a reset during SEL assertion. A timed repeat places nominal reset epochs around41–44ms after start, about6–8ms after expected frame completion. This is read-completion time minus fresh SYS uptime, assuming running clocks and small reply latency; it is not a direct electrical latch. SEL deassertion remains a possible trigger. Conservative analysis counts observed counter decreases only; original helper host-time flags after delayed reads are superseded by raw recomputation.

SEL, SCK and MOSI remain coupled by the resident SPI master. These results narrow the observed trigger to SPI traffic but do not establish an individual faulty wire, MCU action or PLL cause. Polling does not exclude short OE/ready or electrical glitches. MMCM LOCK is not directly recorded in this image. Independent SPI pin control requires a gateware extension; Kasli was not reloaded because its outputs are required to remain driven.

Final checks: original mode/divider restored, driven SPI retained, power healthy, TDC frontend0xfff and a fresh physical ADC bankA capture passed on all12 channels. Raw traces, executed sources and final ADC buffers: `system/evidence-selective-20261114`.



Figure 1: Long SPI reads. Raised blue levels denote sampled master busy; red crosses mark nominal reset epochs from fresh uptime. Host polling is not a continuous electrical measurement.

2 Correct-clock TDC repeat: 14 November 2026

Board `PM12-K7-3A17A1B402212223`; MCU1803. Current SRAM diagnostic: `c7531465`. TDC channels9–12 pass the corrected-clock repeat. Earlier measurements remain historical observations.

2.1 Clock correction and procedure

The audited MCLK is40.0787MHz, rather than the previously assumed120MHz. The TDC MMCM input divider was changed from3 to1, retaining multiplier30 and output dividers2/2/4: VCO1202.361MHz, fast601.1805MHz and coarse300.59025MHz. Corresponding MCLK and TDC-reference constraints were corrected. No external PLL, MCU, CPLD or nonvolatile image was changed. Implementation WNS is0.085ns and WHS0.066ns; no DRC errors or critical warnings.

For each channel: quiet window,16-event discovery, count tests1/64/512/1024, 25 serializer positions with128 events each, then another quiet window. All128 captures were read back as raw13-bit words and independently checked. All19,268 events were conserved; eight quiet windows were empty and there were no cross-channel records. Capture generations were consecutive.

Channel	Fine slope [codes/ns]	Coarse gain [codes/bin]	Result
9	-46.196	1.25231	PASS
10	-46.164	1.25160	PASS
11	-46.160	1.25111	PASS
12	-46.184	1.24769	PASS

The predicted coarse gain is1.25. Fine slopes are relative raw-code responses, not calibrated TDC bin sizes or intrinsic jitter measurements. Coarse values are unwrapped across their empty arc and folded modulo10, one Kasli word, to remove whole-word start branches; this does not establish absolute LHC trigger alignment.

2.2 Regression and remaining findings

Digital ADC acquisition passed on all12 channels and both banks using physical and bit-exact synthetic captures. This is not new analog qualification. HDMI/SPI signatures were correct, but8 of12 reads restarted the SYS counter; that separate fault remains open. Kasli slot6 SPI outputs remained driven. The CFD11 MUX7 divider finding also remains open. USB stimulus remains on10. Channels1–8 TDC require their own stimulated repeat. The historical campaign coverage table is unchanged; these are a separate, versioned qualification series, not retroactive edits to its source contract. Raw data, source snapshots, bitstream, validation and hashes are retained in `tdc_live120/evidence-clock40-20261114`.

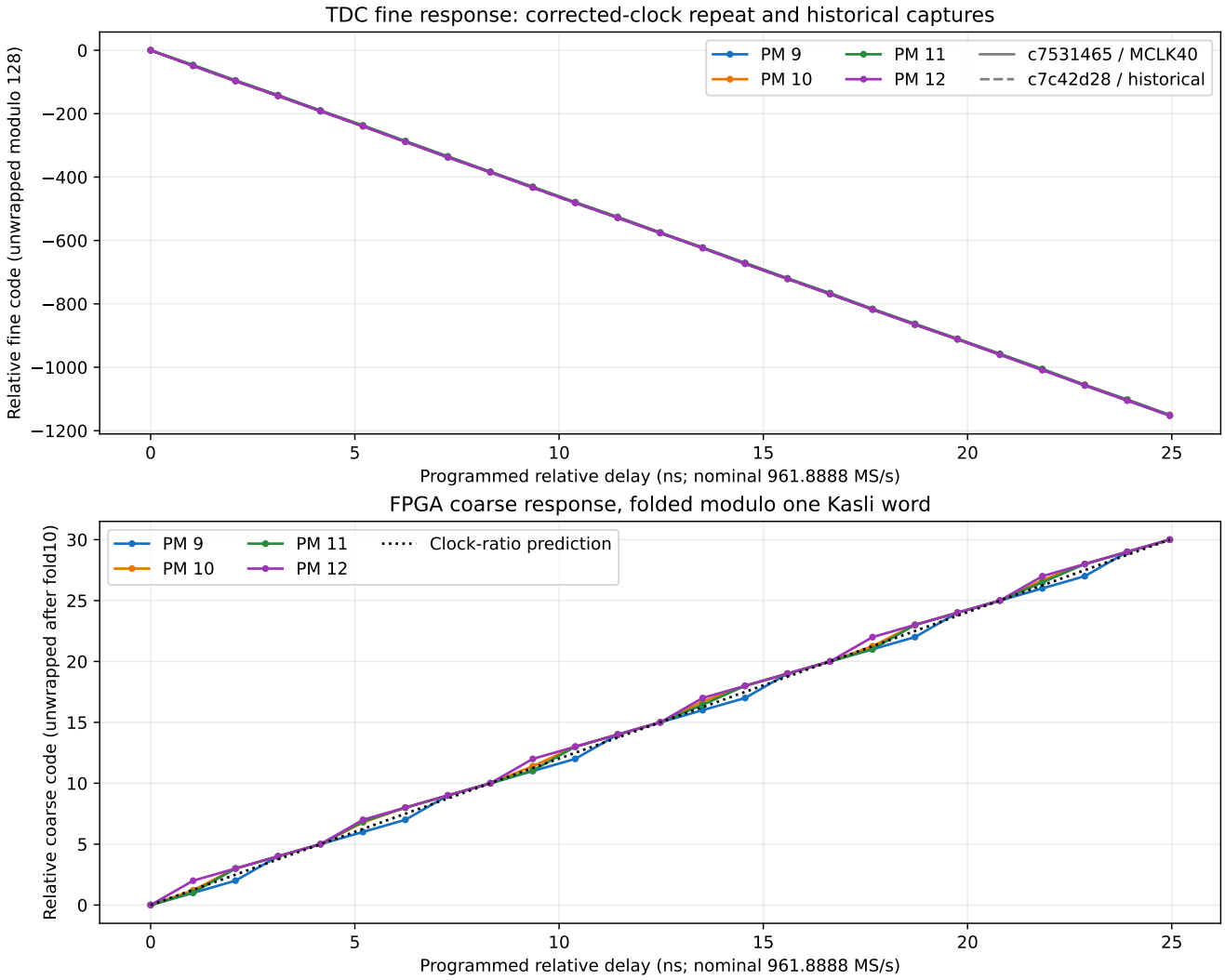


Figure 2: Correct-clock relative fine and coarse response; all four channels overlaid.

3 Direct SYS clock-loss capture after SPI

A temporary PM SRAM diagnostic recorded the SPI inputs, fixed SYS MMCM LOCK, CAP MMCM LOCK and SYS reset on a clock independent of SYS. It survived SYS reset. One idle control and five signature reads yielded two captured reset events (after reads1 and0). Every transmitted frame contains64sampled SCK transitions. No output release or card-direction change occurred, and no UART/PLL command was sent during these captures.

In both events fixed SYS MMCM LOCK drops and SYS reset asserts. CAP MMCM LOCK remains high. Fixed LOCK returns and SYS reset releases two watch-clock ticks later. This directly establishes the clock-domain reset path; it does not establish a FIN defect, MCU action, or external PLL failure. Narrower input glitches can be missed.

The watch clock is approximately SYS frequency (nominal40.0787MHz), despite120MHz source/CSR metadata. This follows independently from the idle watch/SYS counter ratio and200watch ticks per600Kasli-tick SPI half-cycle. This is not counter calibration. With that nominal scale the recorded LOCK-low intervals are8.483/8.508us, after9.826/9.513ms of running watch time from HSEL deassertion. A simultaneous stop of watch and MGT clocks is not represented in those times. Clock input assumptions need separate audit before rated-throughput or absolute timing qualification; no CDCE output settings were changed.

The DIO schematic identifies FIN1101K8X repeaters and100ohm terminations on both sides. Open-circuit fail-safe is not proof of a defined disconnected-but-terminated input. Correct3.3V supply alone is not a CMOS/LVDS mismatch. Oscilloscope observation at the FIN/HDMI path and MGT clock is still needed to identify the electrical cause.

Exact golden c7c42d28 SRAM was restored and physical ADC, stable SYS and PGOOD checked. Kasli SPI remains driven at idle0 with slot6mask0x00b0. Existing campaign case statuses remain unchanged. Raw records, HDL, simulation, source/bit archive and checksum qualification are retained in system/evidence-clock-observer-20261113.

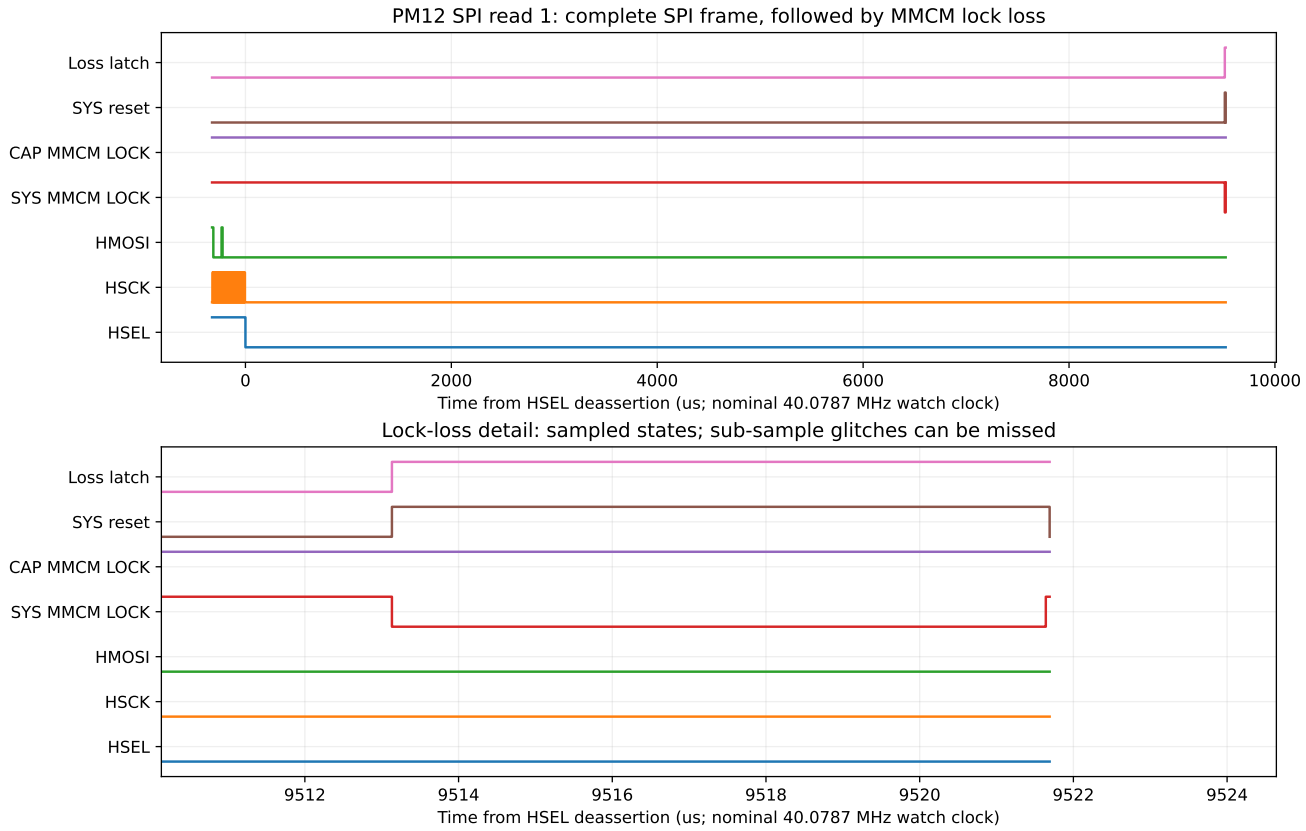


Figure 3: SPI read1 and later SYS MMCM lock loss. Time scale assumes a continuously running40.0787MHz observation clock.

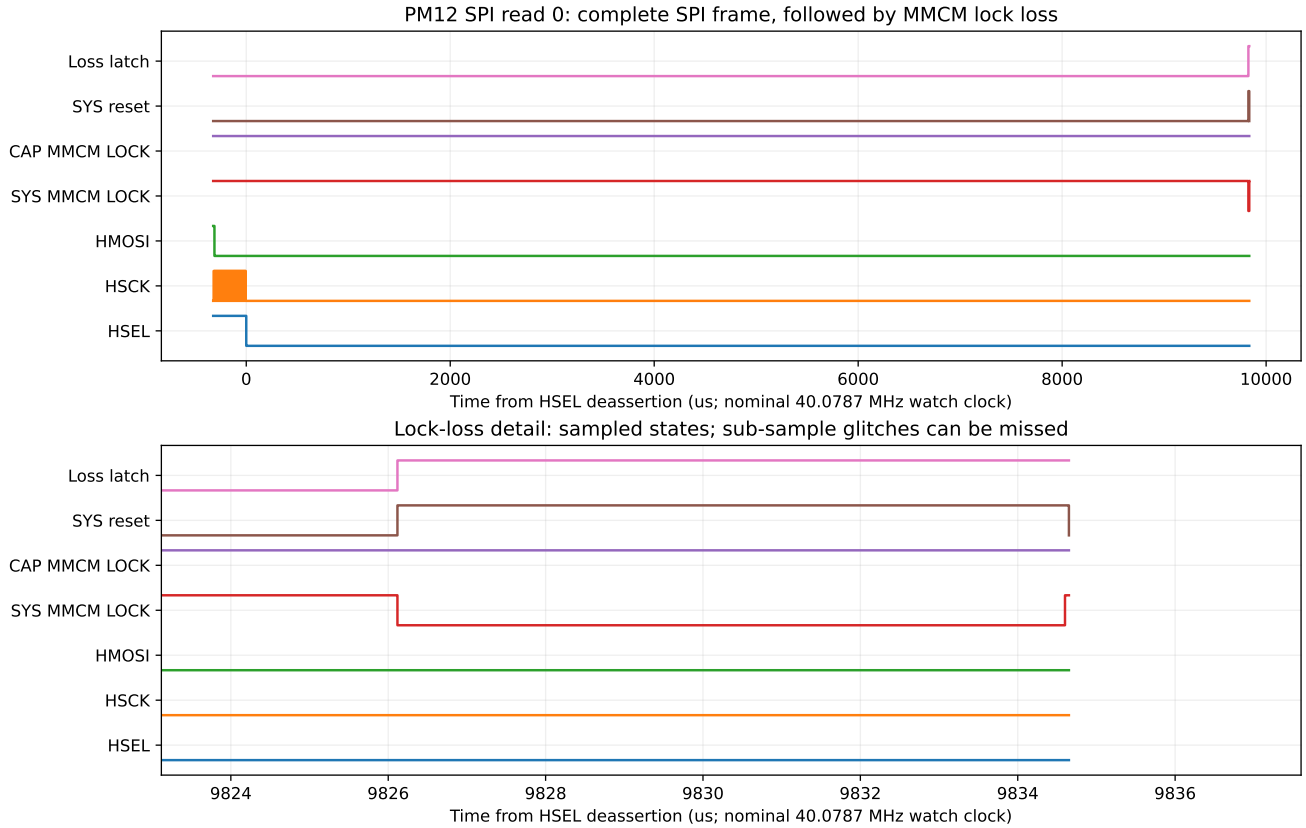


Figure 4: Independent repeat after SPI read0. No release of Kasli outputs occurs.

4 Later follow-up: SPI outputs held driven

The earlier causal attribution to FPGA pad release is superseded. It remains a timestamped observation in the historical results below, not an established root cause. Full HDMI remains FAIL; campaign coverage is unchanged.

Three repeated HDMI tests with continuously enabled Kasli outputs still restart the PM SYS counter. A separate experiment with unchanged card directions and output enable confirms that twelve HDMI mode changes and idle intervals are monotonic. Delayed resets follow SPI reads1 and2 without any output release. All three signature values remain correct.

The MCU–FPGA SPI registers retain scratch0 and write count33 across SYS restarts. The generated c7c42d28 design resets SYS from loss of its fixed-clock MMCM LOCK; HDMI SPI has no logical reset connection to SYS. These observations support a clock-domain reset rather than whole-chip reconfiguration. They do not identify why the clock path is disturbed or electrically prove the state of PROGRAM/INIT/DONE.

Current state supersedes the earlier cleanup state: Kasli HDMI output enable1, slot6 direction mask0x00b0, SEL/SCK/MOSI driven idle0, HDMI mode0. Slot7 and analog stimuli were not changed. A final twelve-sample held-idle check is monotonic. Do not use the legacy integrated runner or global STOP/input-card cleanup on this held setup.

Raw evidence and executed scripts are in the sibling system/evidence-held-spi-20261113 directory in the repository. The current station entry uses hold_spi.py and retains output enable and card directions. No programming, persistent writes or clock-profile changes were made.

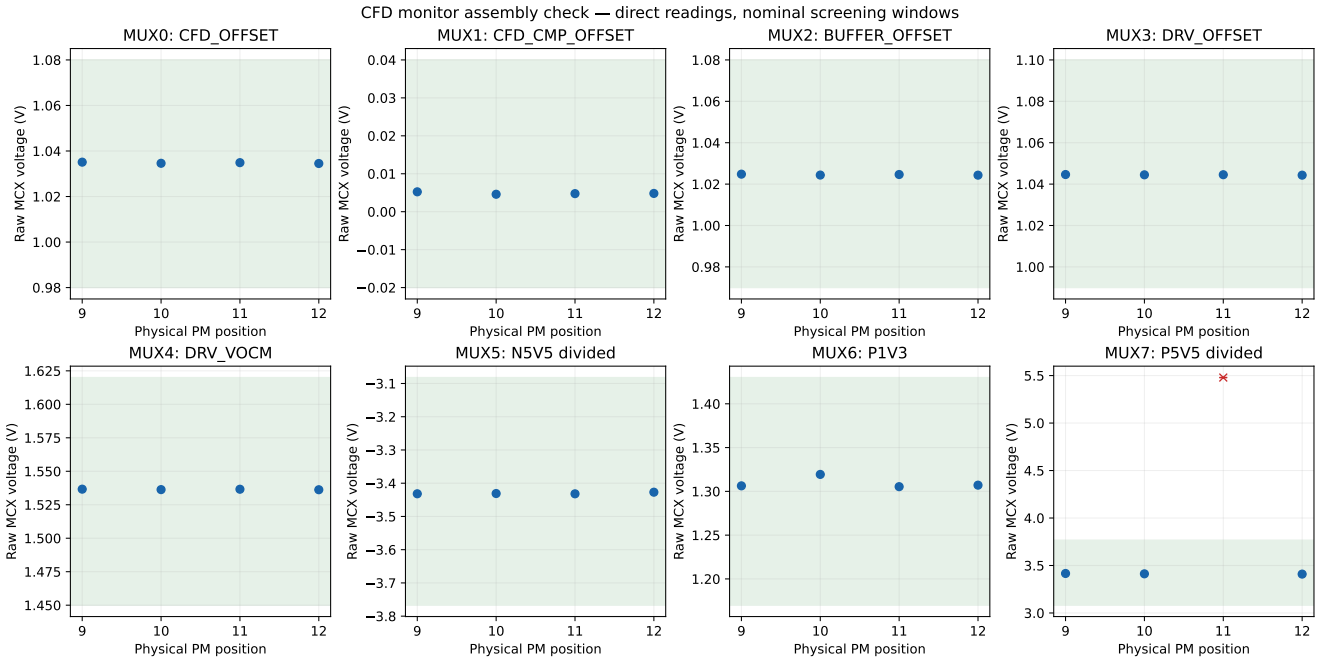
5 Assembly verification: current fitted group9–12

The present campaign is a functional assembly check. Unmeasured fixtures or missing stimuli do not receive PASS. The current group has two observed discrepancies; this document is not an unconditional board release.

Case	Observed result	Disposition
CFD monitor9/10/12	All eight inputs pass	PASS
CFD monitor11	Raw MUX7=5.479V; other modules about3.41V	FAIL: fixture branch/variant
JTAG/static links	168/168 in each independent path	PASS: scoped static check
CPU/FPGA and peripherals	32walking patterns restored; IDs/PGOOD correct	PASS
I2C rails and telemetry	Eight populated rail voltages in nominal windows	PASS; nominal currents
Ethernet/IPbus	393216ADC payload bytes checked; stable ADC/TDC transfers	PASS: functional transport
HDMI data lanes	Walking, slow/fast PRBS and idle frames correct	Data checks PASS
HDMI shared SPI release	SYS resets after pad release following reads0/1/2	Overall HDMI FAIL
Analog channel9	USB remains on10	NOT MEASURED; owner move pending

5.1 CFD monitor discrepancy is on the tester

The emulator at PM position11 (provisional CFD02) gives about5.479V directly at MUX7. Independent revisits reproduce it; position12 gives about3.410V. Check CFD SIM R3=33kOhm to ground, R4=20kOhm from P5V5 and their joints, or confirm an intended assembly variant. The nominal53/33 correction would yield8.80V, but that is not a measured8.80V PM supply. The independent PM INA bus reading is5.496V.



PM position	Mux	Signal	Raw MCX(V)	Nominal equivalent(V)	Screen
9	0	CFD_ OFFSET	1.035097	1.035097	within window
10	0	CFD_ OFFSET	1.034600	1.034600	within window
11	0	CFD_ OFFSET	1.034880	1.034880	within window
12	0	CFD_ OFFSET	1.034510	1.034510	within window
9	1	CFD_CMP_ OFFSET	0.005240	0.005240	within window
10	1	CFD_CMP_ OFFSET	0.004597	0.004597	within window
11	1	CFD_CMP_ OFFSET	0.004763	0.004763	within window
12	1	CFD_CMP_ OFFSET	0.004817	0.004817	within window
9	2	BUFFER_ OFFSET	1.024807	1.024807	within window
10	2	BUFFER_ OFFSET	1.024370	1.024370	within window
11	2	BUFFER_ OFFSET	1.024637	1.024637	within window
12	2	BUFFER_ OFFSET	1.024320	1.024320	within window
9	3	DRV_ OFFSET	1.044630	1.044630	within window
10	3	DRV_ OFFSET	1.044490	1.044490	within window
11	3	DRV_ OFFSET	1.044540	1.044540	within window
12	3	DRV_ OFFSET	1.044317	1.044317	within window
9	4	DRV_VOCM	1.536640	1.536640	within window
10	4	DRV_VOCM	1.536307	1.536307	within window
11	4	DRV_VOCM	1.536590	1.536590	within window
12	4	DRV_VOCM	1.536213	1.536213	within window
9	5	N5V5 divided	-3.431790	-5.511663	within window
10	5	N5V5 divided	-3.431030	-5.510442	within window
11	5	N5V5 divided	-3.431913	-5.511861	within window
12	5	N5V5 divided	-3.427043	-5.504039	within window
9	6	P1V3	1.306307	1.306307	within window
10	6	P1V3	1.319360	1.319360	within window
11	6	P1V3	1.305370	1.305370	within window
12	6	P1V3	1.307083	1.307083	within window
9	7	P5V5 divided	3.416020	5.486335	within window
10	7	P5V5 divided	3.412337	5.480419	within window
11	7	P5V5 divided	5.479303	8.800093	OUTSIDE WINDOW
12	7	P5V5 divided	3.409380	5.475671	within window

Nominal equivalent uses53/33 only for the divided rails. On the failed branch that assumption is suspect; use the direct reading for diagnosis. This is an assembly window check, not traceable calibration or a

measurement of analog off-resistance.

5.2 HDMI reset isolation and restored operating state

Both full integrated runs retain the unexpected reset. Controlled isolation observes no reset during an idle-only enable/release or during the signature reads themselves. The counter clears when FPGA pads are released after SPI reads, before the DIO direction restoration. The exact electrical/MCU interaction remains unresolved; it must not be called a PM assembly defect solely from this test. Full HDMI remains FAIL until resolved. No arbitrary SPI writes were used.

Isolation event	Raw64-bit SYS uptime
initial	18902352588
before FPGA pad release	18902357244
after FPGA pad release	18903171692
outputs enabled	18918616796
idle inside	18918621884
before FPGA pad release	18918626578
after FPGA pad release	18919443448
after idle direction restore	18930711667
before FPGA pad release	18930722089
after FPGA pad release	18931546991
outputs enabled	18947031189
before read 0	18947035621
after read 0	18947072587
before read 1	18947077350
after read 1	18947107457
before read 2	18947110108
after read 2	18947137010
before FPGA pad release	18947139746
after FPGA pad release	40569538
after SPI direction restore	51888040
before FPGA pad release	51894323
after FPGA pad release	52713683

After isolation, fresh ADC and TDC transport checks passed with no SYS reset. The JTAG wrapper retains its ERROR from the combined HDMI guard; JTAG PASS here comes from independently rebuilt static vectors and checksum-verified restored SRAM followed by those stable checks. The separate HDMI failure remains visible.

Firmware still prints the old R15-not-fitted label for P3V8. The owner confirmed R15 fitted; the schematic specifies1mOhm. Host conversion of the retained signed shunt code at40uV/LSB gives about3.12A, uncalibrated. The original firmware text is retained. No MCU firmware was changed; route6/channel3 is unpopulated and excluded. Power-sequence fault injection is not performed.

Final state: qualified c7c42d28 diagnostic in FPGA SRAM, private PM bridge active, HDMI/CFD outputs released, DIO cards input, ADC bankA physical capture and frozen quiet TDC. SMU OFF/HIZ at0A, all four monitor muxes inhibited and TCA masks0. USB cables stay on10. Prior analog measurements retain their own acquisition settings; they are not silently remeasured after reconfiguration.

6 Scope and recorded coverage

Board PM12-K7-3A17A1B402212223. Campaign 5ef2e2b4-c04a-4fad-b86f-a23be621d7a7.

Recorded status: INCOMPLETE. Missing tests are NOT MEASURED. Complete selected-protocol coverage is not a calibrated whole-board release.

Channel	cfd-monitor	cfd-dacs	gate-phase	tdc	adc-dc	adc-lines	integrator
1	NM	NM	NM	NM	NM	NM	NM
2	NM	NM	NM	NM	NM	NM	NM
3	NM	NM	NM	NM	NM	NM	NM
4	NM	NM	NM	NM	NM	NM	NM
5	NM	NM	NM	NM	NM	NM	NM
6	NM	NM	NM	NM	NM	NM	NM
7	NM	NM	NM	NM	NM	NM	NM
8	NM	NM	NM	NM	NM	NM	NM
9	PASS	PASS	PASS	PASS	NM	NM	NM
10	PASS	PASS	PASS	PASS	PASS	PASS	PASS
11	FAIL	PASS	PASS	PASS	PASS	PASS	PASS
12	PASS	PASS	PASS	PASS	PASS	PASS	PASS

NM = not measured; INC = incomplete; ERR = acquisition error. No missing series is invented.

6.1 Board-level tests

Test	Status
Boundary scan and CPLD–FPGA connections	PASS
MCU, CPU–FPGA communication and peripheral IDs	PASS
I2C sensors, currents and voltages	PASS
HDMI digital link	FAIL
Ethernet/bridge diagnostic	PASS

7 Fitted batches and fixture provenance

Batch attempt	PM channel	CFD label	Declared connection	Fitting
ch09-12-attempt001	12	CFD01	slot 7, RJ45 0	Identity/fitting confirmed
ch09-12-attempt001	11	CFD02	slot 7, RJ45 1	Identity/fitting confirmed
ch09-12-attempt001	10	CFD03	slot 7, RJ45 2	Identity/fitting confirmed
ch09-12-attempt001	9	CFD04	slot 7, RJ45 3	Identity/fitting confirmed

Fixture labels CFD01–CFD04 are provisional bench labels. Factory identities and photos remain unknown until supplied; a label is not proof of module identity. Declared cable routes must be checked physically.

Raw inputs and selected result IDs are retained in inputs/. Repeated measurements are never silently substituted.

8 Retained attempts and explicit selection

Result ID prefix	Test	Channel	Fitting	Status	Selected
result-b9126235	adc-dc	12	ch09-12-attempt001	PASS	yes
result-ee0613ea	adc-lines	12	ch09-12-attempt001	PASS	yes
result-217f5cb4	integrator	12	ch09-12-attempt001	PASS	yes
result-b99efe3a	cfd-dacs	9	ch09-12-attempt001	INCOMPLETE	no
result-ef50493e	cfd-dacs	10	ch09-12-attempt001	INCOMPLETE	no
result-c4800a66	cfd-dacs	11	ch09-12-attempt001	INCOMPLETE	no
result-c80f3b9e	cfd-dacs	12	ch09-12-attempt001	INCOMPLETE	no
result-9e64b43e	cfd-dacs	9	ch09-12-attempt001	PASS	yes
result-0fcaadd7	cfd-dacs	10	ch09-12-attempt001	PASS	yes
result-8b6b8aa7	cfd-dacs	11	ch09-12-attempt001	PASS	yes
result-b4c6c497	cfd-dacs	12	ch09-12-attempt001	PASS	yes
result-aa2f2b7c	adc-dc	11	ch09-12-attempt001	PASS	yes
result-fa68f809	adc-lines	11	ch09-12-attempt001	PASS	yes
result-021f9e97	integrator	11	ch09-12-attempt001	PASS	yes
result-317b8d60	adc-dc	10	ch09-12-attempt001	PASS	yes
result-0386d766	adc-lines	10	ch09-12-attempt001	PASS	yes
result-e87c1c5e	integrator	10	ch09-12-attempt001	PASS	yes
result-1642153e	tdc	12	ch09-12-attempt001	PASS	yes
result-5f833ba0	tdc	11	ch09-12-attempt001	PASS	yes
result-cdf2341e	tdc	10	ch09-12-attempt001	PASS	yes
result-5e3a4c64	tdc	9	ch09-12-attempt001	PASS	yes
result-d2e5cd6c	gate-phase	12	ch09-12-attempt001	PASS	yes
result-769a5717	gate-phase	11	ch09-12-attempt001	PASS	yes
result-5b389adb	gate-phase	10	ch09-12-attempt001	PASS	yes
result-a00721bb	gate-phase	9	ch09-12-attempt001	PASS	yes
result-a2a60491	cfd-monitor	9	ch09-12-attempt001	PASS	yes

Result ID prefix	Test	Channel	Fitting	Status	Selected
result-161d19ce	cfd-monitor	10	ch09-12-attempt001	PASS	yes
result-fb9647b0	cfd-monitor	11	ch09-12-attempt001	FAIL	yes
result-88ebf7d4	cfd-monitor	12	ch09-12-attempt001	PASS	yes
result-4f22ebeb	jtag	board	ch09-12-attempt001	PASS	yes
result-5ab3262e	mcu	board	ch09-12-attempt001	PASS	yes
result-0a5dcd9c	i2c	board	ch09-12-attempt001	PASS	yes
result-6231a067	ethernet	board	ch09-12-attempt001	PASS	yes
result-10b214db	hdmi	board	ch09-12-attempt001	FAIL	yes

All attempts, including failures and rejected repeats, remain in the archived inputs. The coverage matrix uses only the explicitly selected attempt.

9 Selected measurement conditions and limits

Test	Physical channel	Confirmed stimulus connection
adc-dc	12	{"dac_channels": [0, 1], "stimulus_channel": 12, "usb_input": "J8"}
adc-lines	12	{"dac_channels": [0, 1], "stimulus_channel": 12, "usb_input": "J8"}
integrator	12	{"dac_channels": [2, 3], "stimulus_channel": 12, "usb_input": "J6"}
cfd-dacs	9	{"selection": "one monitor mux enabled; all others inhibited", "shared_monitor_channels": [9, 10, 11, 12]}
cfd-dacs	10	{"selection": "one monitor mux enabled; all others inhibited", "shared_monitor_channels": [9, 10, 11, 12]}
cfd-dacs	11	{"selection": "one monitor mux enabled; all others inhibited", "shared_monitor_channels": [9, 10, 11, 12]}
cfd-dacs	12	{"selection": "one monitor mux enabled; all others inhibited", "shared_monitor_channels": [9, 10, 11, 12]}

Test	Physical channel	Confirmed stimulus connec tion
adc-dc	11	{”dac_channels”: [0, 1], ”stimulus_channel”: 11, ”usb_input”: ”J8”}
adc-lines	11	{”dac_channels”: [0, 1], ”stimulus_channel”: 11, ”usb_input”: ”J8”}
integrator	11	{”dac_channels”: [2, 3], ”stimulus_channel”: 11, ”usb_input”: ”J6”}
adc-dc	10	{”dac_channels”: [0, 1], ”stimulus_channel”: 10, ”usb_input”: ”J8”}
adc-lines	10	{”dac_channels”: [0, 1], ”stimulus_channel”: 10, ”usb_input”: ”J8”}
integrator	10	{”dac_channels”: [2, 3], ”stimulus_channel”: 10, ”usb_input”: ”J6”}
tdc	12	{”phase_lanes”: [8, 9], ”rj45_port”: 0, ”slot”: 7, ”stimulus_dio”: 0}
tdc	11	{”phase_lanes”: [10, 11], ”rj45_port”: 1, ”slot”: 7, ”stimulus_dio”: 4}
tdc	10	{”phase_lanes”: [12, 13], ”rj45_port”: 2, ”slot”: 7, ”stimulus_dio”: 8}
tdc	9	{”phase_lanes”: [14, 15], ”rj45_port”: 3, ”slot”: 7, ”stimulus_dio”: 12}
gate-phase	12	{”phase_lanes”: [8, 9], ”rj45_port”: 0, ”slot”: 7, ”stimulus_dio”: 0}
gate-phase	11	{”phase_lanes”: [10, 11], ”rj45_port”: 1, ”slot”: 7, ”stimulus_dio”: 4}
gate-phase	10	{”phase_lanes”: [12, 13], ”rj45_port”: 2, ”slot”: 7, ”stimulus_dio”: 8}
gate-phase	9	{”phase_lanes”: [14, 15], ”rj45_port”: 3, ”slot”: 7, ”stimulus_dio”: 12}
cfd-monitor	9	{”selection”: ”one of four muxes active at a time”, ”shared_monitor_channels”: [9, 10, 11, 12]}
cfd-monitor	10	{”selection”: ”one of four muxes active at a time”, ”shared_monitor_channels”: [9, 10, 11, 12]}

Test	Physical channel	Confirmed stimulus connec tion
cfd-monitor	11	{”selection”: ”one of four muxes active at a time”, ”shared_monitor_channels”: [9, 10, 11, 12]}
cfd-monitor	12	{”selection”: ”one of four muxes active at a time”, ”shared_monitor_channels”: [9, 10, 11, 12]}
jtag	board	{}
mcu	board	{}
i2c	board	{}
ethernet	board	{}
hdmi	board	{}

9.1 Common hardware configuration

Setting	Recorded value
clock_profile.adc_rate	2x reference
clock_profile.kasli_rate	3x reference
clock_profile.reference_mhz	40.0787
cpld_gate_mode	original INTA/INTB enabled, nomina l20MHz
kasli_build.ABI	CFD4
kasli_build.expected_bit_sha256	a71a08a258620b0e5a52c5d03aefd8c69040 c9d15e8de6391c484ac182b7173b
pm_build	c7c42d28

9.2 adc-dc

Setting	Recorded value
method	”direct-ADC-DC-v1”
stimulus_points_per_bank	257
commanded_range_V	[-1.2, 1.2]
common_mode_V	1.0
dac_channels	[0, 1]
fit_interval_V	[-0.8, 0.8]
fft_sample_rate_assumed_Hz	80157400
integrator_input_parked_V	[2.0480001, 1.02400005]

9.3 adc-lines

Setting	Recorded value
method	”adjacent-pair-states-v1”
stimulus_points_per_bank	257
commanded_range_V	[-1.2, 1.2]
common_mode_V	1.0
dac_channels	[0, 1]

Setting	Recorded value
fit_interval_V	[-0.8, 0.8]
fft_sample_rate_assumed_Hz	80157400
integrator_input_parked_V	[2.0480001, 1.02400005]

9.4 cfd-dacs

Setting	Recorded value
method	”shared-monitor-SMU-BCD-v1”
instrument_identity	”Keysight Technologies,B2910BL,MY6332 0299,5.0.2429.8966”
sense	”Force HI/LO, local”
source_current_A	0
source_range_A	1e-06
voltage_range_V	20
nplc	1
settling_s	0.1
points	101
repeats	3
limits	{”repeat_std_V”: 0.001, ”reverse_step_ V”: 0.0002, ”span_relative_error”: 0.05}
post_scan_verification	”SMU OFF/HIZ, four monitor muxes inhibi ted, explicit TCA masks0”

9.5 cfd-monitor

Setting	Recorded value
method	”eight-monitor-assembly-v1”
windows_V	[[0.98, 1.08], [-0.02, 0.04], [0.97, 1.08], [0.99, 1.1], [1.45, 1.62], [-6.05, -4.95], [1.17, 1.43], [4.95, 6.05]]
relative_offset_windows_V	[[0.005, 0.016], [0.012, 0.03]]
vocm_ratio_window	[1.45, 1.55]
raw_repeat_std_limit_V	0.002
divider_correction	1.606060606060606
settle_s	0.2
samples_per_input	3
source_current_A	0
sense	”local Force HI/LO”
nplc	1
range_V	20

9.6 ethernet

Setting	Recorded value
method	”board-assembly-ethernet-v1”
scope	”separate static-link/CPU/sensor/network/ HDMI cases; shared-SPI clock guard retain ed”

9.7 gate-phase

Setting	Recorded value
method	"DAC-A-ISERDES-gate-phase-v1"
codes	[0, 655, 1311, 1966, 2621, 3277, 3932, 4587, 5243, 5898, 6554, 7209, 7864, 8520, 9175, 9830, 10486, 11141, 11796, 12452, 13107, 13762, 14418, 15073, 15728, 16384, 17039, 17694, 18350, 19005, 19661, 20316, 20971, 21627, 22282, 22937, 23593, 24248, 24903, 25559, 26214, 26869, 27525, 28180, 28835, 29491, 30146, 30801, 31457, 32112, 32768, 33423, 34078, 34734, 35389, 36044, 36700, 37355, 38010, 38666, 39321, 39976, 40632, 41287, 41942, 42598, 43253, 43908, 44564, 45219, 45875, 46530, 47185, 47841, 48496, 49151, 49807, 50462, 51117, 51773, 52428, 53083, 53739, 54394, 55049, 55705, 56360, 57015, 57671, 58326, 58982, 59637, 60292, 60948, 61603, 62258, 62914, 63569, 64224, 64880, 65535]
sample_hz	961888800.0
gate_period_bins	48
capture_words	4095
minimum_functional_excursion_ns	5

9.8 hdmi

Setting	Recorded value
method	"board-assembly-hdmi-v1"
scope	"separate static-link/CPU/sensor/network/HDMI cases; shared-SPI clock guard retained"

9.9 i2c

Setting	Recorded value
method	"board-assembly-i2c-v1"
scope	"separate static-link/CPU/sensor/network/HDMI cases; shared-SPI clock guard retained"

9.10 integrator

Setting	Recorded value
method	"signed-integration-pair-v1"
dac_channels	[2, 3]
base_V	[1.024, 2.048]

Setting	Recorded value
coarse_offsets_V	[-0.09999999999999987, -0.080000000000 00007, -0.05999999999999983, -0.040000 000000000036, -0.019999999999999796, -0.010000000000000009, 0.0, 0.009999999999 999787, 0.02000000000000024, 0.040000 000000000036, 0.060000000000000275, 0.08000000000000007, 0.10000000000000009]
fine_method	”26 points,0.4mV step around coarse estima te”
return_method	”two candidate revisits and +/-5mV offset s;near-zero <=2codes; signed response >4codes”

9.11 jtag

Setting	Recorded value
method	”board-assembly-jtag-v1”
scope	”separate static-link/CPU/sensor/network/ HDMI cases; shared-SPI clock guard retain ed”

9.12 mcu

Setting	Recorded value
method	”board-assembly-mcu-v1”
scope	”separate static-link/CPU/sensor/network/ HDMI cases; shared-SPI clock guard retain ed”

9.13 tdc

Setting	Recorded value
method	”finite-OSERDES-TDC-phase-v1”
phase_slots	[24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48]
sample_hz	961888800.0
events_per_phase	128
discovery_events	16
response_slope_window	[-60, -20]
maximum_residual_fine_codes	10

Commanded DAC voltages, whole connected chain; not calibrated ADC-only INL/noise or electrical continuity proof.

Effective differential whole-chain balance; waveform-based integration pair, not an absolute gate timestamp or calibrated opamp bias.

Fine modulo128 unwrapped by nearest continuation; coarse fields retained without repair. Progra mmed slot1.0396ns assumes40.0787MHz; not calibrated TDC timing or intrinsic jitter.

Functional assembled-chain transfer check: span within5% of schematic scaling, reverse step <=0.2mV, repeat standard deviation <=1mV. Not traceable calibration or exhaustive16-bit INL/DNL. DAC-A unchanged; exercised B/C/D end at0000..

Functional assembly evidence; no supply fault injection, calibrated current accuracy, Ethernet throug hput/BER or unconnected analog-channel acceptance.

Functional assembly windows and relative reference checks; negative/positive supply dividers correc ted by53/33. Three fresh0A-SMU voltage readings per input; no traceable voltage calibration or analog off-resistance claim.

Occupancy-weighted sampled phase centroid; nominal1.0396ns bins and49.9ns gate period. Neares t-continuation unwrap; discontinuities retained. No calibrated sub-bin precision or independent-edge sqrt(N) uncertainty. DAC A commanded0000 at cleanup, without hardware code readback.

integrator; physical channel 10: The coarse waveform assignment produces an additional positive-going sign crossing near +40mV. It was not accepted as balance. Fine acquisition used the central negati ve-going crossing for this reversed input coordinate, then confirmed it with independent candidate revisi ts and +/-5mV controls. The original coarse data and both controller versions remain archived.

cfd-monitor; physical channel 11: Fixture anomaly on CFD emulator mounted at PM11 (provis ional CFD02): raw MUX7 reads5.479V, versus3.41V on9/12, confirmed by two five-reading revisits. Nominal53/33 scaling would give8.80V but this is NOT a measured8.80V PM supply. Check CFD_SIM R3=33kOhm to ground, R4=20kOhm fromP5V5 and their joints/variant. Other monitor inputs and independent board P5V5 bus about5.496V are normal. This FAIL concerns the tester monitor branch, not proof of PM channel11 failure.

jtag; physical channel board: Legacy combined boundary wrapper reports ERROR because its post-restore HDMI check resets SYS. The independently rebuilt338vectors in each path all PASS and checks um-verified SRAM restoration is confirmed by fresh stable ADC/TDC transfers. This JTAG case is scoped to static links and restoration; the separate HDMI case remains FAIL.

i2c; physical channel board: Firmware still labels R15/P3V8 current as not fitted. Owner confirmed R15 fitted; schematic R15=1mOhm, so host converts the retained signed13-bit shunt code at40uV/LSB. Original NA/status text is retained and no MCU firmware was changed. Route6/channel3 is unpopu lated and excluded.

hdmi; physical channel board: All four HDMI data lanes pass walking patterns, slow/fast PRBS and empty frames. Signature reads0/1/2 are correct, but SYS uptime resets when FPGA pads are released after SPI reads, before DIO direction restoration. Idle-only output enable/release does not reset. Two integrated runs and controlled isolation retain the failure; shared-MCU SPI release remains unqualified and is not assigned to a PM assembly defect without further isolation.

ADC total RMS includes the complete connected chain and periodic gate feedthrough. The four-phase-removed RMS removes the mean of each sample phase; it is a residual statistic, not an intrinsic ADC noise specification. FFT and time axes use the declared clock frequency, not an independent frequency measurement.

10 Measurements and common channel plots

Test	Channel	Bank	Observable	Value	Unit
adc-dc	12	A	gain	-1784.205440671235	code/V
adc-dc	12	A	zero_mean	2162.0811339198435	code
adc-dc	12	A	zero_total_rms	56.22754588987761	code
adc-dc	12	A	zero_four_phase_removed_rms	0.5926371250728474	code
adc-dc	12	B	gain	-1784.2694869545635	code/V

Test	Channel	Bank	Observable	Value	Unit
adc-dc	12	B	zero_mean	2165.847507 3313784	code
adc-dc	12	B	zero_total_ rms	64.408946 78040004	code
adc-dc	12	B	zero_four_ phase_remove d_rms	0.562557 1866355403	code
adc-lines	12	A	adjacent_ pair_states	44	state
adc-lines	12	A	visited_codes	3147	code
adc-lines	12	B	adjacent_ pair_states	44	state
adc-lines	12	B	visited_codes	3184	code
integrator	12	A	balance_delta	0.006030 008721686399	V
integrator	12	A	gain	-2530.411960 665549	code/V
integrator	12	A	near_balanc e_change	0.130164 91045236994	code
integrator	12	B	balance_delta	0.007424 191555263186	V
integrator	12	B	gain	-2448.007789 5617496	code/V
integrator	12	B	near_balanc e_change	0.111943 22217075807	code
cfd-dacs	9		dac_B_ minimum_ code_V	1.044726 6666666666	V
cfd-dacs	9		dac_B_ maximum_ code_V	1.004613 3333333334	V
cfd-dacs	9		dac_B_span_ V	-0.040113 33333333322	V
cfd-dacs	9		dac_B_ maximum_ reverse_step_ V	0	V
cfd-dacs	9		dac_B_ maximum_ repeat_std_V	7.023769 168570249e-05	V
cfd-dacs	9		dac_B_interi or_gain_ error_percent	-0.006716 857635891138	%
cfd-dacs	9		dac_C_ minimum_ code_V	1.034956 6666666667	V
cfd-dacs	9		dac_C_ maximum_ code_V	1.014396 6666666666	V
cfd-dacs	9		dac_C_span_ V	-0.020560 000000000134	V

Test	Channel	Bank	Observable	Value	Unit
cfd-dacs	9		dac_C_maximum_reverse_step_V	0	V
cfd-dacs	9		dac_C_maximum_repeat_std_V	9.504384952928754e-05	V
cfd-dacs	9		dac_C_interior_gain_error_percent	0.8518971864797997	%
cfd-dacs	9		dac_D_minimum_code_V	0.0054	V
cfd-dacs	9		dac_D_maximum_code_V	1.06306	V
cfd-dacs	9		dac_D_span_V	1.0576599999999998	V
cfd-dacs	9		dac_D_maximum_reverse_step_V	0	V
cfd-dacs	9		dac_D_maximum_repeat_std_V	7.000000000000657e-05	V
cfd-dacs	9		dac_D_interior_gain_error_percent	-0.026880429186237897	%
cfd-dacs	10		dac_B_minimum_code_V	1.04437	V
cfd-dacs	10		dac_B_maximum_code_V	1.00438	V
cfd-dacs	10		dac_B_span_V	-0.03998999999997	V
cfd-dacs	10		dac_B_maximum_reverse_step_V	0	V
cfd-dacs	10		dac_B_maximum_repeat_std_V	8.082903768656434e-05	V
cfd-dacs	10		dac_B_interior_gain_error_percent	-0.14222304801190244	%
cfd-dacs	10		dac_C_minimum_code_V	1.0346766666666667	V
cfd-dacs	10		dac_C_maximum_code_V	1.0141433333333334	V

Test	Channel	Bank	Observable	Value	Unit
cfd-dacs	10		dac_C_span_V	-0.020533333333292	V
cfd-dacs	10		dac_C_maximum_reverse_step_V	0	V
cfd-dacs	10		dac_C_maximum_repeat_std_V	9.539392 01416794e-05	V
cfd-dacs	10		dac_C_interior_gain_error_percent	0.582829 3358783654	%
cfd-dacs	10		dac_D_minimum_code_V	0.004546 666666666667	V
cfd-dacs	10		dac_D_maximum_code_V	1.069293 3333333334	V
cfd-dacs	10		dac_D_span_V	1.064746 6666666667	V
cfd-dacs	10		dac_D_maximum_reverse_step_V	0	V
cfd-dacs	10		dac_D_maximum_repeat_std_V	7.000000 000003828e-05	V
cfd-dacs	10		dac_D_interior_gain_error_percent	0.042824 877925129456	%
cfd-dacs	11		dac_B_minimum_code_V	1.044666 6666666666	V
cfd-dacs	11		dac_B_maximum_code_V	1.00461	V
cfd-dacs	11		dac_B_span_V	-0.040056 66666666663	V
cfd-dacs	11		dac_B_maximum_reverse_step_V	0	V
cfd-dacs	11		dac_B_maximum_repeat_std_V	7.549834 435276095e-05	V
cfd-dacs	11		dac_B_interior_gain_error_percent	-0.174320 7749736131	%
cfd-dacs	11		dac_C_minimum_code_V	1.034903 3333333333	V

Test	Channel	Bank	Observable	Value	Unit
cfd-dacs	11		dac_C_maximum_code_V	1.014386666666665	V
cfd-dacs	11		dac_C_span_V	-0.02051666666674	V
cfd-dacs	11		dac_C_maximum_reverse_step_V	0	V
cfd-dacs	11		dac_C_maximum_repeat_std_V	7.234178138066471e-05	V
cfd-dacs	11		dac_C_interior_gain_error_percent	0.17720494086375105	%
cfd-dacs	11		dac_D_minimum_code_V	0.00485	V
cfd-dacs	11		dac_D_maximum_code_V	1.0636333333333334	V
cfd-dacs	11		dac_D_span_V	1.0587833333333334	V
cfd-dacs	11		dac_D_maximum_reverse_step_V	0	V
cfd-dacs	11		dac_D_maximum_repeat_std_V	6.110100926609052e-05	V
cfd-dacs	11		dac_D_interior_gain_error_percent	0.026777441441017125	%
cfd-dacs	12		dac_B_minimum_code_V	1.04463	V
cfd-dacs	12		dac_B_maximum_code_V	1.0044266666666666	V
cfd-dacs	12		dac_B_span_V	-0.040203333333337	V
cfd-dacs	12		dac_B_maximum_reverse_step_V	0	V
cfd-dacs	12		dac_B_maximum_repeat_std_V	4.9328828623155544e-05	V
cfd-dacs	12		dac_B_interior_gain_error_percent	0.06510873936658168	%

Test	Channel	Bank	Observable	Value	Unit
cfd-dacs	12		dac_C_minimum_code_V	1.0348066666666667	V
cfd-dacs	12		dac_C_maximum_code_V	1.0144333333333333	V
cfd-dacs	12		dac_C_span_V	-0.020373333333333354	V
cfd-dacs	12		dac_C_maximum_reverse_step_V	3.3333333335772153e-06	V
cfd-dacs	12		dac_C_maximum_repeat_std_V	0.00011846237095938458	V
cfd-dacs	12		dac_C_interior_gain_error_percent	-1.3950279374863153	%
cfd-dacs	12		dac_D_minimum_code_V	0.00498	V
cfd-dacs	12		dac_D_maximum_code_V	1.06321	V
cfd-dacs	12		dac_D_span_V	1.05823	V
cfd-dacs	12		dac_D_maximum_reverse_step_V	0	V
cfd-dacs	12		dac_D_maximum_repeat_std_V	8.020806277010361e-05	V
cfd-dacs	12		dac_D_interior_gain_error_percent	-0.03786962571341594	%
adc-dc	11	A	gain	-1783.3085324525614	code/V
adc-dc	11	A	zero_mean	2163.592375366569	code
adc-dc	11	A	zero_total_rms	61.45588679712764	code
adc-dc	11	A	zero_four_phase_removed_rms	0.5394259839466975	code
adc-dc	11	B	gain	-1776.8598609607993	code/V
adc-dc	11	B	zero_mean	2158.097751710655	code
adc-dc	11	B	zero_total_rms	59.53654688741656	code

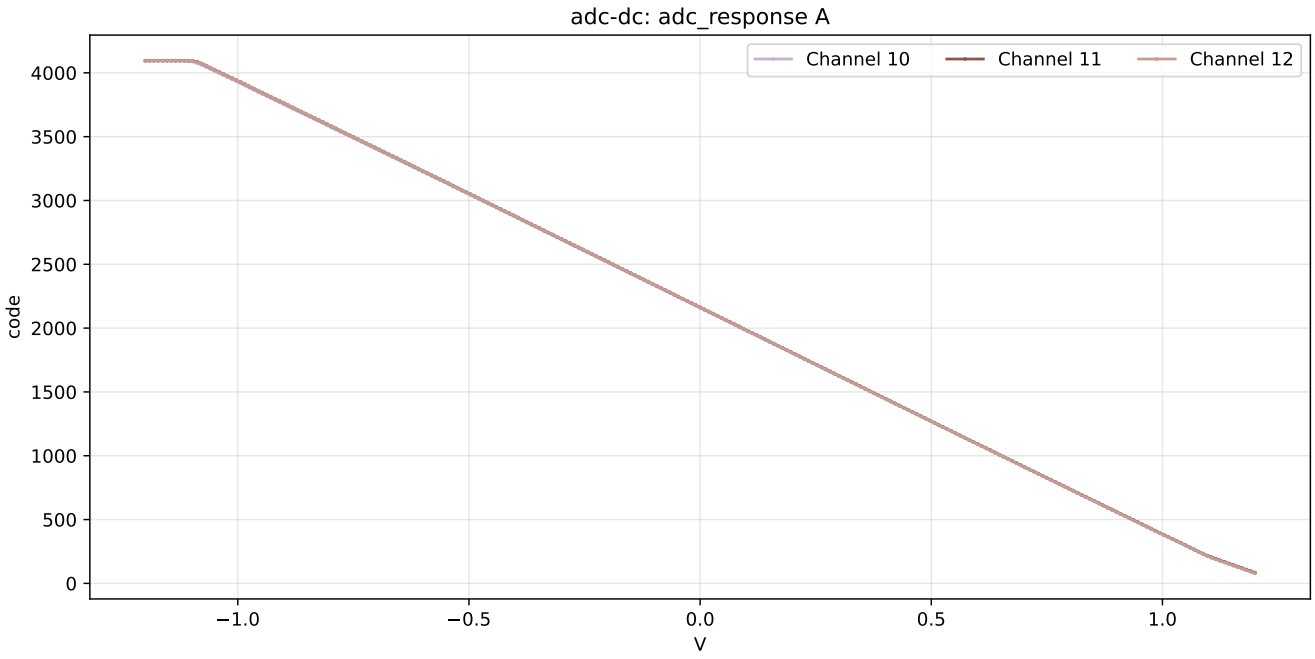
Test	Channel	Bank	Observable	Value	Unit
adc-dc	11	B	zero_four_phase_removed_rms	0.5120989067289364	code
adc-lines	11	A	adjacent_pair_states	44	state
adc-lines	11	A	visited_codes	3175	code
adc-lines	11	B	adjacent_pair_states	44	state
adc-lines	11	B	visited_codes	3237	code
integrator	11	A	balance_delta	0.003963723409567792	V
integrator	11	A	gain	-2558.7140130296752	code/V
integrator	11	A	near_balance_change	-0.13615726802532035	code
integrator	11	B	balance_delta	0.009665397533032197	V
integrator	11	B	gain	-2553.295843675655	code/V
integrator	11	B	near_balance_change	0.1115210552725027	code
adc-dc	10	A	gain	-1777.6734930206578	code/V
adc-dc	10	A	zero_mean	2159.386119257087	code
adc-dc	10	A	zero_total_rms	52.46655577574308	code
adc-dc	10	A	zero_four_phase_removed_rms	0.5128491252775638	code
adc-dc	10	B	gain	-1778.2659098179804	code/V
adc-dc	10	B	zero_mean	2165.9296187683285	code
adc-dc	10	B	zero_total_rms	59.04859019865752	code
adc-dc	10	B	zero_four_phase_removed_rms	0.5082101785128094	code
adc-lines	10	A	adjacent_pair_states	44	state
adc-lines	10	A	visited_codes	3220	code
adc-lines	10	B	adjacent_pair_states	44	state
adc-lines	10	B	visited_codes	3266	code
integrator	10	A	balance_delta	0.0021254808876183984	V
integrator	10	A	gain	-2581.6928210142723	code/V
integrator	10	A	near_balance_change	-0.2340184498996449	code

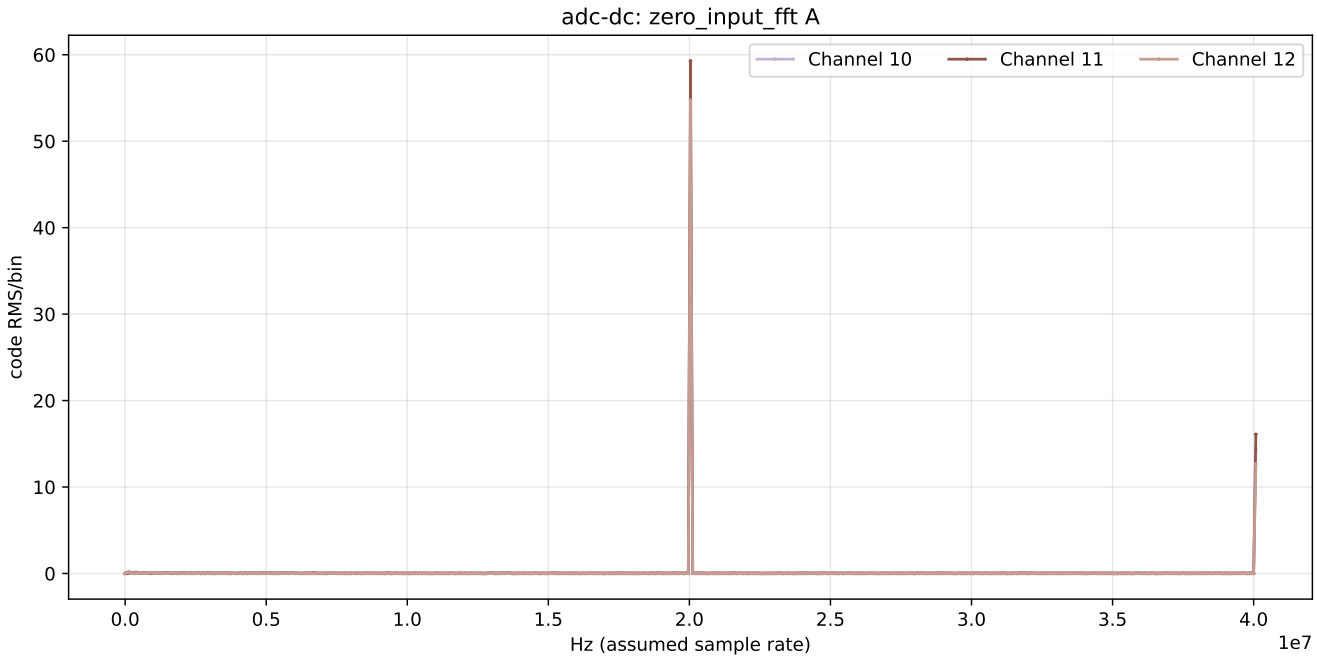
Test	Channel	Bank	Observable	Value	Unit
integrator	10	B	balance_delta	0.008891470489998301	V
integrator	10	B	gain	-2528.717823356551	code/V
integrator	10	B	near_balance_change	0.19092461787864753	code
tdc	12		slope	-46.175872088649626	fine code/ns
tdc	12		max_fit_residual	1.7276416408801651	fine code
tdc	12		max_circular_spread	1.0733590480934954	fine code
tdc	12		events	3200	event
tdc	11		slope	-46.175079944169404	fine code/ns
tdc	11		max_fit_residual	1.6139100538513276	fine code
tdc	11		max_circular_spread	1.1070618583155902	fine code
tdc	11		events	3200	event
tdc	10		slope	-46.17427221876848	fine code/ns
tdc	10		max_fit_residual	1.1208303844473448	fine code
tdc	10		max_circular_spread	1.3046642418206191	fine code
tdc	10		events	3200	event
tdc	9		slope	-46.17113568153328	fine code/ns
tdc	9		max_fit_residual	1.5841465962344046	fine code
tdc	9		max_circular_spread	1.0614654925174123	fine code
tdc	9		events	3200	event
gate-phase	12	A	valid_points	101	point
gate-phase	12	A	observed_excursion	19.75280302671161	ns
gate-phase	12	A	end_to_end	-19.75280302671161	ns
gate-phase	12	A	largest_step	1.0396212119321913	ns
gate-phase	12	A	max_spread	0.4376432508078598	ns
gate-phase	12	B	valid_points	101	point
gate-phase	12	B	observed_excursion	19.75280302671161	ns
gate-phase	12	B	end_to_end	-19.75280302671161	ns
gate-phase	12	B	largest_step	1.0396212119321913	ns
gate-phase	12	B	max_spread	0.5191642461901361	ns
gate-phase	11	A	valid_points	101	point

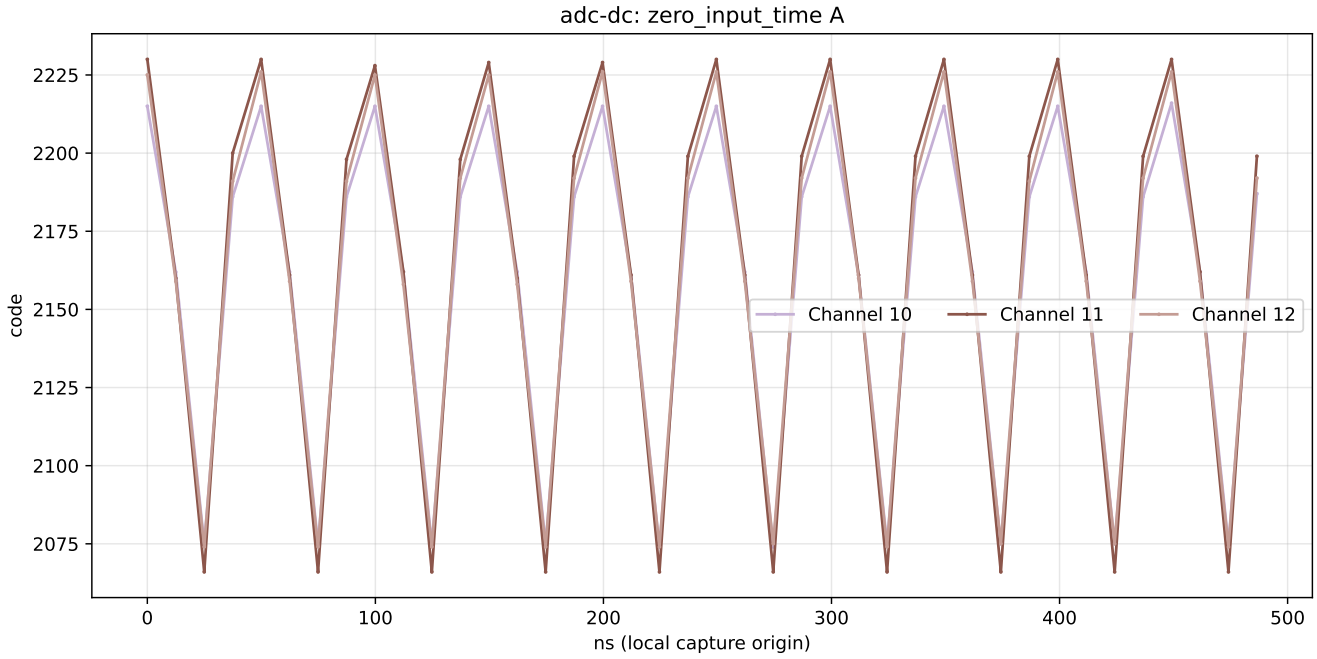
Test	Channel	Bank	Observable	Value	Unit
gate-phase	11	A	observed_excursion	20.905062554431478	ns
gate-phase	11	A	end_to_end	-20.905062554431478	ns
gate-phase	11	A	largest_step	1.0396212119321913	ns
gate-phase	11	A	max_spread	0.5173708034731153	ns
gate-phase	11	B	valid_points	101	point
gate-phase	11	B	observed_excursion	20.7924242386438	ns
gate-phase	11	B	end_to_end	-20.7924242386438	ns
gate-phase	11	B	largest_step	1.0396212119321913	ns
gate-phase	11	B	max_spread	0.5150593173424455	ns
gate-phase	10	A	valid_points	101	point
gate-phase	10	A	observed_excursion	21.832045450575993	ns
gate-phase	10	A	end_to_end	-21.832045450575993	ns
gate-phase	10	A	largest_step	1.039621211932193	ns
gate-phase	10	A	max_spread	0.5197010721490758	ns
gate-phase	10	B	valid_points	101	point
gate-phase	10	B	observed_excursion	20.7924242386438	ns
gate-phase	10	B	end_to_end	-20.7924242386438	ns
gate-phase	10	B	largest_step	1.0396212119321913	ns
gate-phase	10	B	max_spread	0.5017937590614953	ns
gate-phase	9	A	valid_points	101	point
gate-phase	9	A	observed_excursion	20.7924242386438	ns
gate-phase	9	A	end_to_end	-20.7924242386438	ns
gate-phase	9	A	largest_step	1.0396212119321913	ns
gate-phase	9	A	max_spread	0.4529892999602304	ns
gate-phase	9	B	valid_points	101	point
gate-phase	9	B	observed_excursion	20.7924242386438	ns
gate-phase	9	B	end_to_end	-20.7924242386438	ns
gate-phase	9	B	largest_step	1.0396212119321913	ns
gate-phase	9	B	max_spread	0.5175945457149657	ns

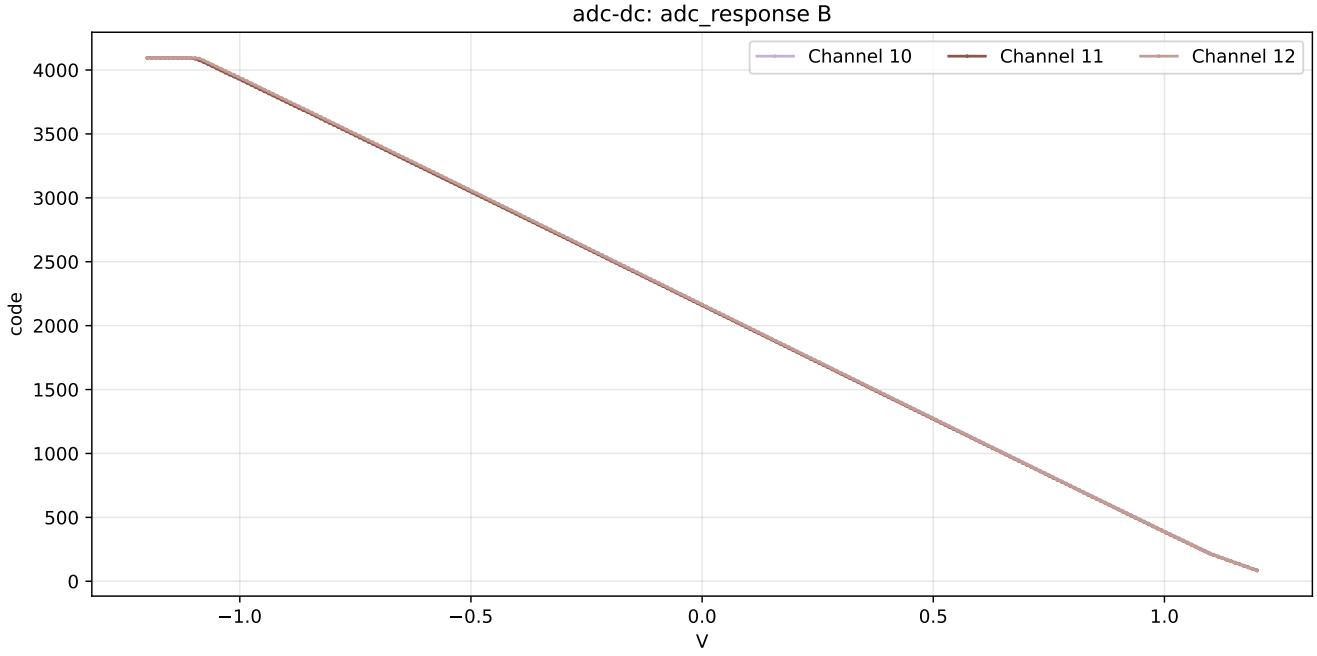
Test	Channel	Bank	Observable	Value	Unit
cfd-monitor	9		CFD_ OFFSET	1.035096 6666666668	V
cfd-monitor	9		CFD_CMP_ OFFSET	0.00524	V
cfd-monitor	9		BUFFER_ OFFSET	1.024806 6666666666	V
cfd-monitor	9		DRV_ OFFSET	1.04463	V
cfd-monitor	9		DRV_VOCM	1.53664	V
cfd-monitor	9		N5V5_PM12	-5.511662 727272727	V
cfd-monitor	9		P1V3	1.306306 6666666667	V
cfd-monitor	9		P5V5_PM12	5.486335 151515151	V
cfd-monitor	10		CFD_ OFFSET	1.0346	V
cfd-monitor	10		CFD_CMP_ OFFSET	0.004596 6666666666664	V
cfd-monitor	10		BUFFER_ OFFSET	1.02437	V
cfd-monitor	10		DRV_ OFFSET	1.04449	V
cfd-monitor	10		DRV_VOCM	1.536306 6666666667	V
cfd-monitor	10		N5V5_PM12	-5.510442 121212121	V
cfd-monitor	10		P1V3	1.31936	V
cfd-monitor	10		P5V5_PM12	5.480419 494949494	V
cfd-monitor	12		CFD_ OFFSET	1.03451	V
cfd-monitor	12		CFD_CMP_ OFFSET	0.004816 666666666666	V
cfd-monitor	12		BUFFER_ OFFSET	1.02432	V
cfd-monitor	12		DRV_ OFFSET	1.044316 6666666668	V
cfd-monitor	12		DRV_VOCM	1.536213 3333333334	V
cfd-monitor	12		N5V5_PM12	-5.504039 292929292	V
cfd-monitor	12		P1V3	1.307083 3333333334	V
cfd-monitor	12		P5V5_PM12	5.475670 909090909	V
jtag	board		links_per_ path	168	connection
jtag	board		vectors_per_ path	338	vector
mcu	board		walking_ patterns	32	pattern

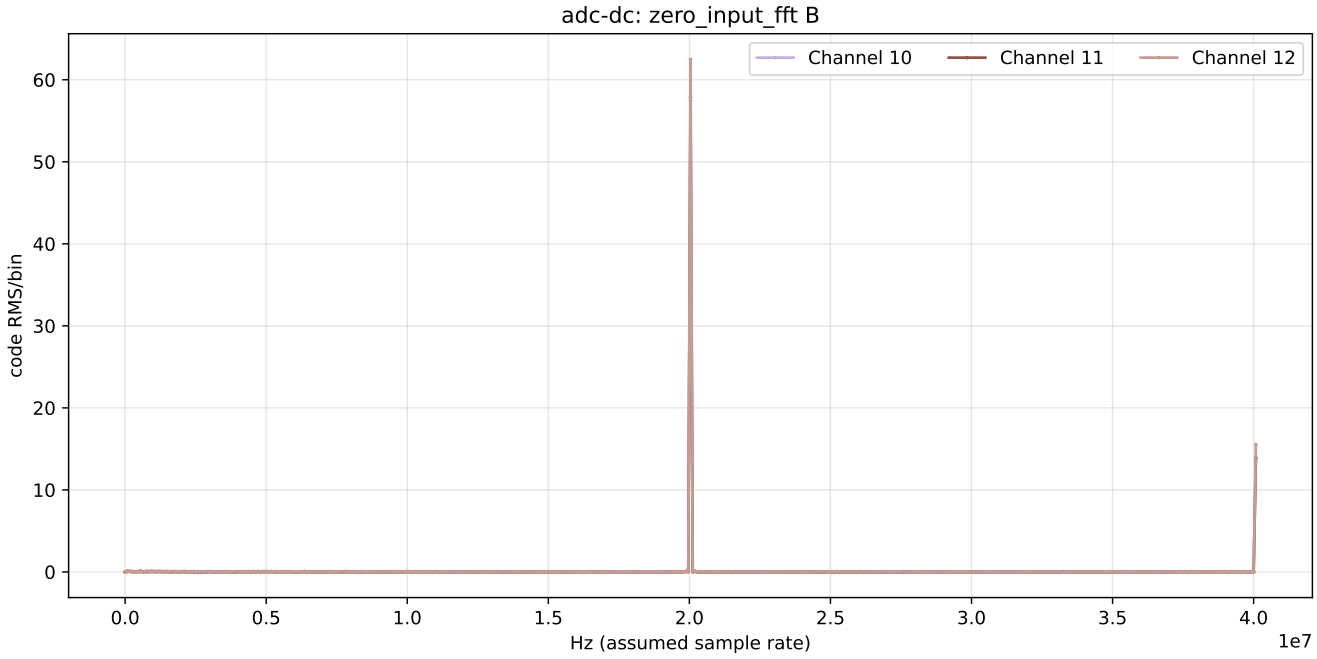
Test	Channel	Bank	Observable	Value	Unit
i2c	board		P1V0_LDO_voltage	1.0	V
i2c	board		P1V0_LDO_nominal_current	0.124	A
i2c	board		P1V2_LDO_voltage	1.2	V
i2c	board		P1V2_LDO_nominal_current	0.092	A
i2c	board		P1V8_LDO_voltage	1.792	V
i2c	board		P1V8_LDO_nominal_current	0.004	A
i2c	board		P3V8_voltage	3.92	V
i2c	board		P3V8_nominal_current	3.12	A
i2c	board		P3V3_voltage	3.328	V
i2c	board		P3V3_nominal_current	4.880000000000001	A
i2c	board		P1V8_voltage	1.776	V
i2c	board		P1V8_nominal_current	0.9600000000000001	A
i2c	board		P1V0_voltage	1.008	V
i2c	board		P1V0_nominal_current	0.32	A
i2c	board		P5V5_voltage	5.496	V
i2c	board		P5V5_nominal_current	0.28	A
ethernet	board		verified_ADC_payload	393216	byte

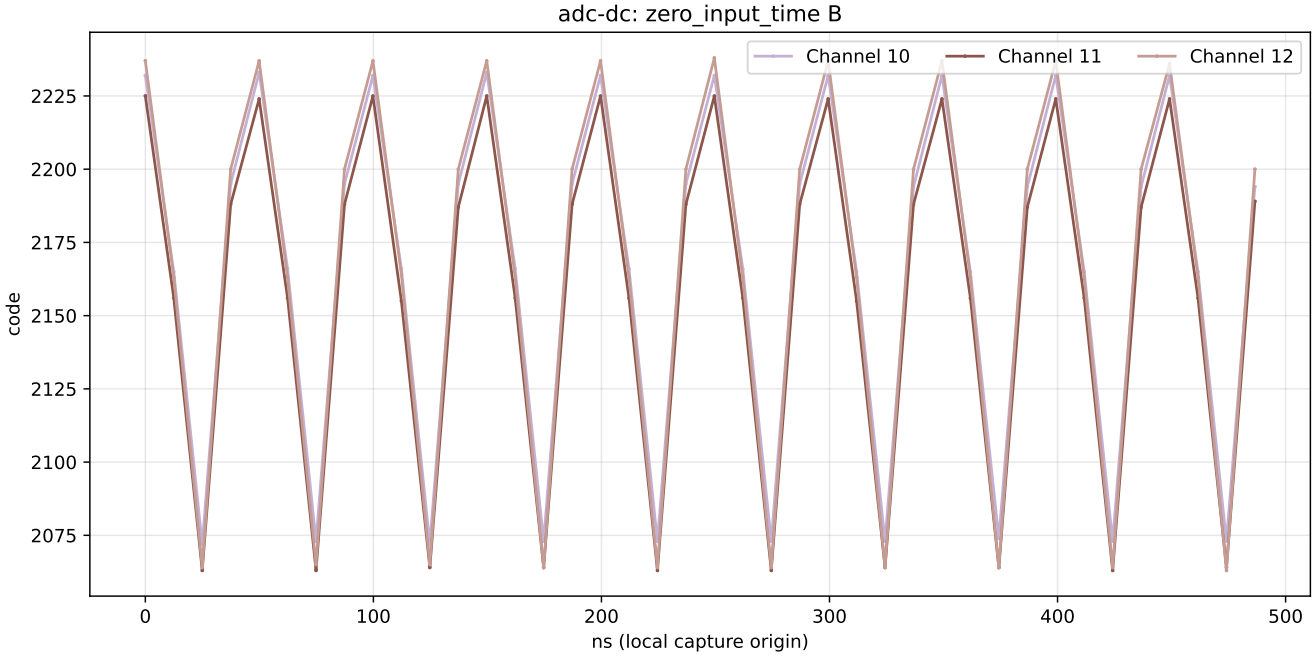


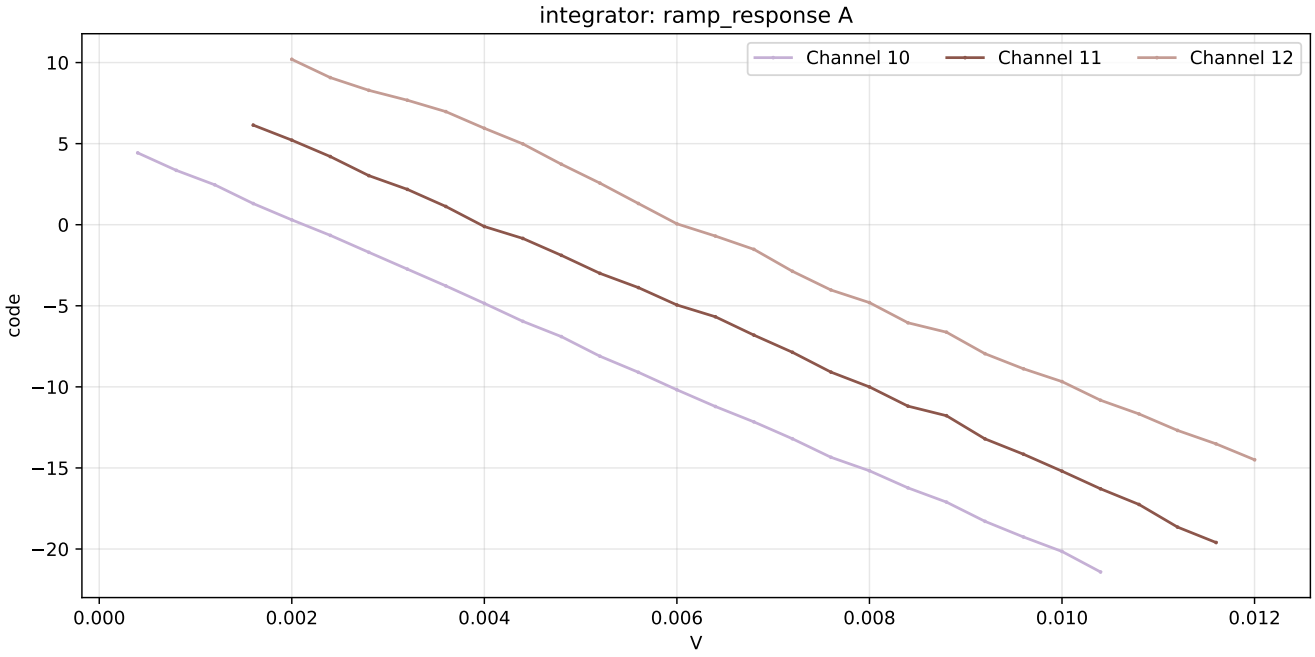

adc-dc: zero_input_fft A
Channel 10
Channel 11
Channel 12
code RMS/bin
60
50
40
30
20
10
0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Hz (assumed sample rate)
1e7

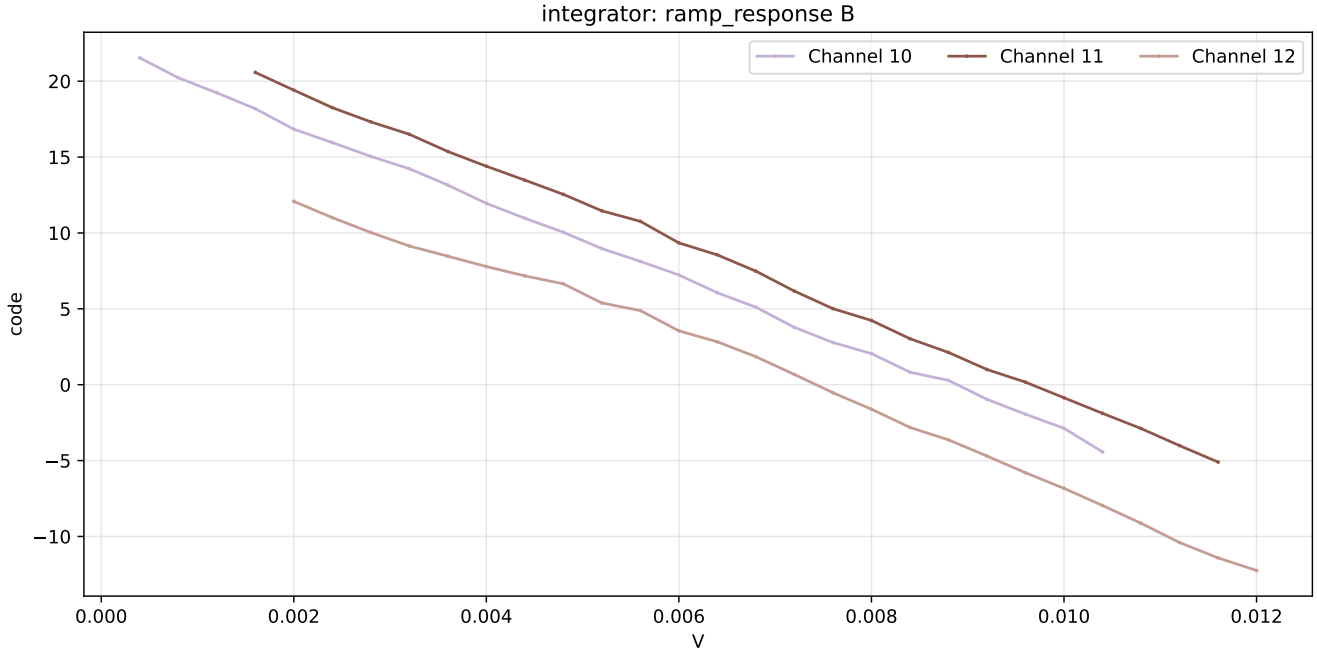

adc-dc: zero_input_time A
Channel 10
Channel 11
Channel 12
code
2225
2200
2175
2150
2125
2100
2075
0
100
200
300
400
500
ns (local capture origin)

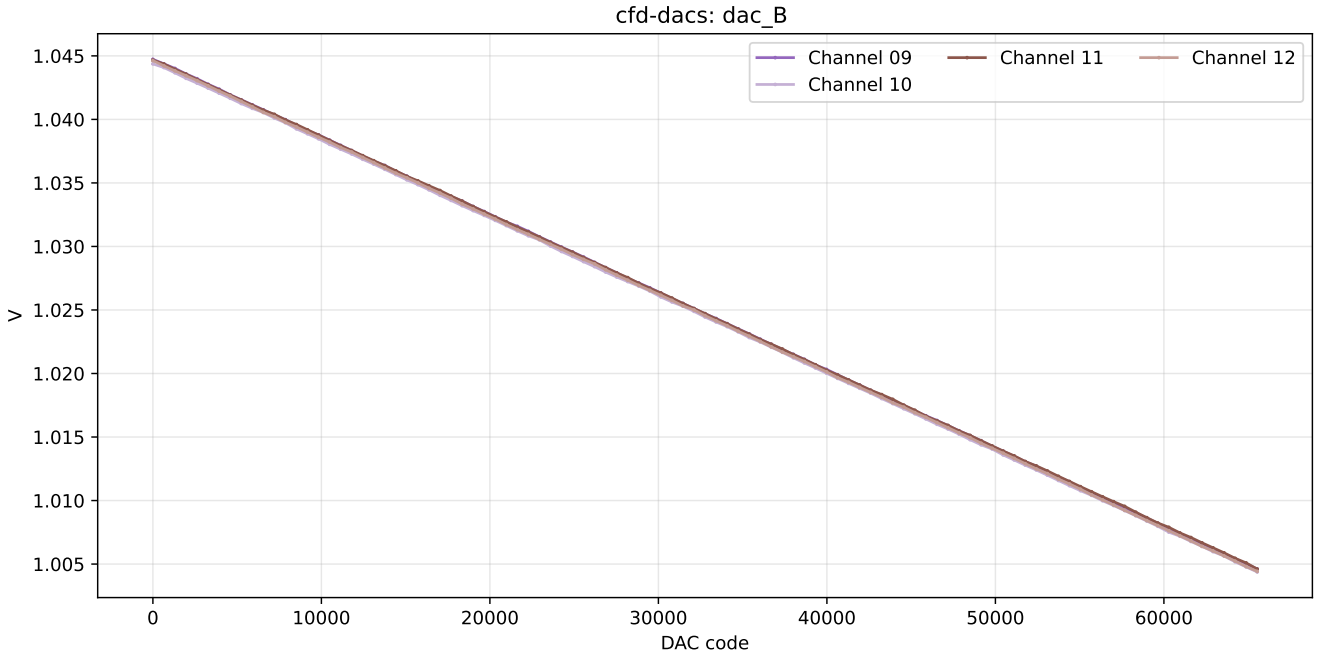

adc-dc: adc_response B
Channel 10
Channel 11
Channel 12
code
V

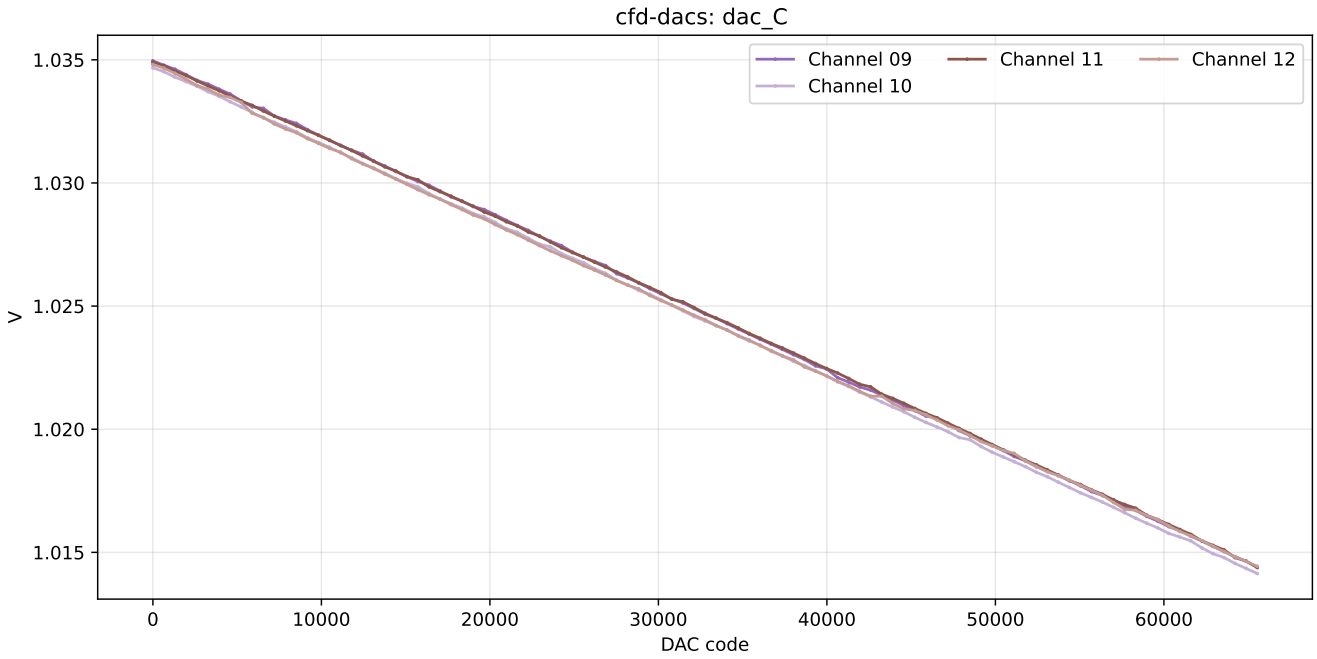

adc-dc: zero_input_fft B
Channel 10
Channel 11
Channel 12
code RMS/bin
Hz (assumed sample rate)
1e7

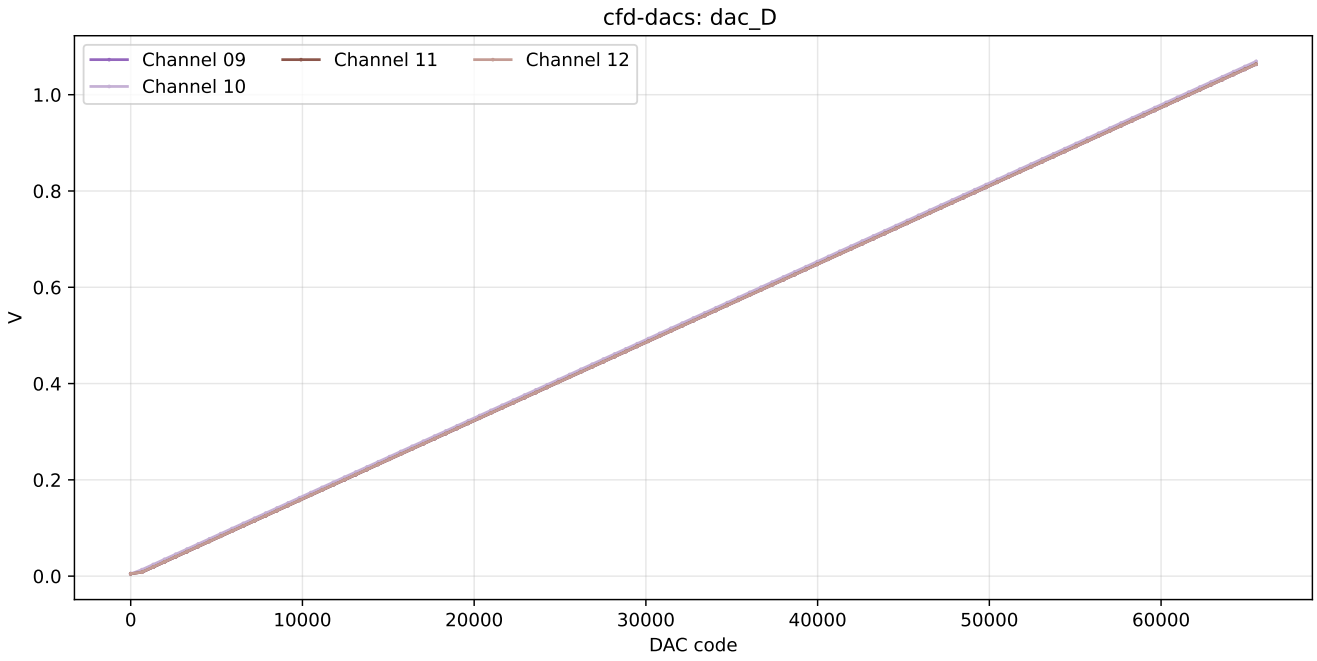

adc-dc: zero_input_time B
Channel 10
Channel 11
Channel 12
code
2225
2200
2175
2150
2125
2100
2075
0
100
200
300
400
500
ns (local capture origin)

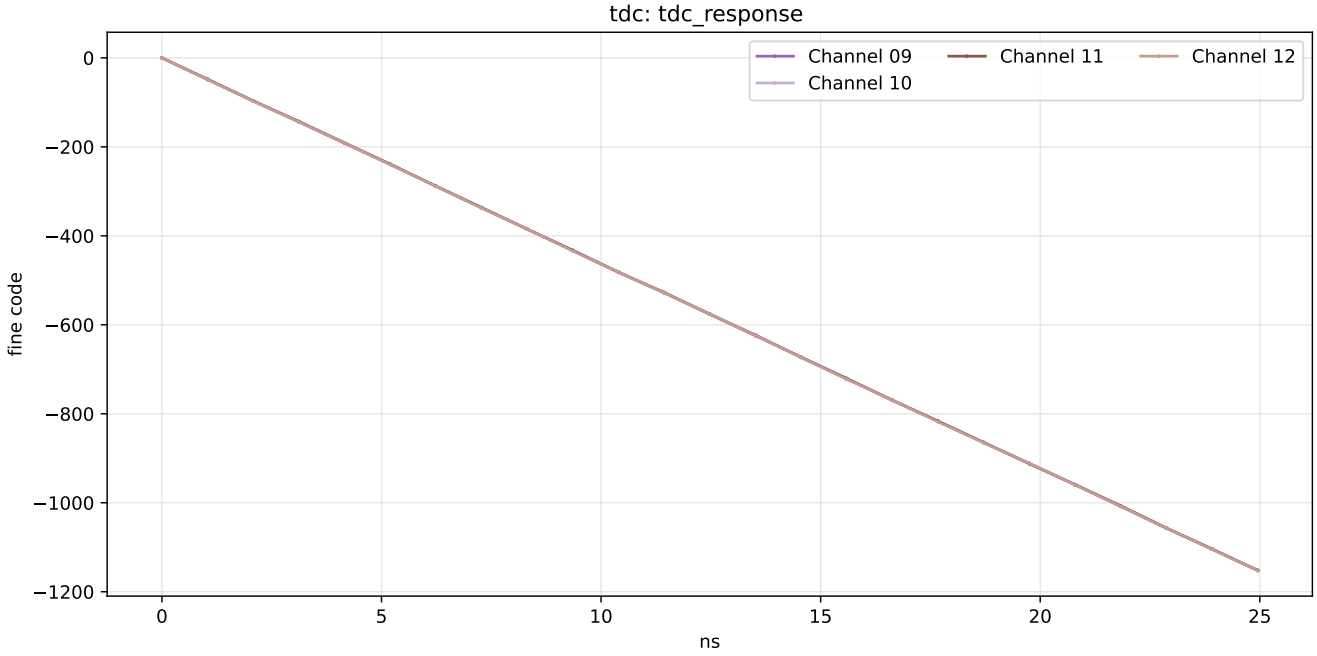

integrator: ramp_response A
Channel 10
Channel 11
Channel 12
code
10
5
0
−5
−10
−15
−20
0.000
0.002
0.004
0.006
0.008
0.010
0.012
V

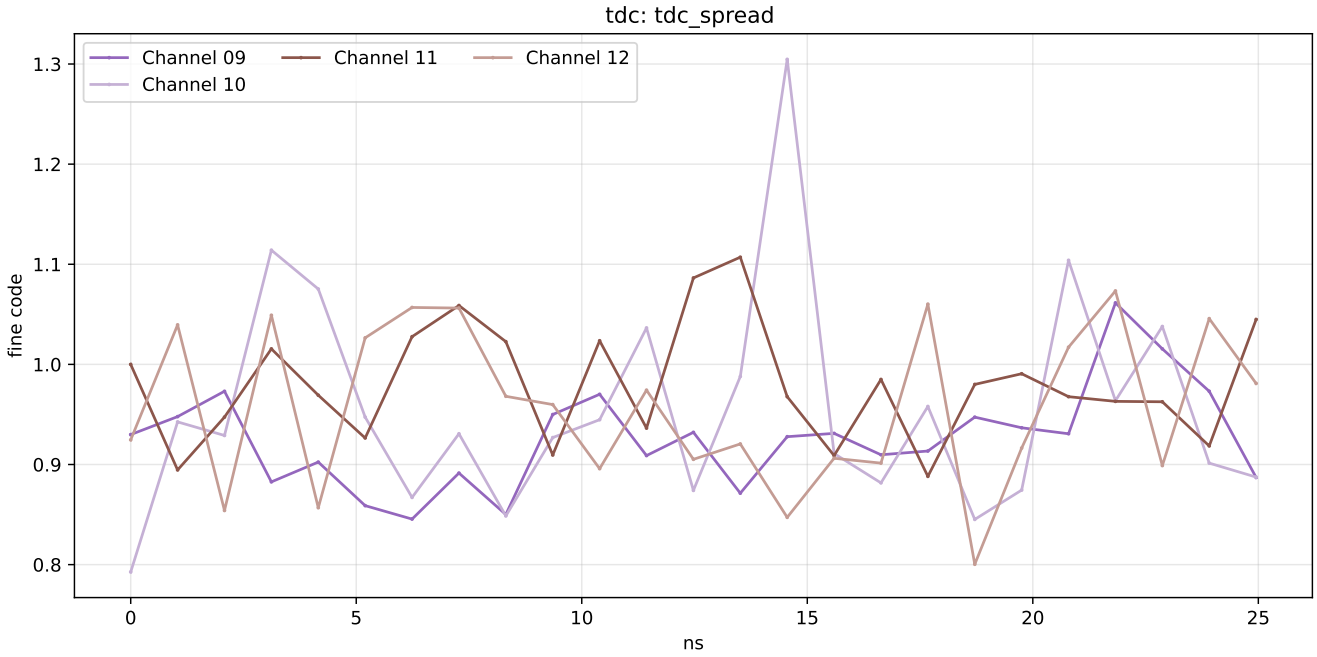

integrator: ramp_response B
Channel 10
Channel 11
Channel 12
code
V
20
15
10
5
0
−5
−10
0.000
0.002
0.004
0.006
0.008
0.010
0.012

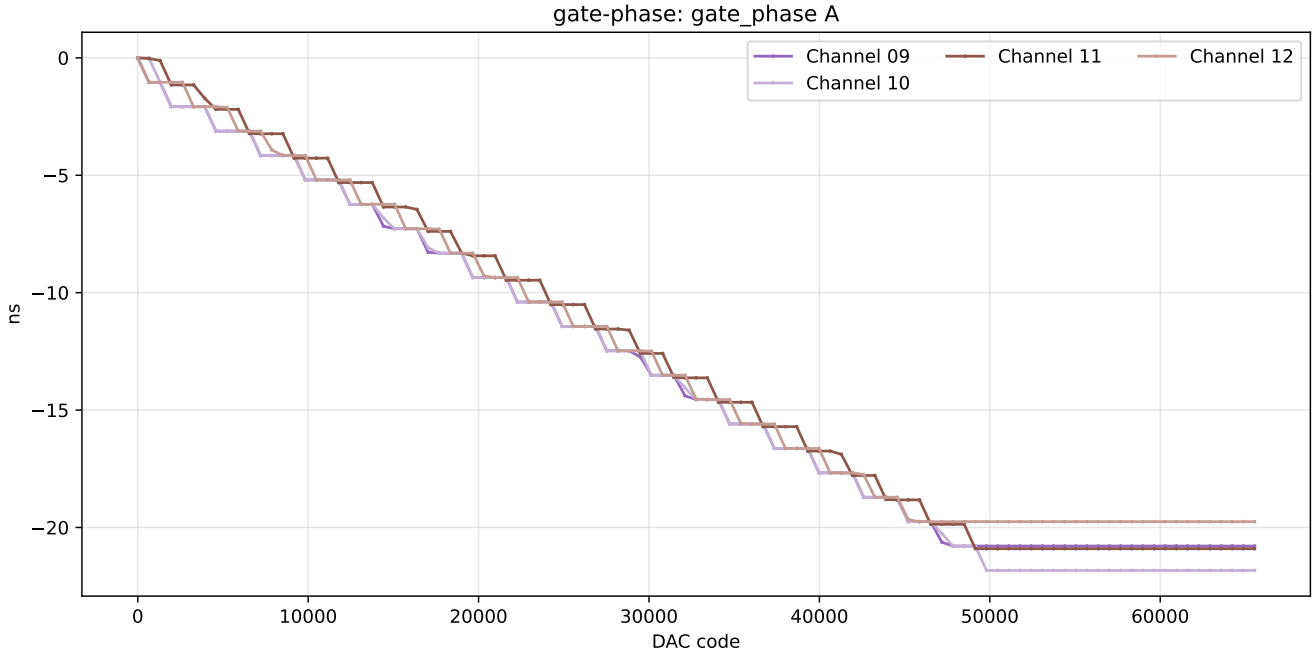

cfd-dacs: dac_B
Channel 09
Channel 10
Channel 11
Channel 12
V
DAC code
1.045
1.040
1.035
1.030
1.025
1.020
1.015
1.010
1.005
0
10000
20000
30000
40000
50000
60000

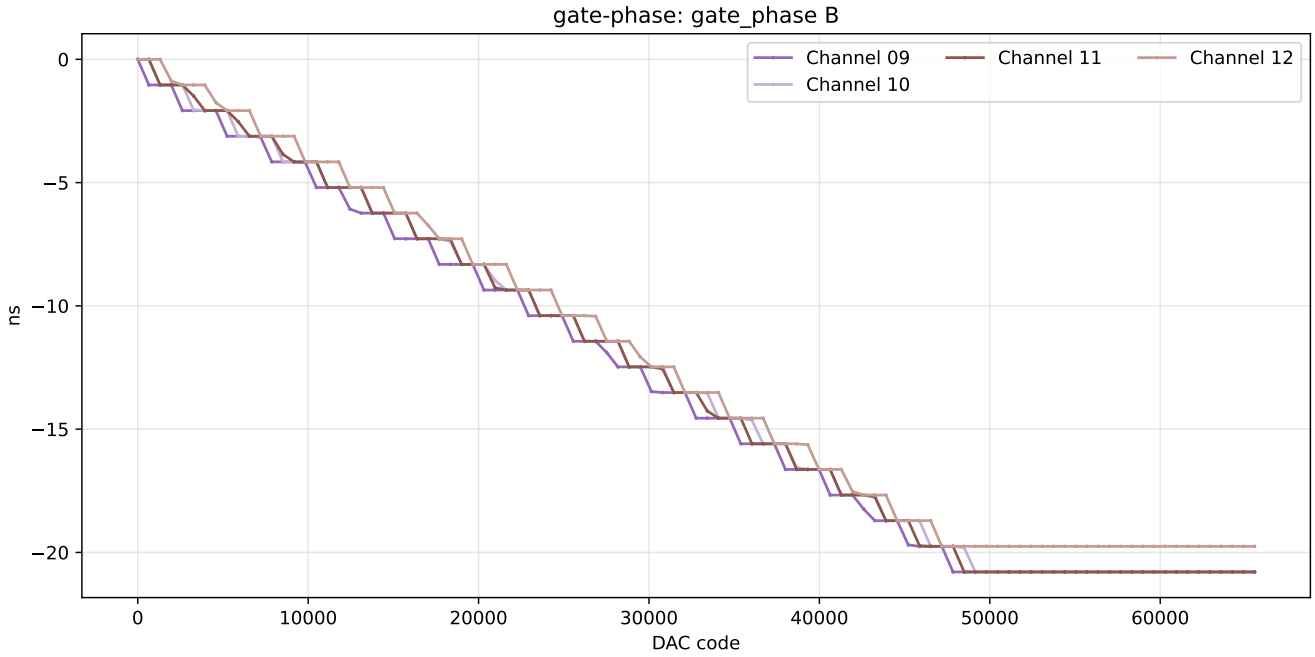

cfd-dacs: dac_C
Channel 09
Channel 10
Channel 11
Channel 12
1.035
1.030
1.025
1.020
1.015
V
0
10000
20000
30000
40000
50000
60000
DAC code

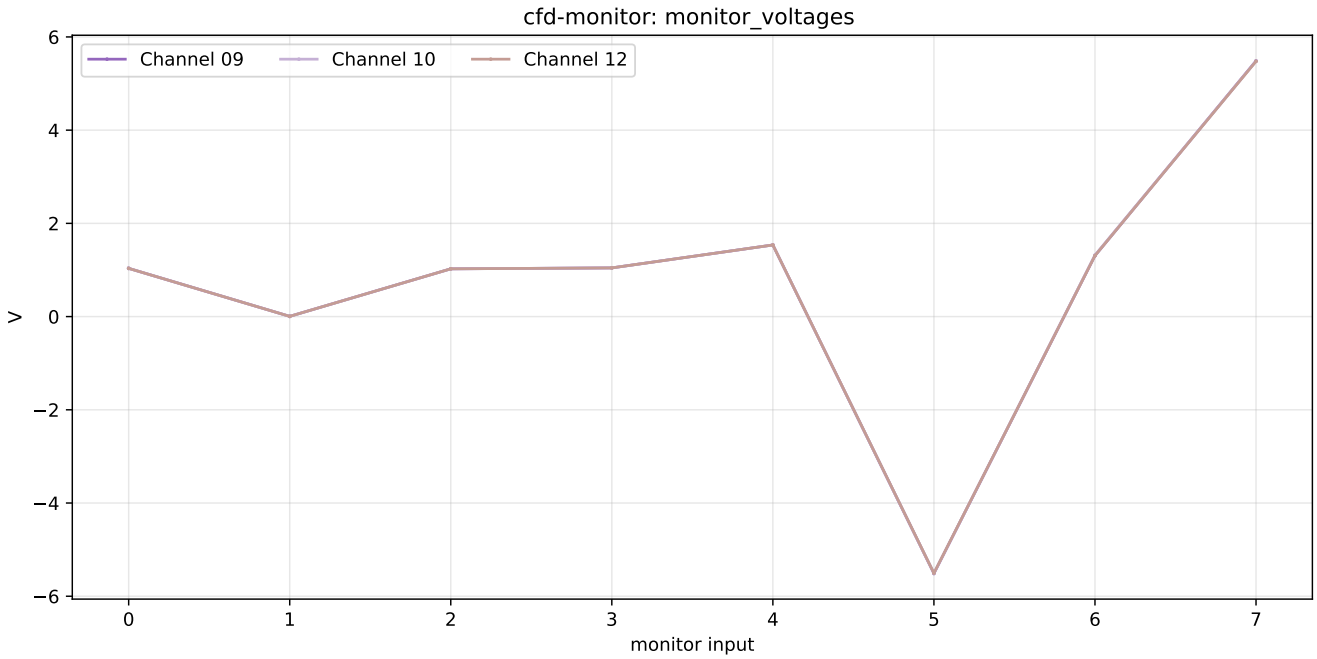

cfd-dacs: dac_D
Channel 09
Channel 10
Channel 11
Channel 12
1.0
0.8
0.6
0.4
0.2
0.0
V
0
10000
20000
30000
40000
50000
60000
DAC code

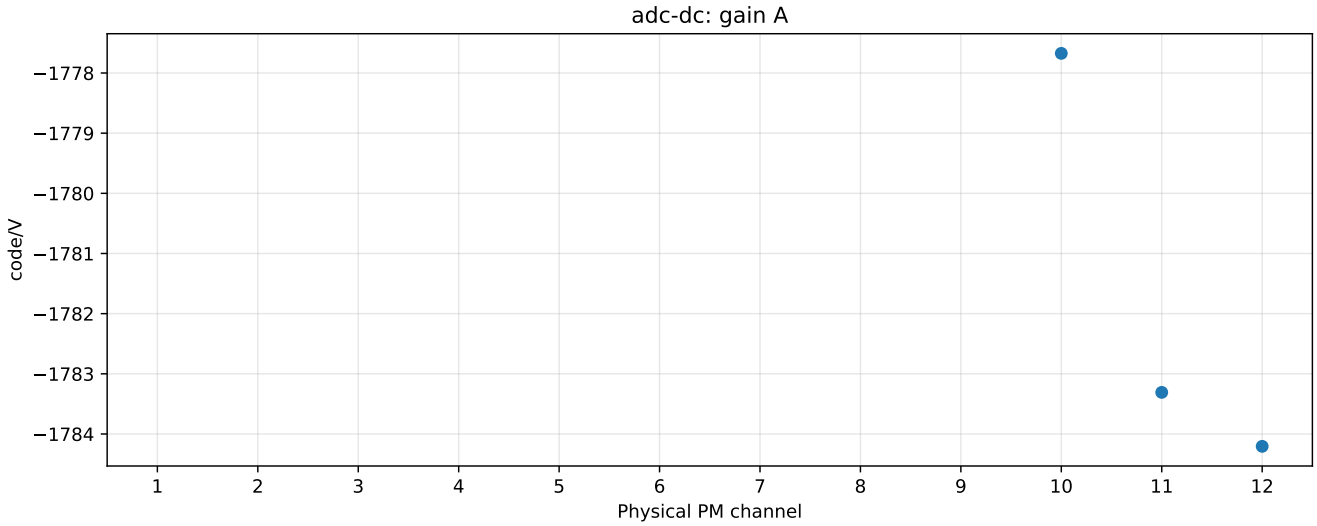

tdc: tdc_response
Channel 09
Channel 10
Channel 11
Channel 12
fine code
ns
0
−200
−400
−600
−800
−1000
−1200
0
5
10
15
20
25

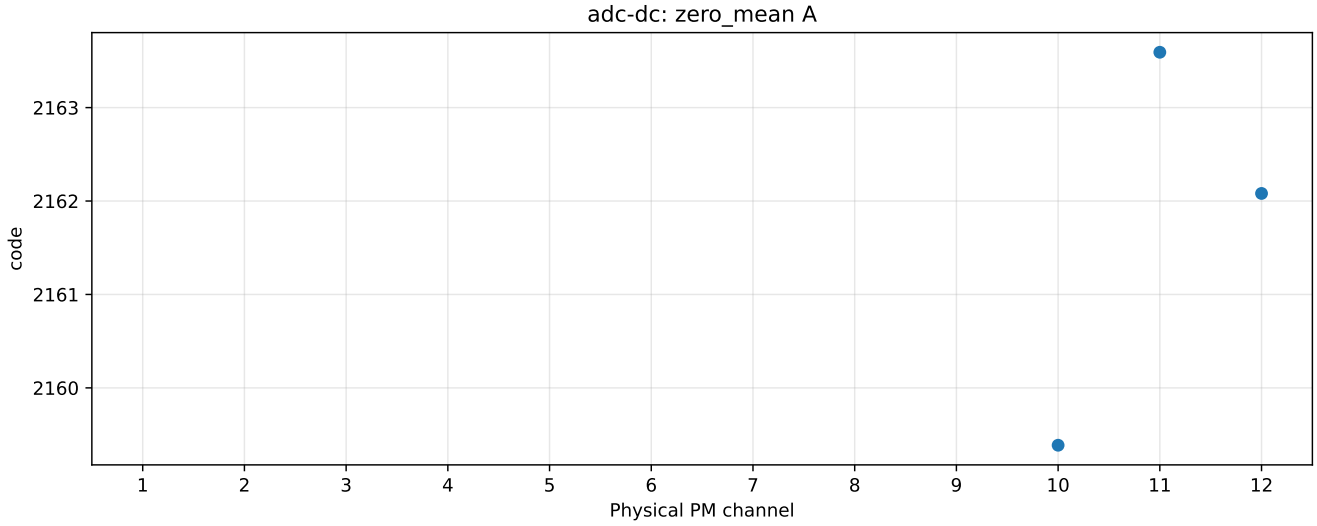

tdc: tdc_spread
Channel 09
Channel 10
Channel 11
Channel 12
fine code
ns
1.3
1.2
1.1
1.0
0.9
0.8
0
5
10
15
20
25

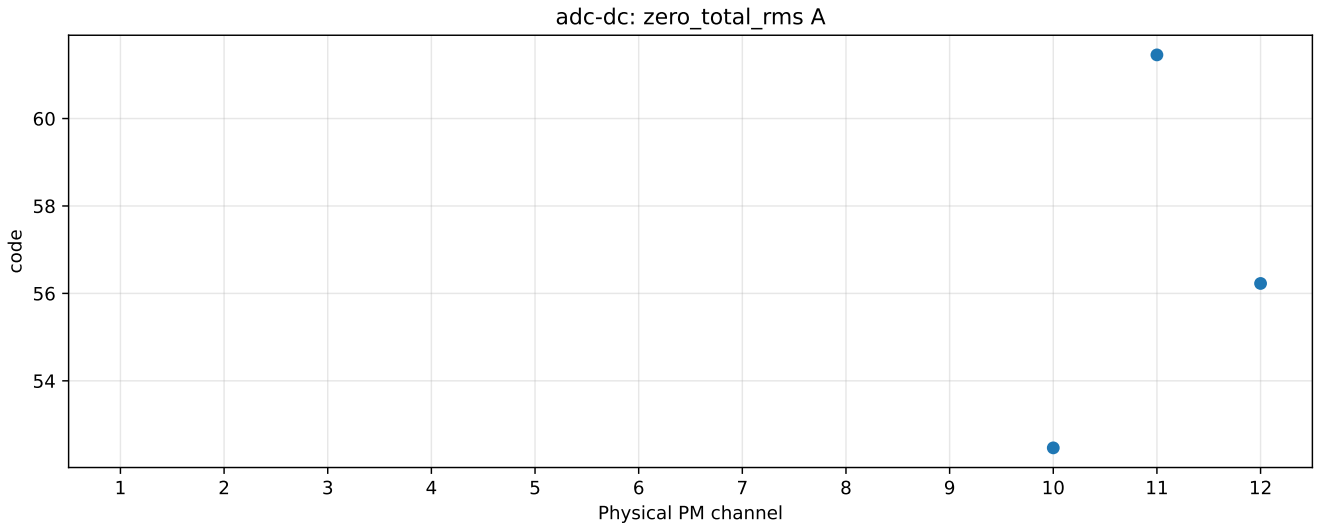

gate-phase: gate_phase A
Channel 09
Channel 10
Channel 11
Channel 12
0
−5
−10
−15
−20
ns
0
10000
20000
30000
40000
50000
60000
DAC code

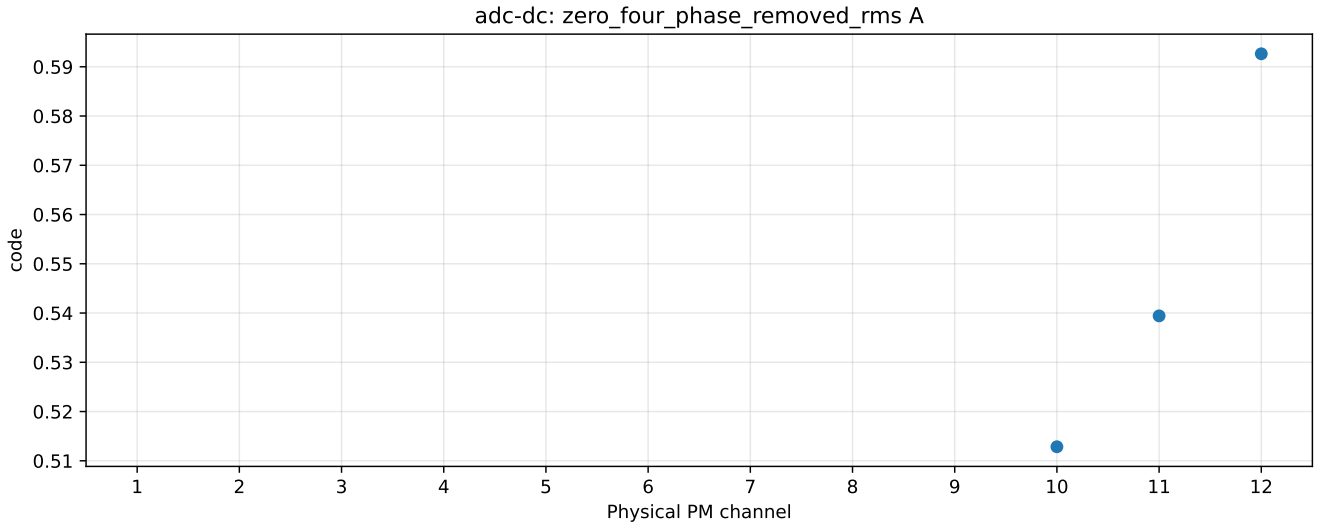

gate-phase: gate_phase B
Channel 09
Channel 10
Channel 11
Channel 12
0
−5
−10
−15
−20
ns
0
10000
20000
30000
40000
50000
60000
DAC code

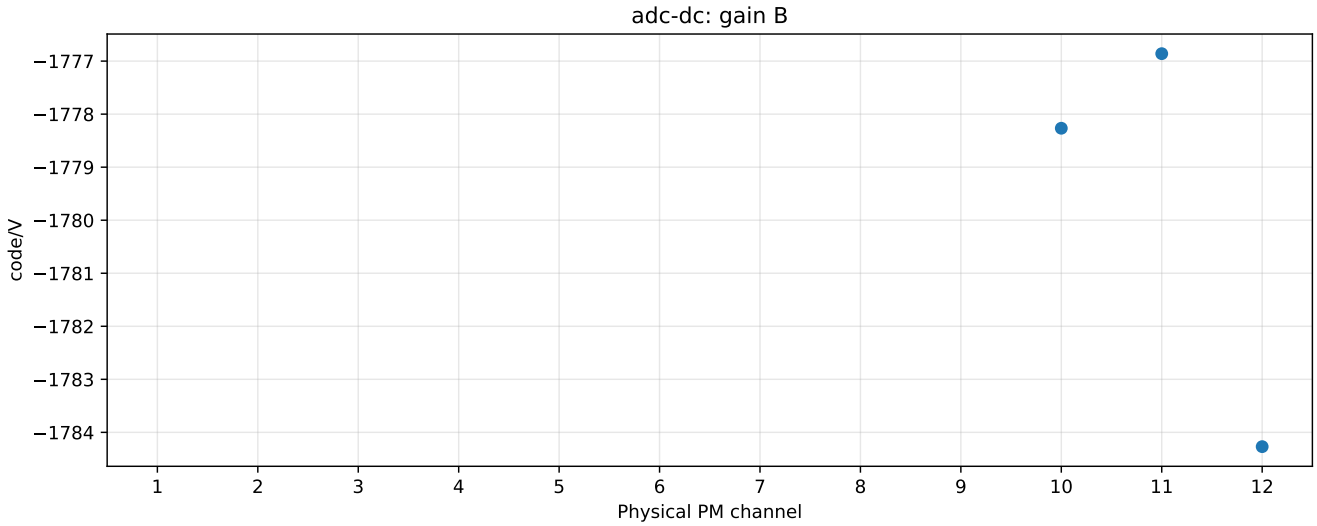

cfd-monitor: monitor_voltages
Channel 09
Channel 10
Channel 12
V
monitor input

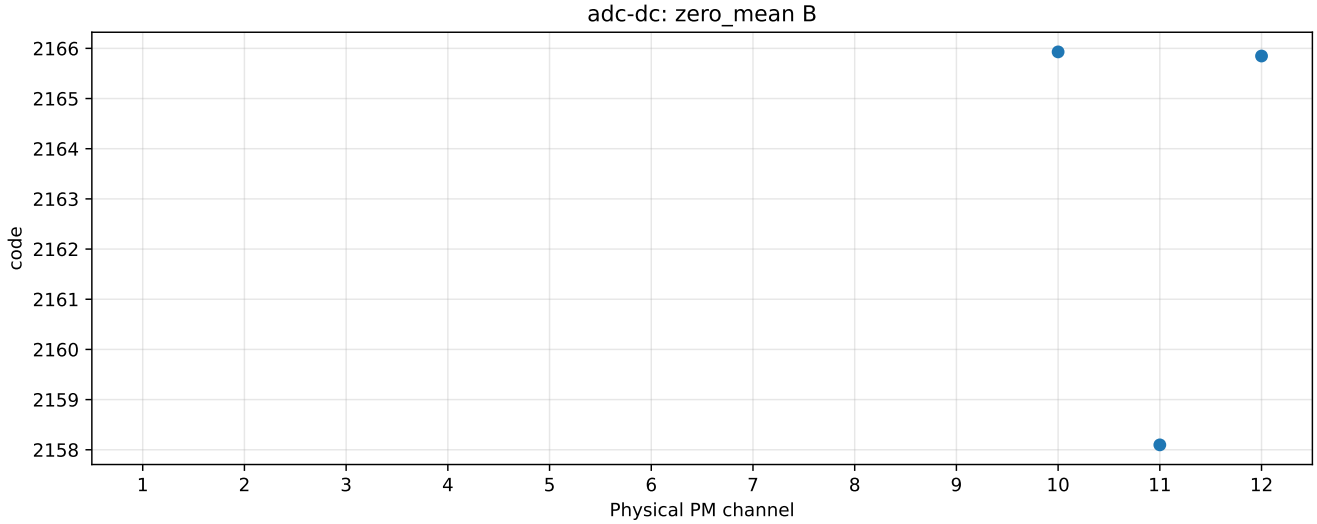

adc-dc: gain A
code/V
Physical PM channel

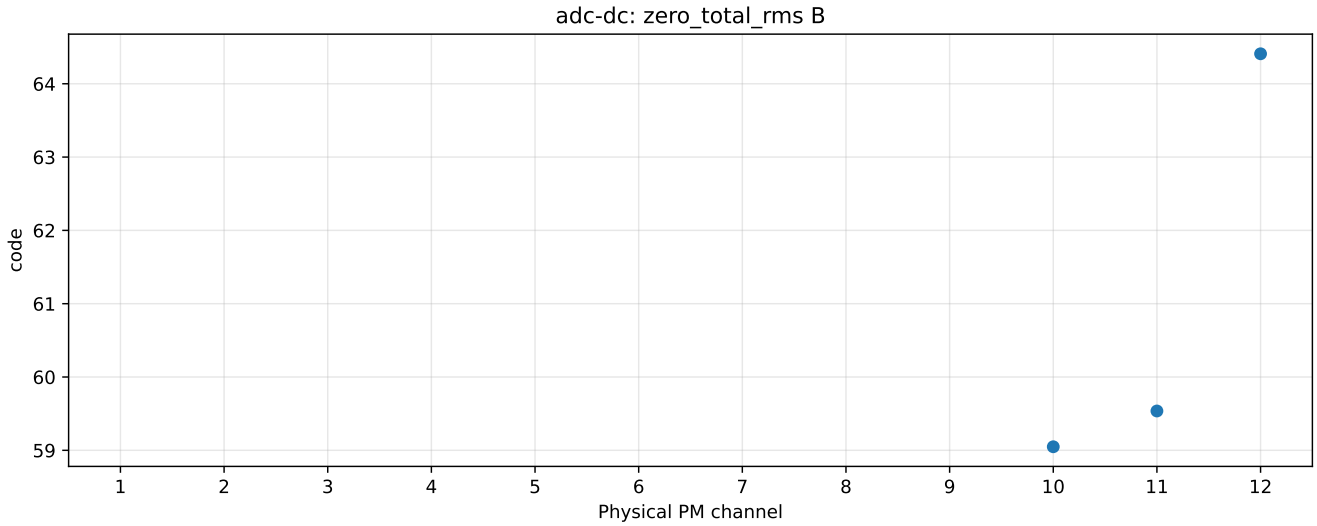

adc-dc: zero_mean A
code
Physical PM channel

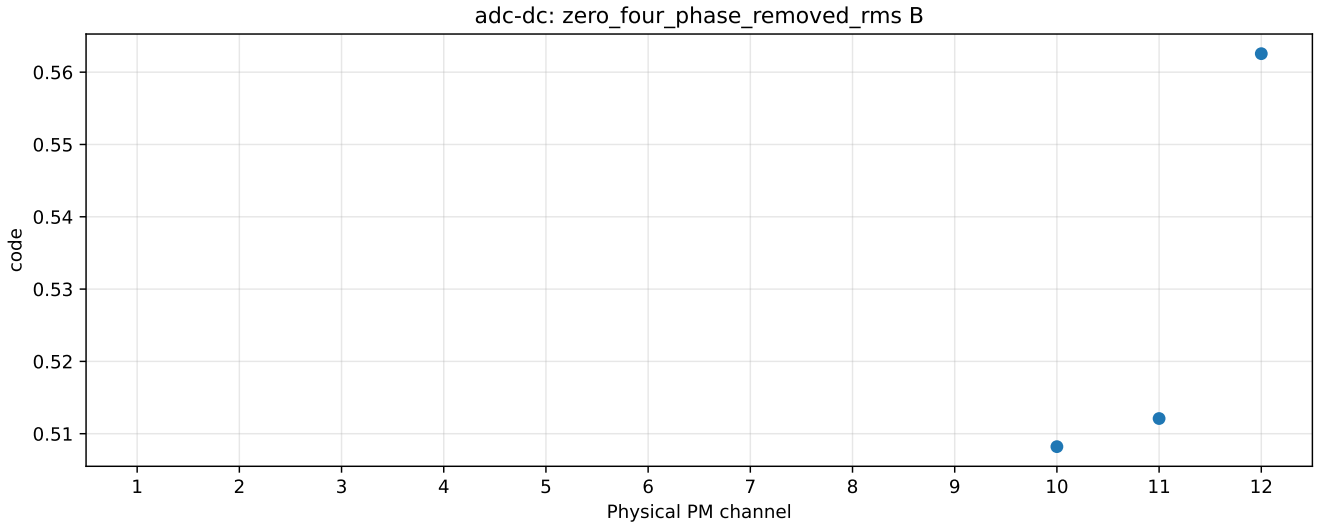

adc-dc: zero_total_rms A
code
60
58
56
54
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

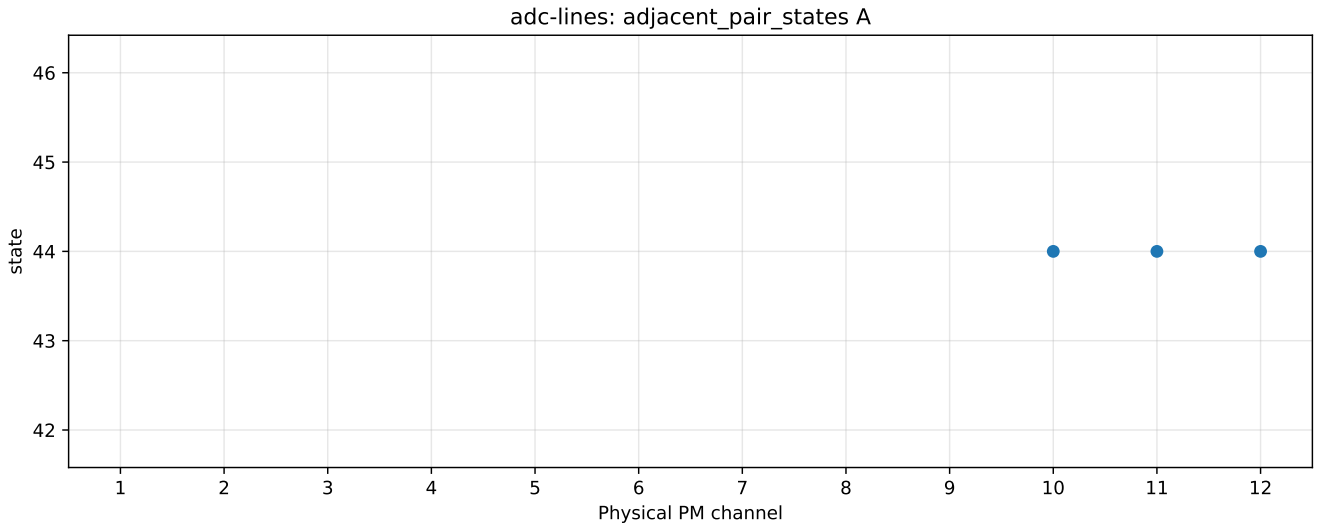

adc-dc: zero_four_phase_removed_rms A
code
0.59
0.58
0.57
0.56
0.55
0.54
0.53
0.52
0.51
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

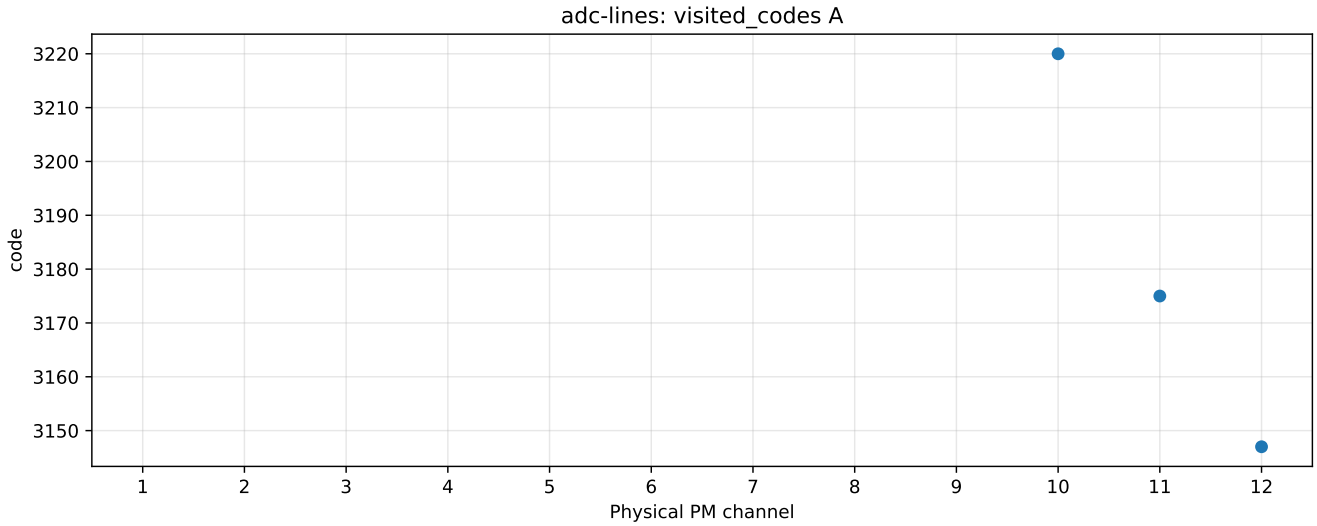

adc-dc: gain B
code/V
−1777
−1778
−1779
−1780
−1781
−1782
−1783
−1784
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

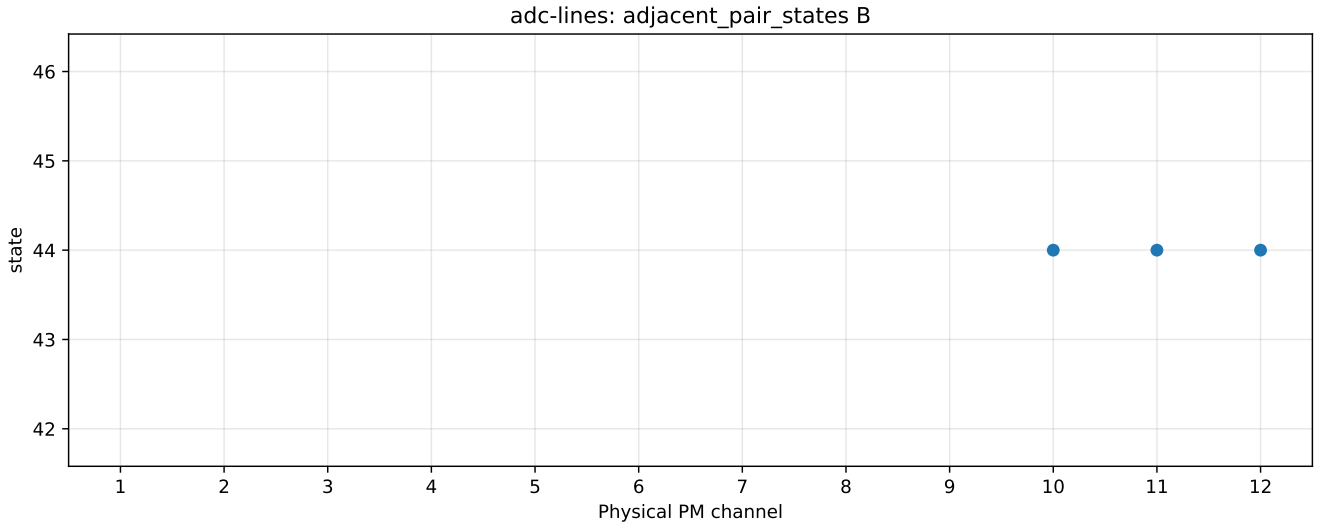

adc-dc: zero_mean B
code
2166
2165
2164
2163
2162
2161
2160
2159
2158
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

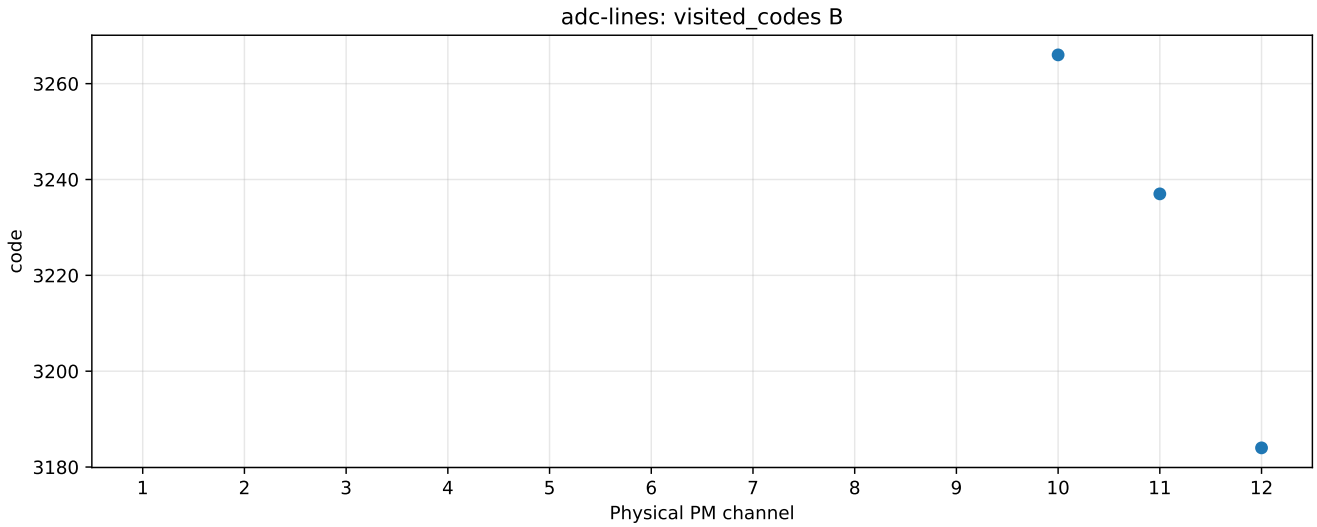

adc-dc: zero_total_rms B
code
64
63
62
61
60
59
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

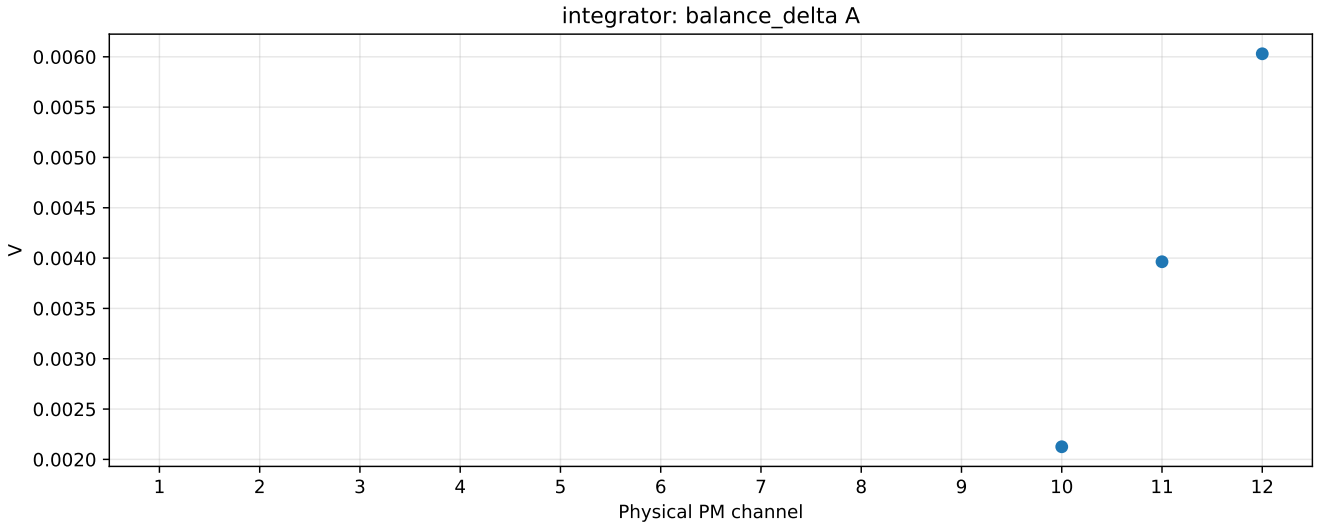

adc-dc: zero_four_phase_removed_rms B
code
0.56
0.55
0.54
0.53
0.52
0.51
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

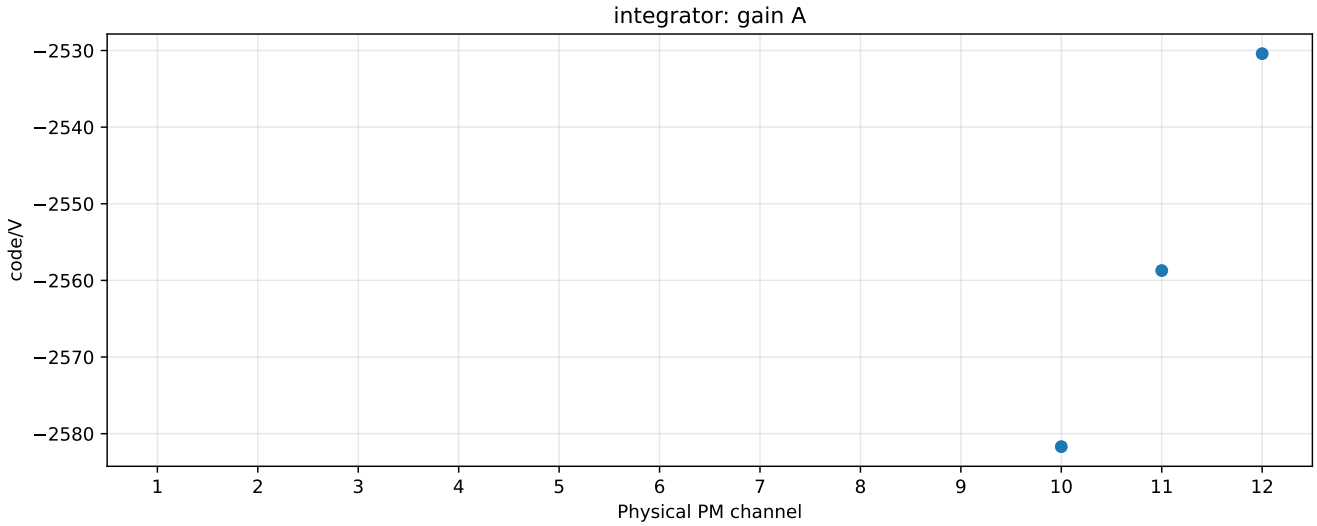

adc-lines: adjacent_pair_states A
state
46
45
44
43
42
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

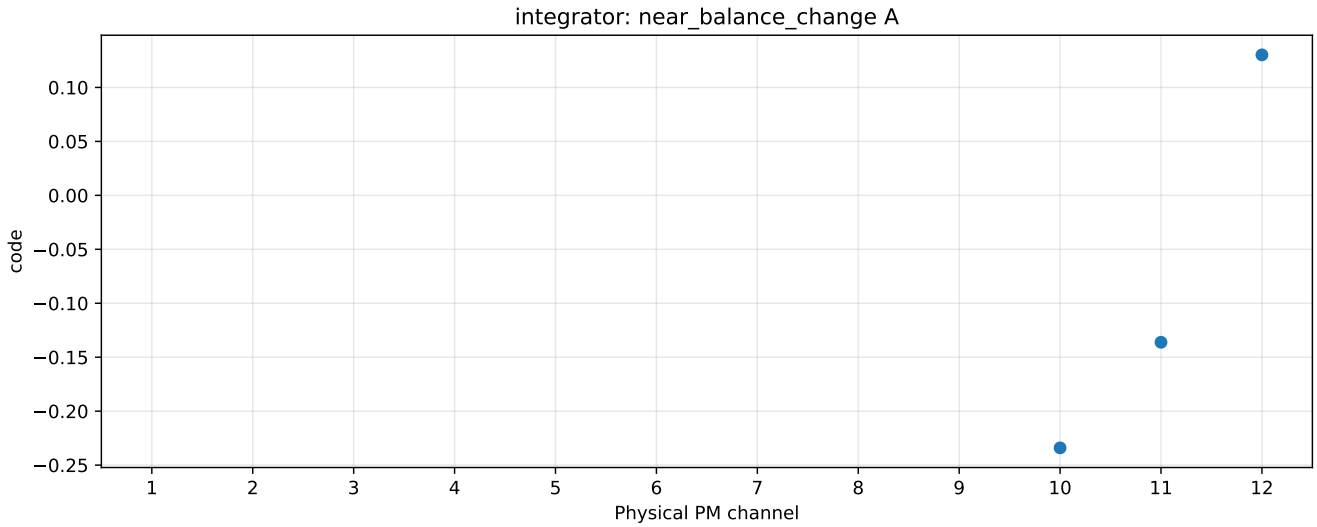

adc-lines: visited_codes A
code
3220
3210
3200
3190
3180
3170
3160
3150
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

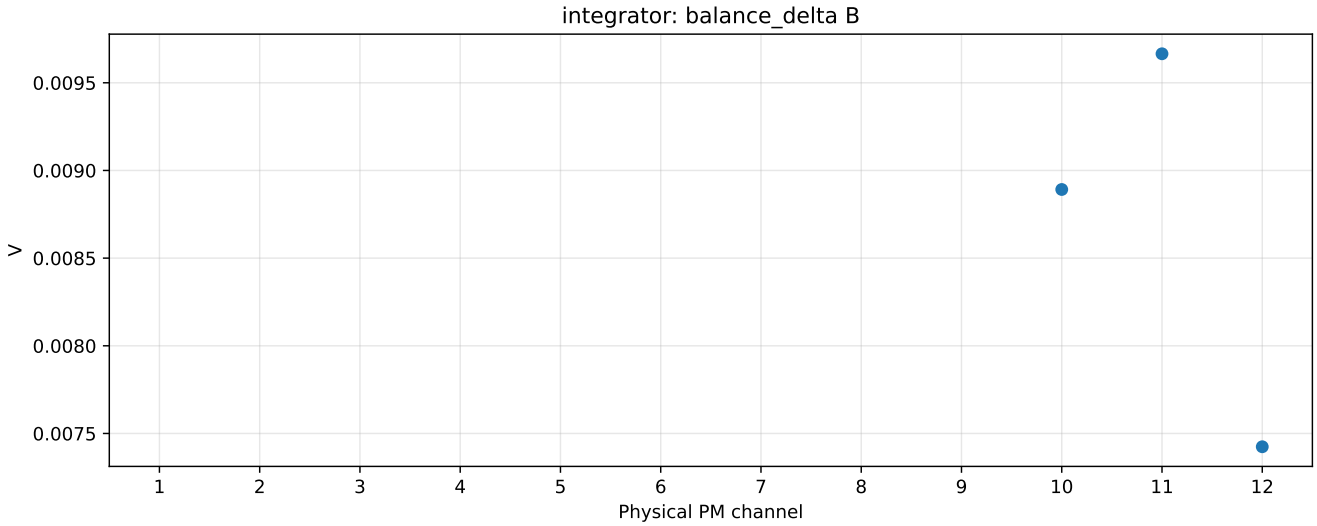

adc-lines: adjacent_pair_states B
state
46
45
44
43
42
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

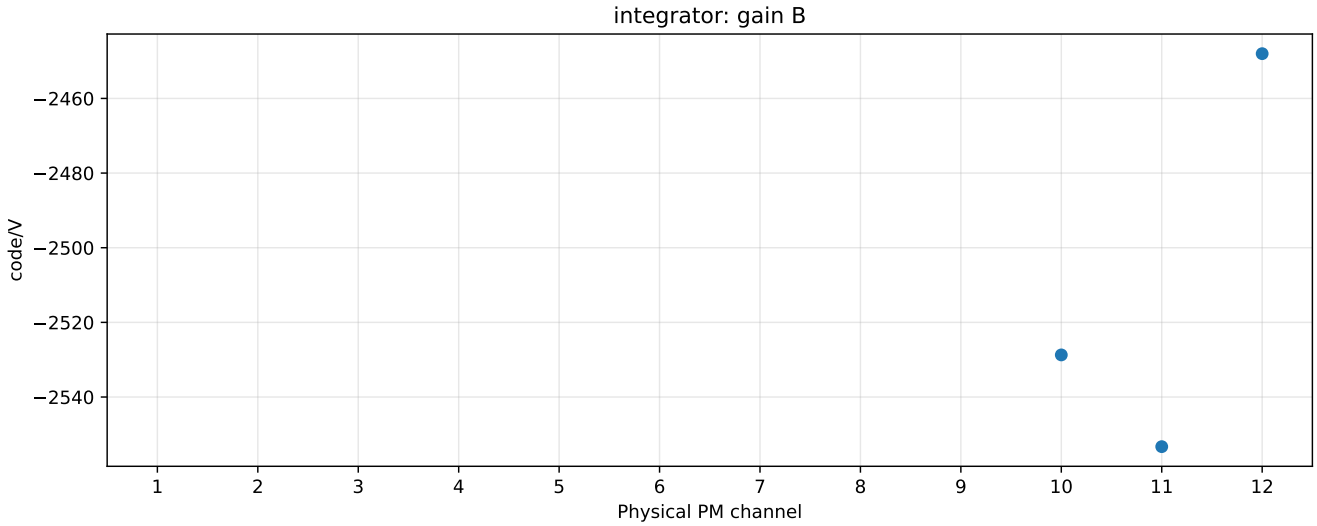

adc-lines: visited_codes B
code
3260
3240
3220
3200
3180
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

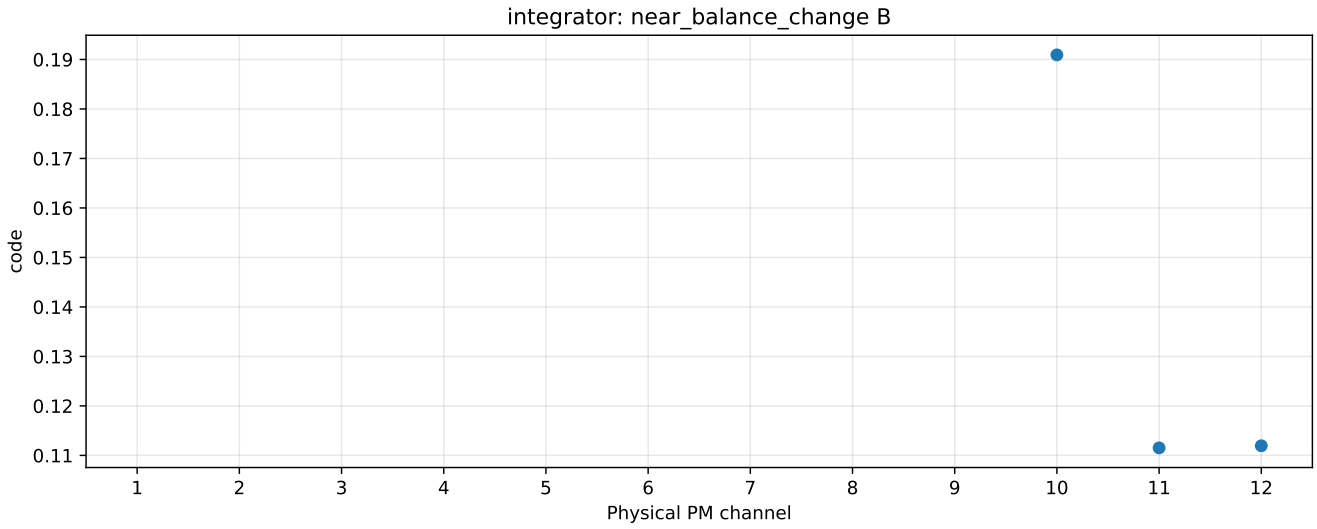

integrator: balance_delta A
V
0.0060
0.0055
0.0050
0.0045
0.0040
0.0035
0.0030
0.0025
0.0020
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

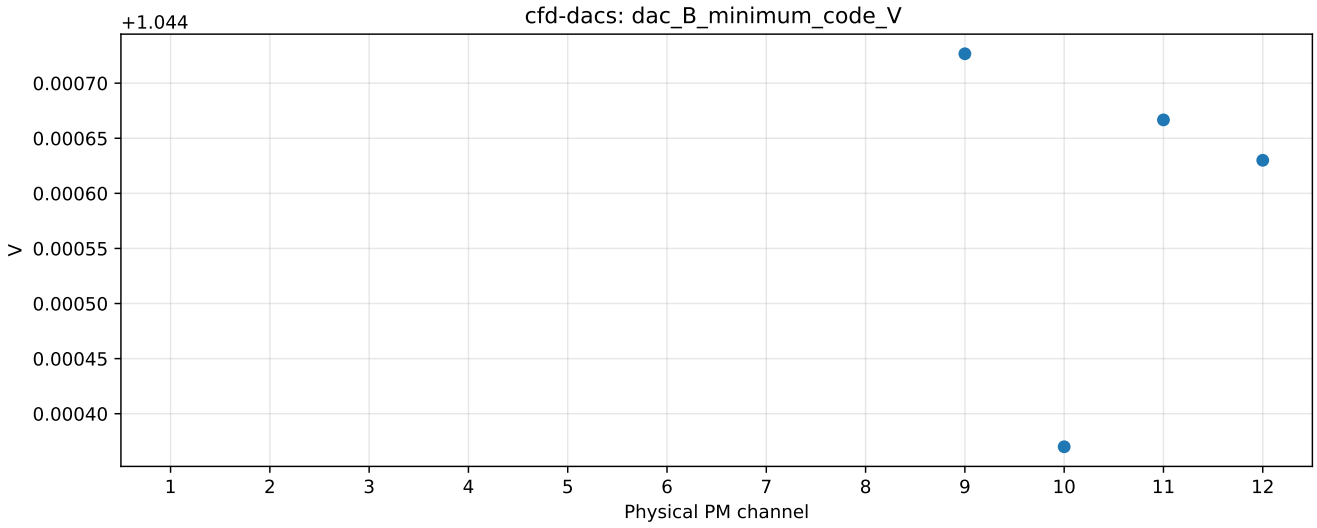

integrator: gain A
code/V
−2530
−2540
−2550
−2560
−2570
−2580
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

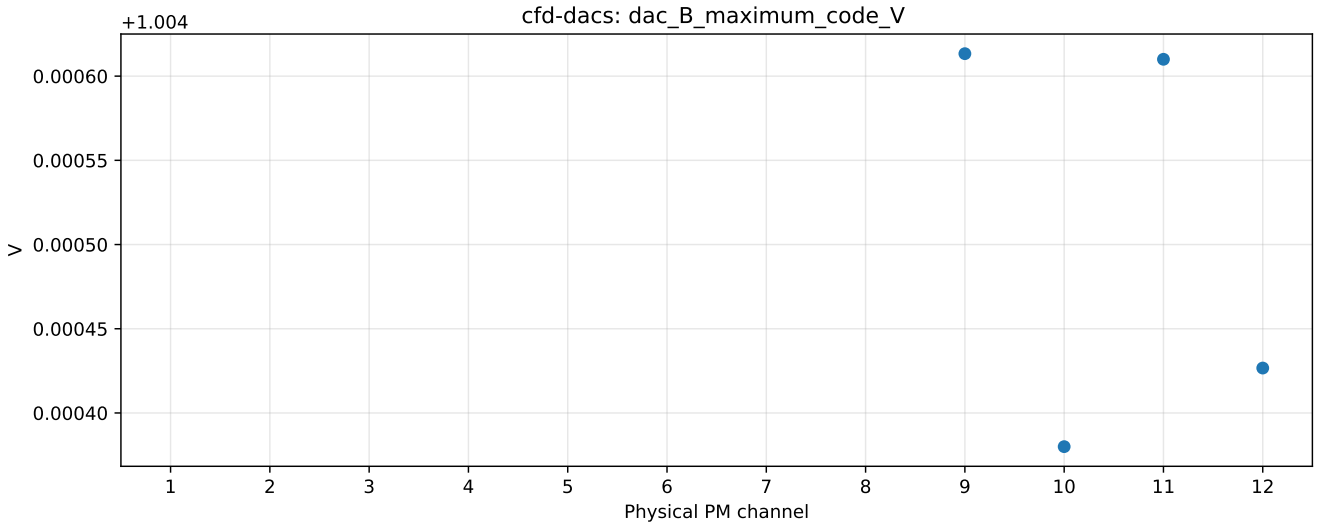

integrator: near_balance_change A
code
0.10
0.05
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

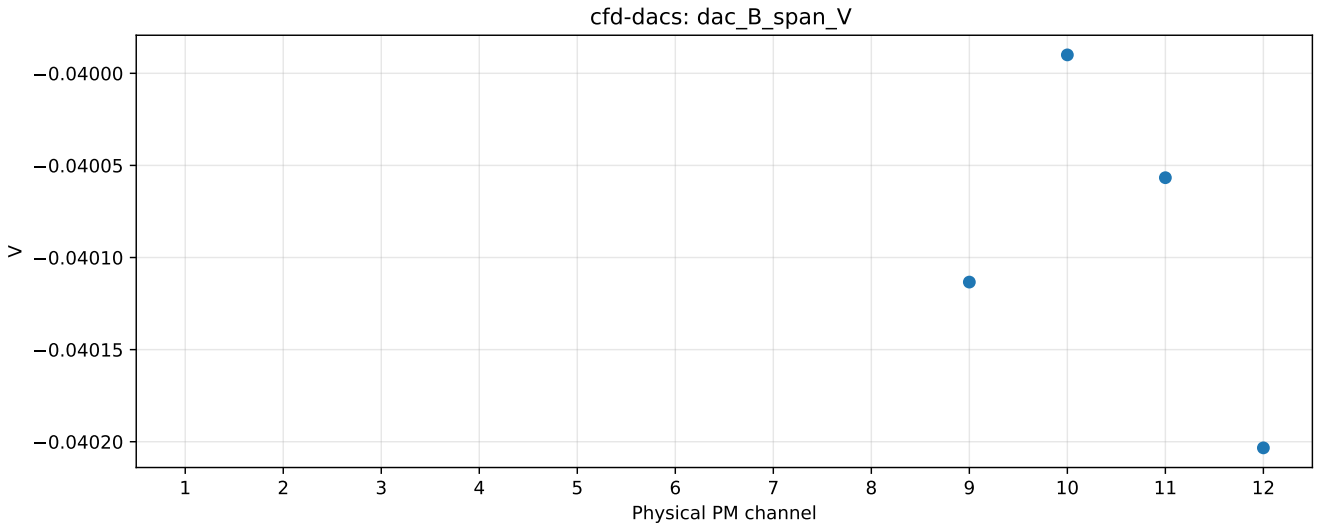

integrator: balance_delta B
V
0.0095
0.0090
0.0085
0.0080
0.0075
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

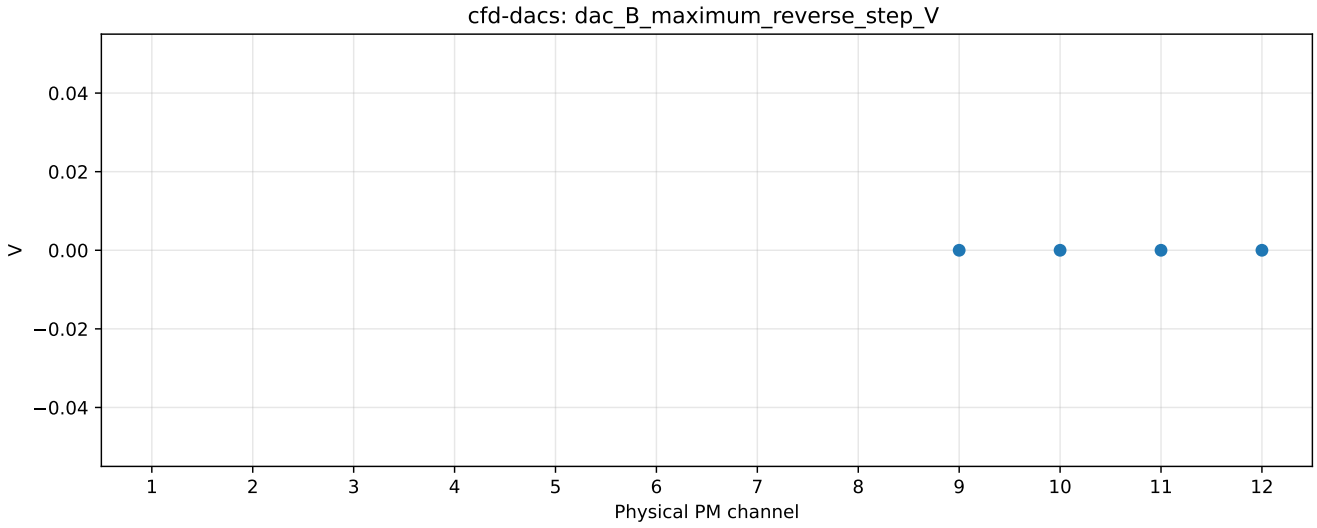

integrator: gain B
code/V
−2460
−2480
−2500
−2520
−2540
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

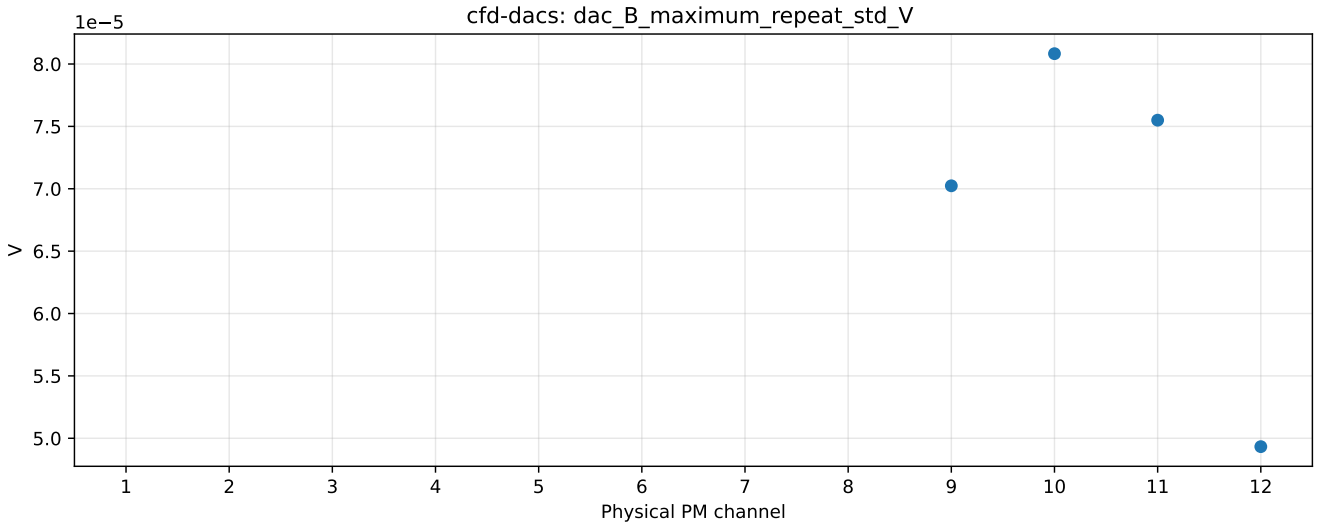

integrator: near_balance_change B
code
0.19
0.18
0.17
0.16
0.15
0.14
0.13
0.12
0.11
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

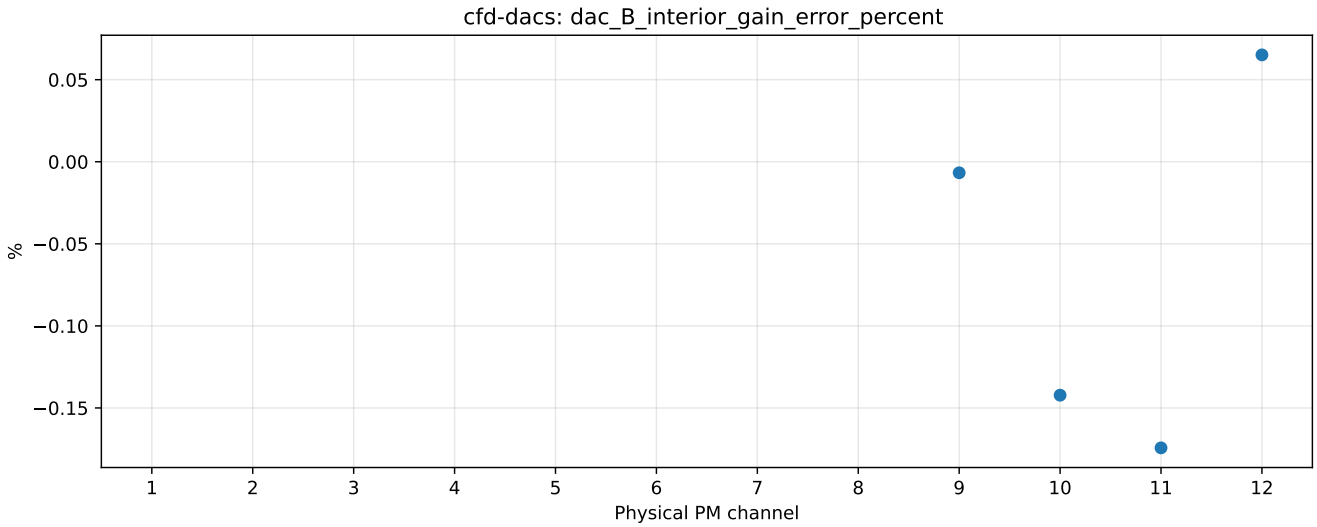

cfd-dacs: dac_B_minimum_code_V
+1.044
V
0.00070
0.00065
0.00060
0.00055
0.00050
0.00045
0.00040
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

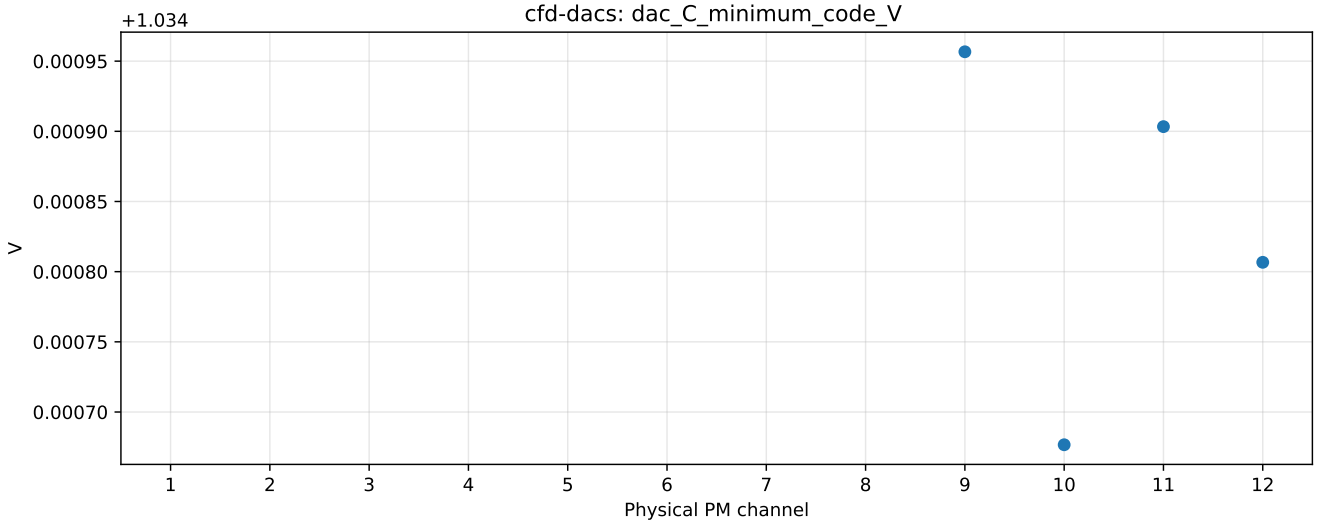

cfd-dacs: dac_B_maximum_code_V
+1.004
V
0.00060
0.00055
0.00050
0.00045
0.00040
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

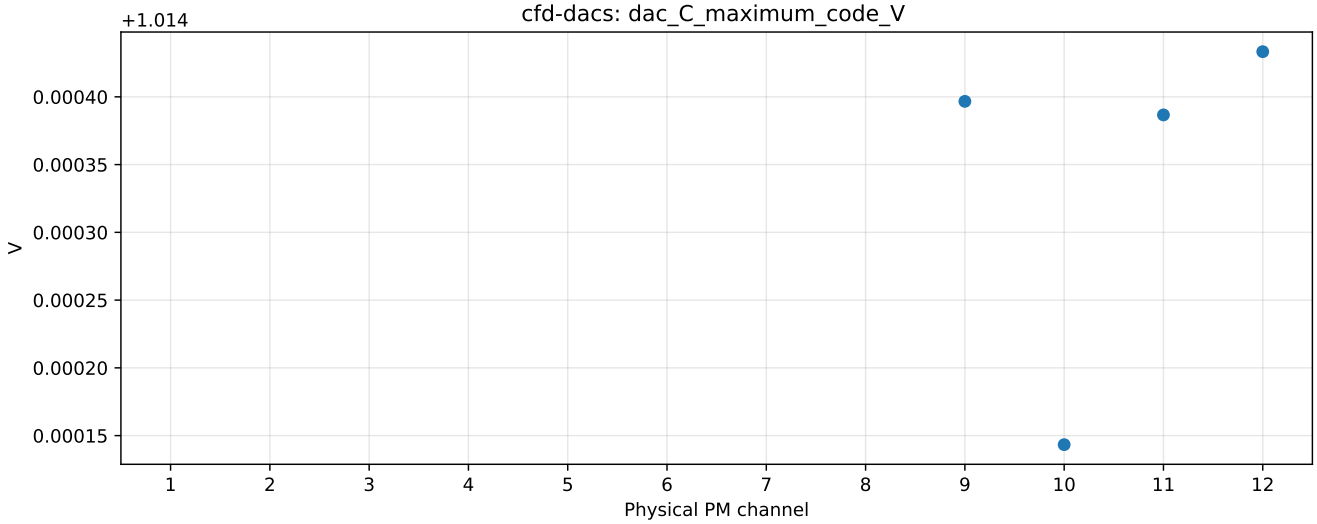

cfd-dacs: dac_B_span_V
−0.04000
−0.04005
−0.04010
−0.04015
−0.04020
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

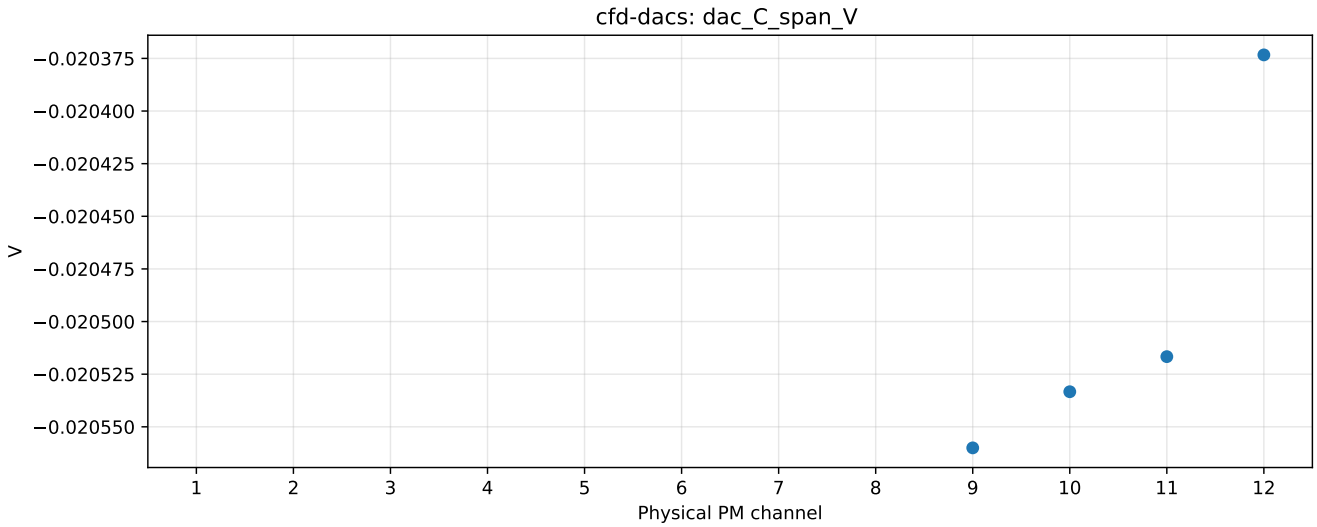

cfd-dacs: dac_B_maximum_reverse_step_V
0.04
0.02
0.00
−0.02
−0.04
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

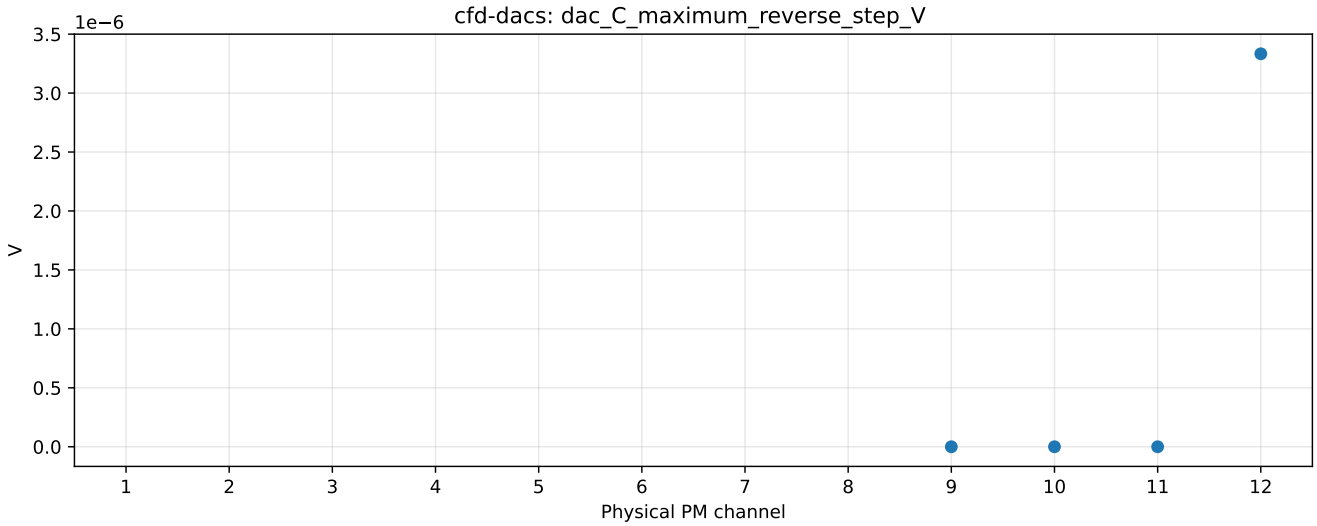

cfd-dacs: dac_B_maximum_repeat_std_V
1e−5
8.0
7.5
7.0
6.5
6.0
5.5
5.0
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

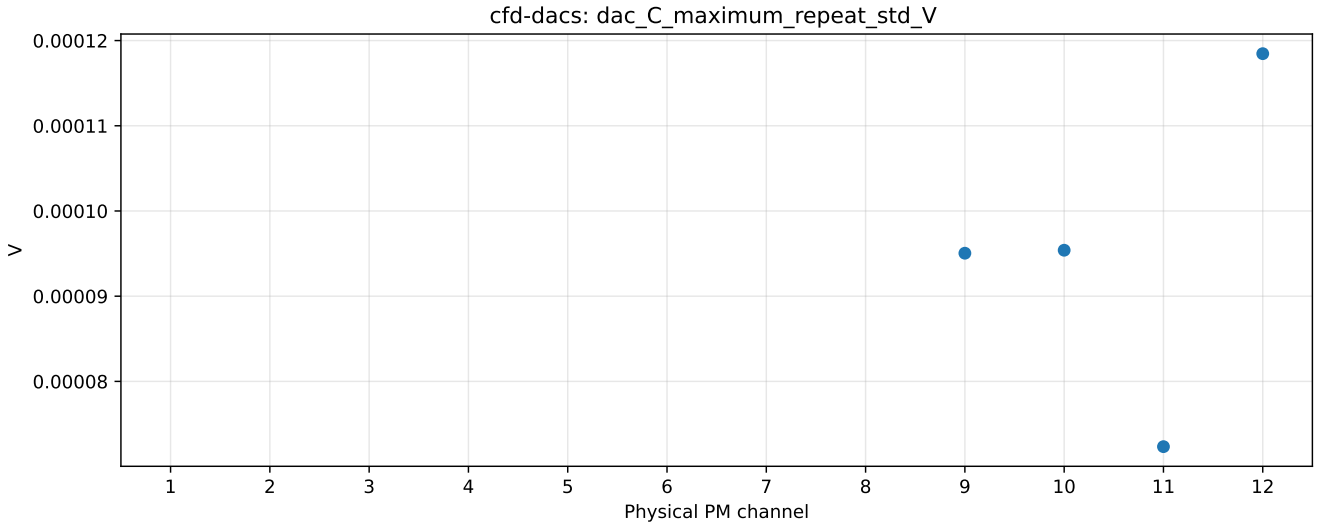

cfd-dacs: dac_B_interior_gain_error_percent
0.05
0.00
−0.05
−0.10
−0.15
%
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

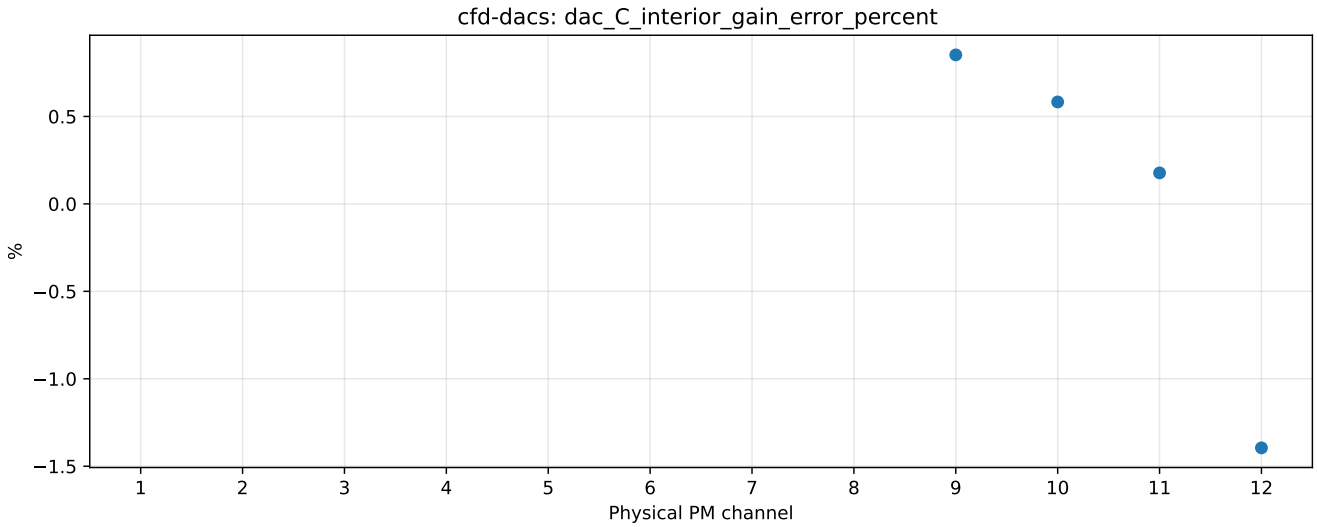

cfd-dacs: dac_C_minimum_code_V
+1.034
0.00095
0.00090
0.00085
0.00080
0.00075
0.00070
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

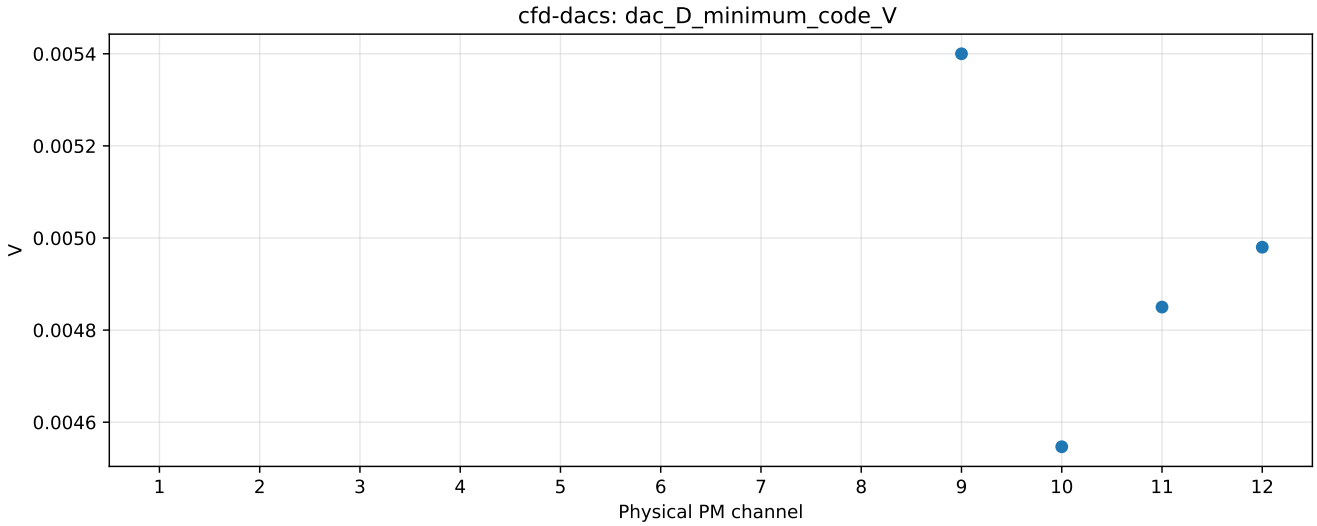

cfd-dacs: dac_C_maximum_code_V
+1.014
0.00040
0.00035
0.00030
0.00025
0.00020
0.00015
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

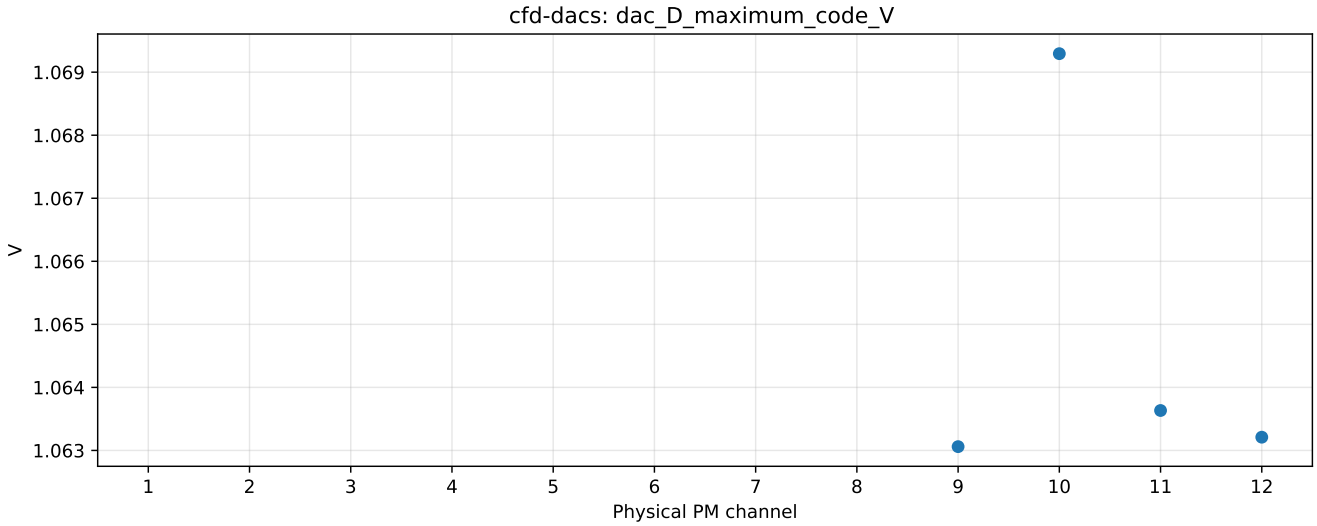

cfd-dacs: dac_C_span_V
−0.020375
−0.020400
−0.020425
−0.020450
−0.020475
−0.020500
−0.020525
−0.020550
V
1 2 3 4 5 6 7 8 9 10 11 12
Physical PM channel

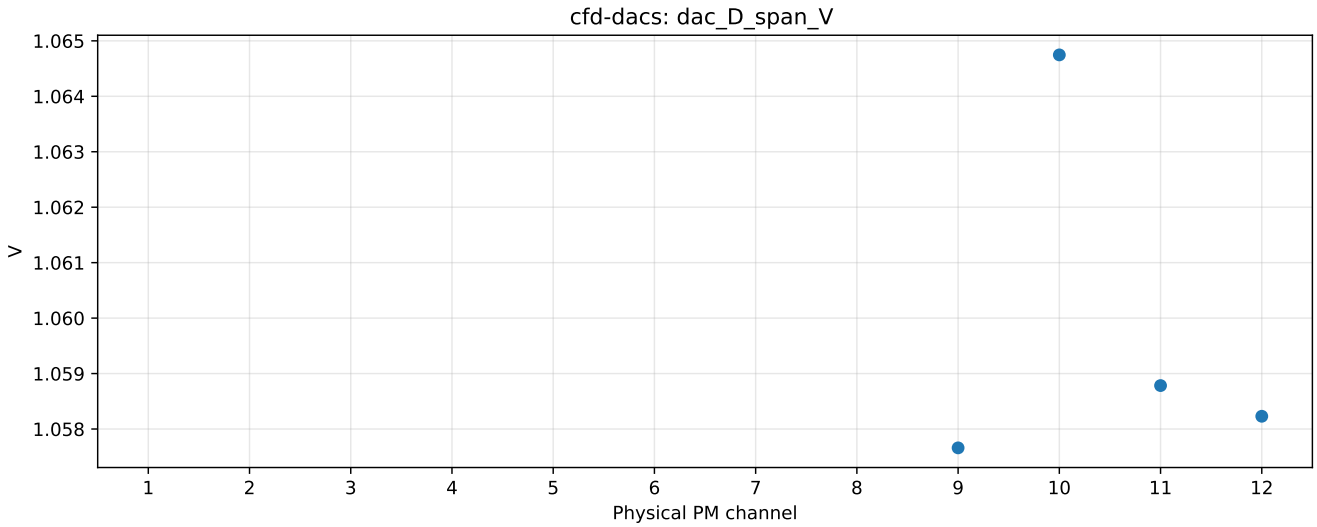

cfd-dacs: dac_C_maximum_reverse_step_V
1e−6
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
V
1 2 3 4 5 6 7 8 9 10 11 12
Physical PM channel

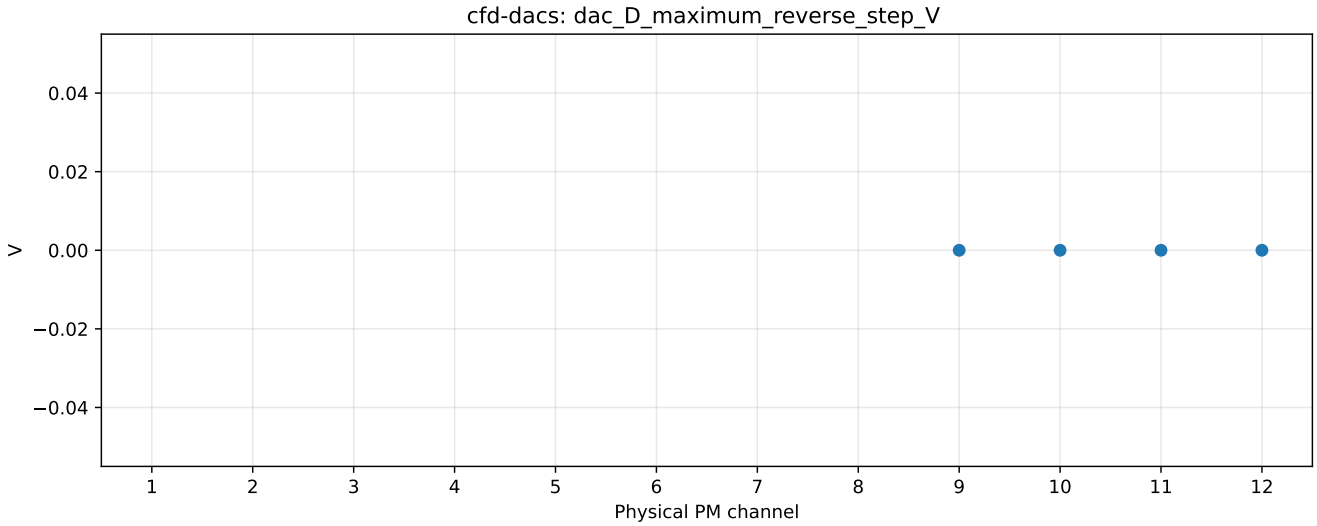

cfd-dacs: dac_C_maximum_repeat_std_V
0.00012
0.00011
0.00010
0.00009
0.00008
V
1 2 3 4 5 6 7 8 9 10 11 12
Physical PM channel

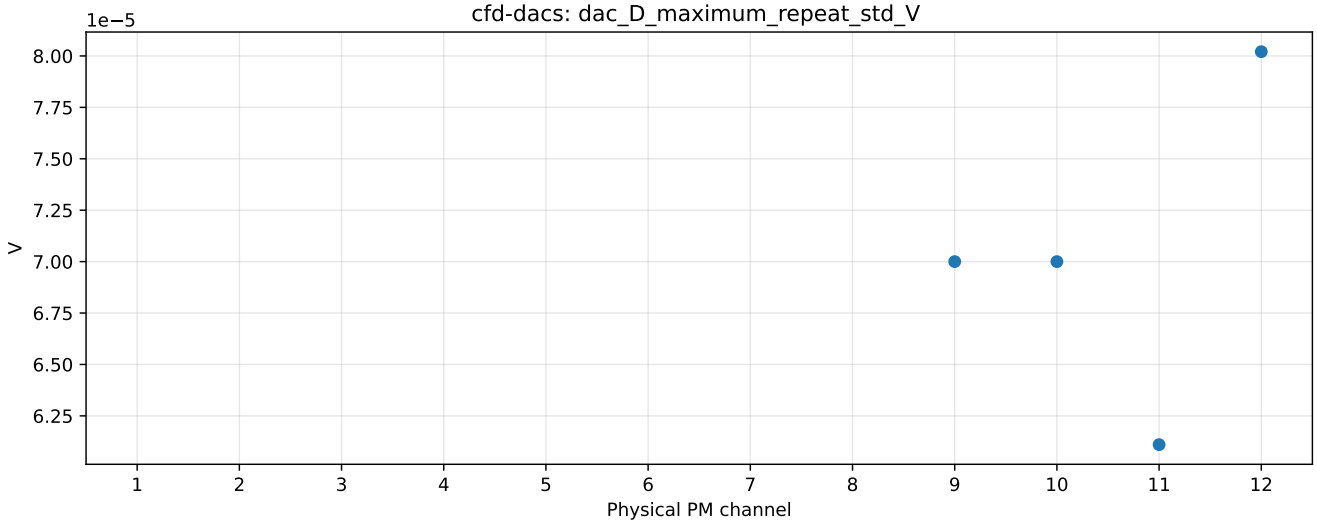

cfd-dacs: dac_C_interior_gain_error_percent
%
0.5
0.0
−0.5
−1.0
−1.5
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

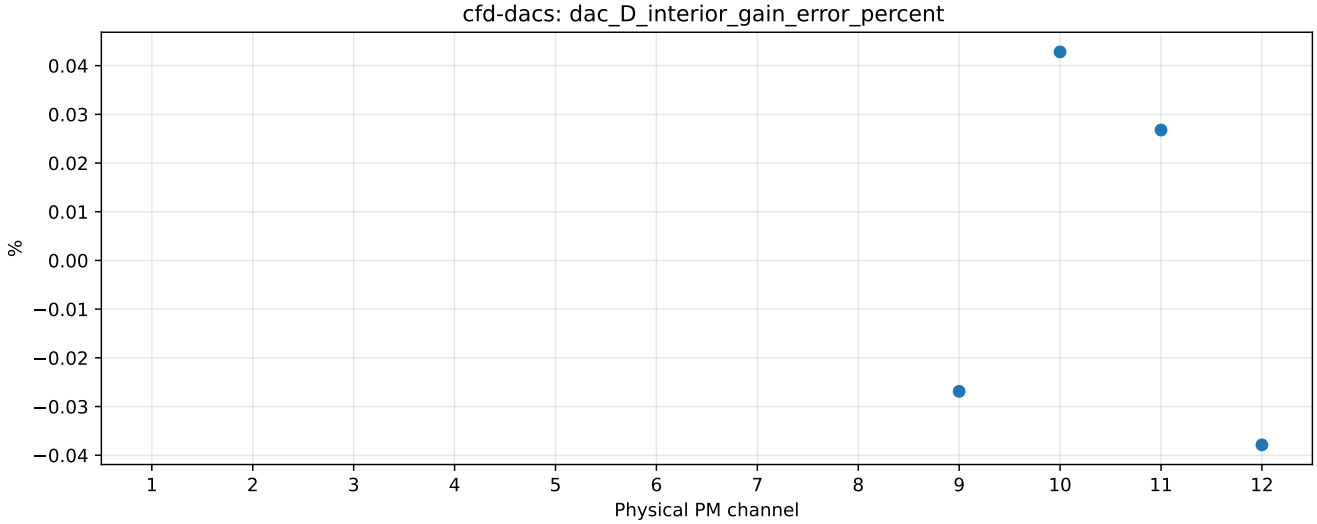

cfd-dacs: dac_D_minimum_code_V
V
0.0054
0.0052
0.0050
0.0048
0.0046
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

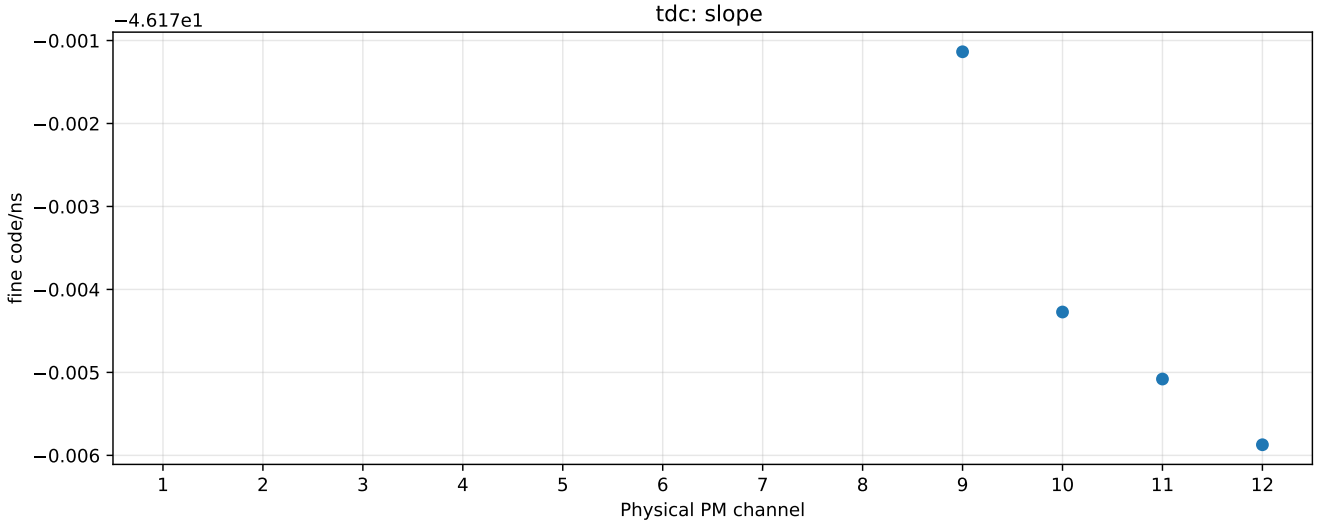

cfd-dacs: dac_D_maximum_code_V
V
1.069
1.068
1.067
1.066
1.065
1.064
1.063
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

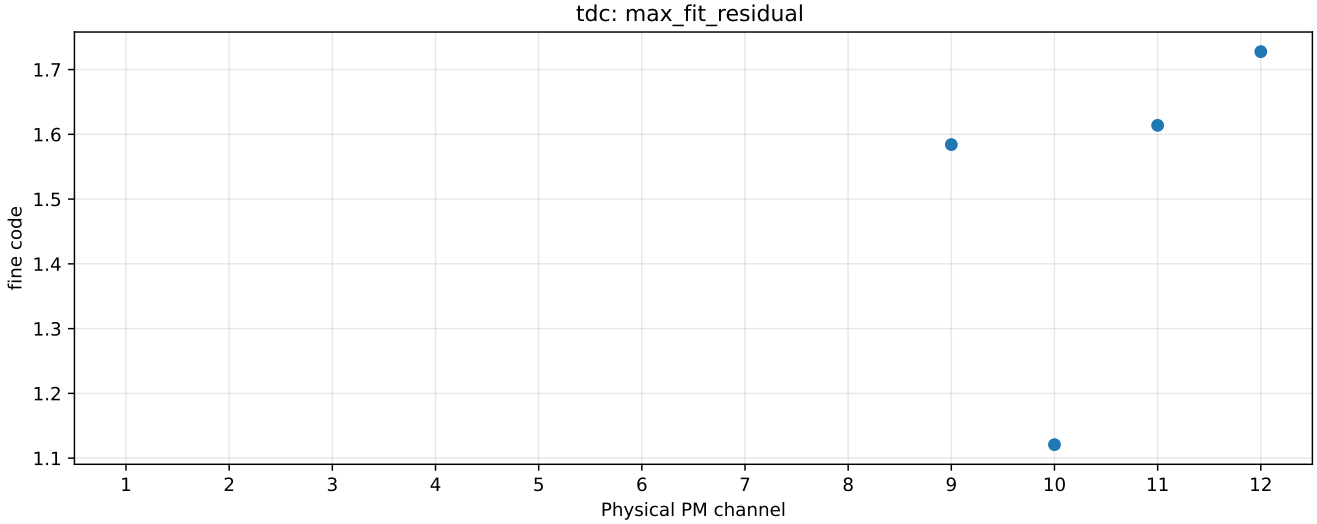

cfd-dacs: dac_D_span_V
1.065
1.064
1.063
1.062
1.061
1.060
1.059
1.058
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

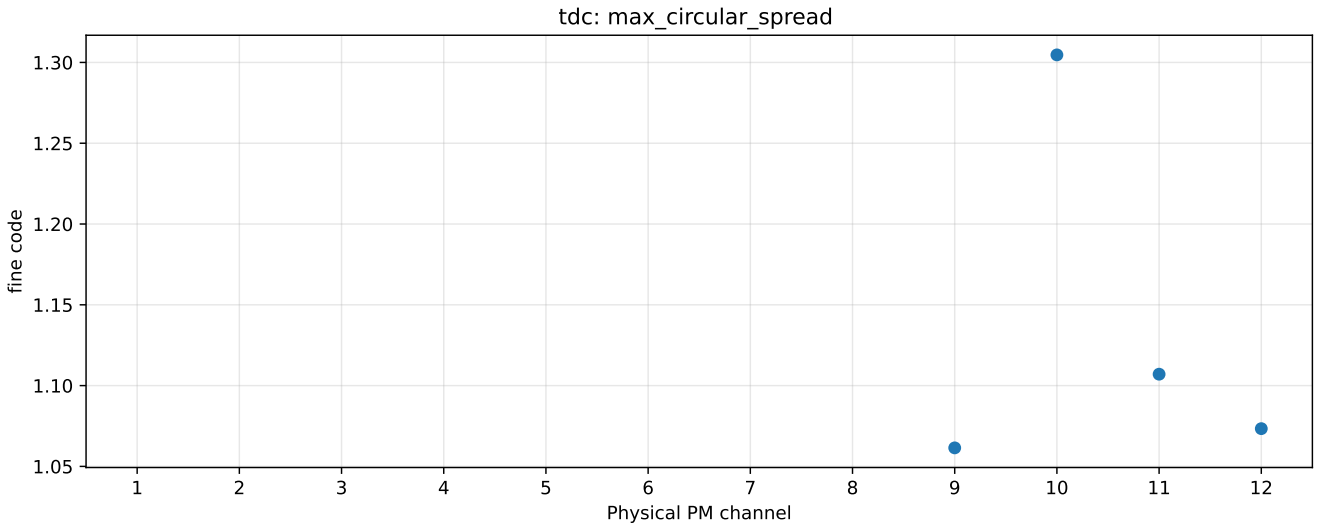

cfd-dacs: dac_D_maximum_reverse_step_V
0.04
0.02
0.00
−0.02
−0.04
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

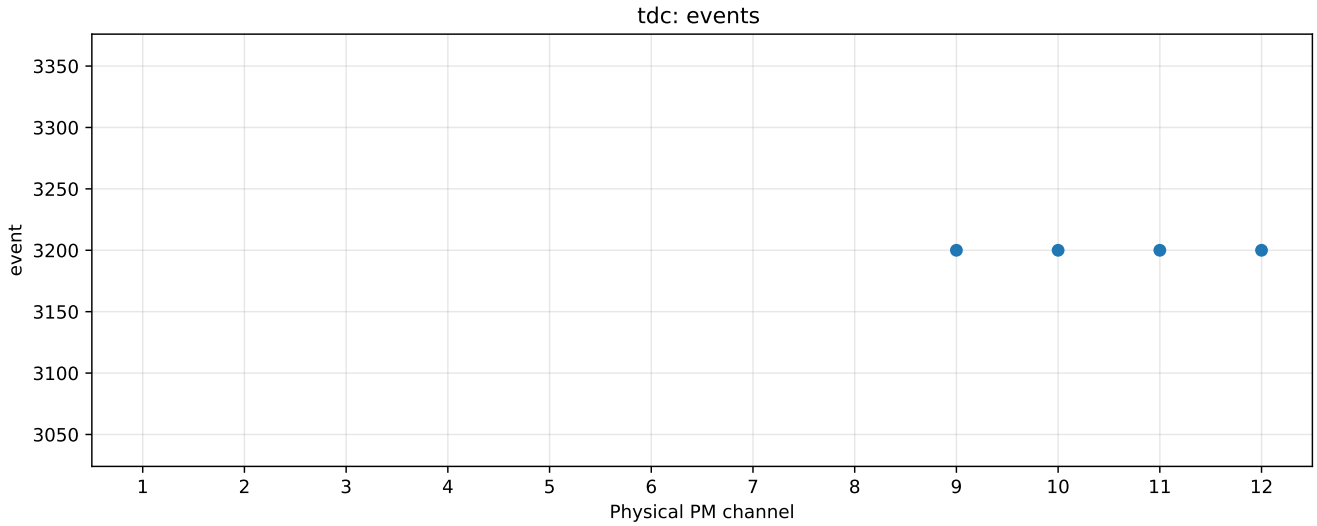

cfd-dacs: dac_D_maximum_repeat_std_V
1e−5
8.00
7.75
7.50
7.25
7.00
6.75
6.50
6.25
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

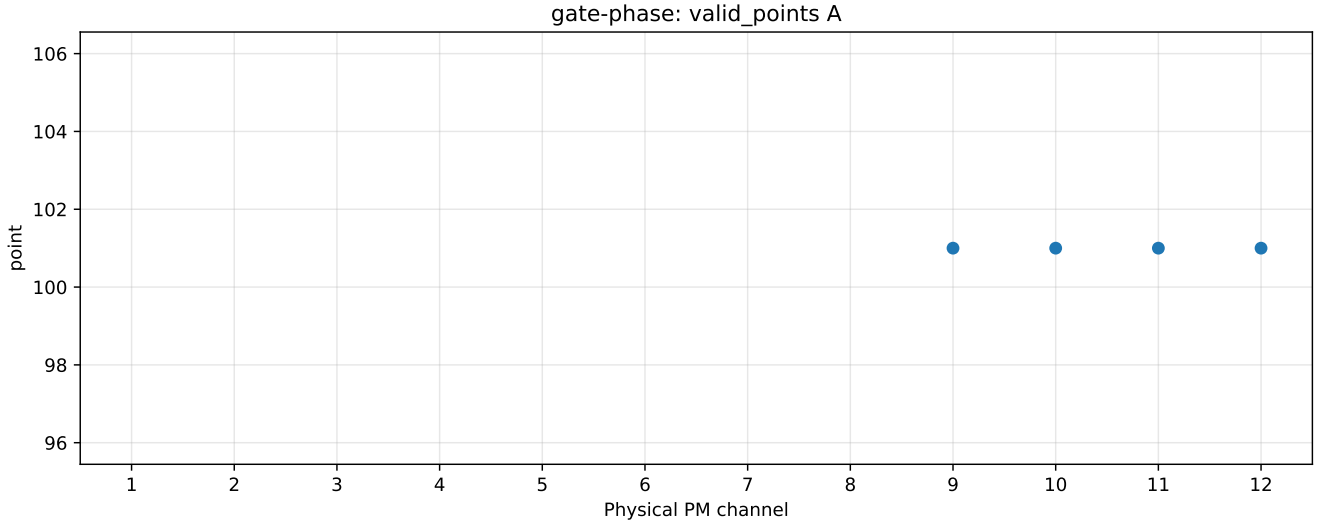

cfd-dacs: dac_D_interior_gain_error_percent
%
0.04
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
−0.04
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

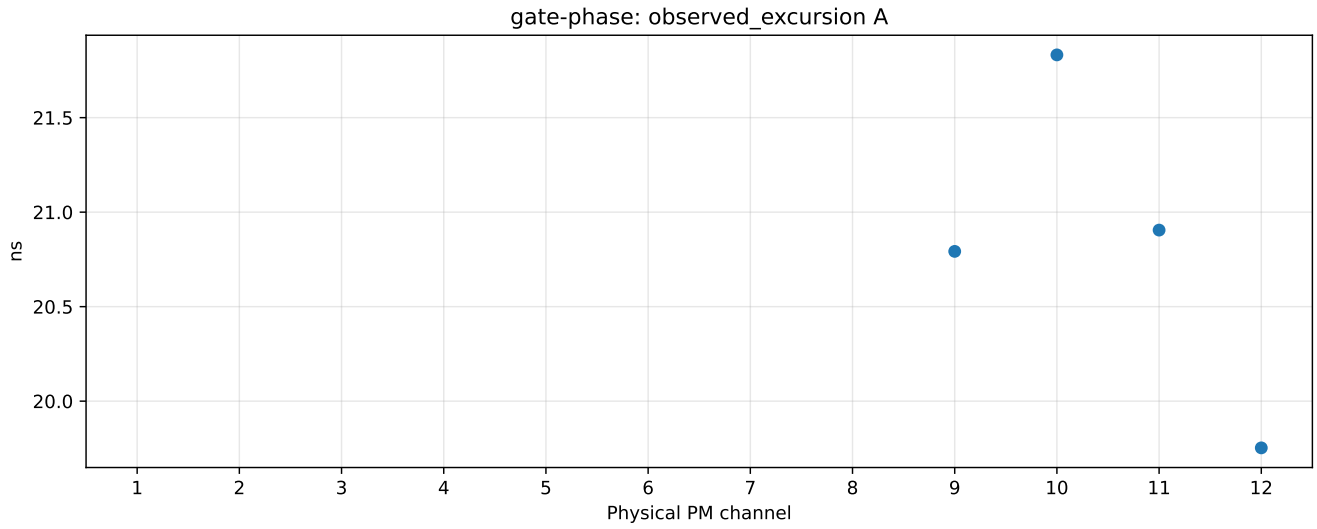

tdc: slope
−4.617e1
fine code/ns
−0.001
−0.002
−0.003
−0.004
−0.005
−0.006
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

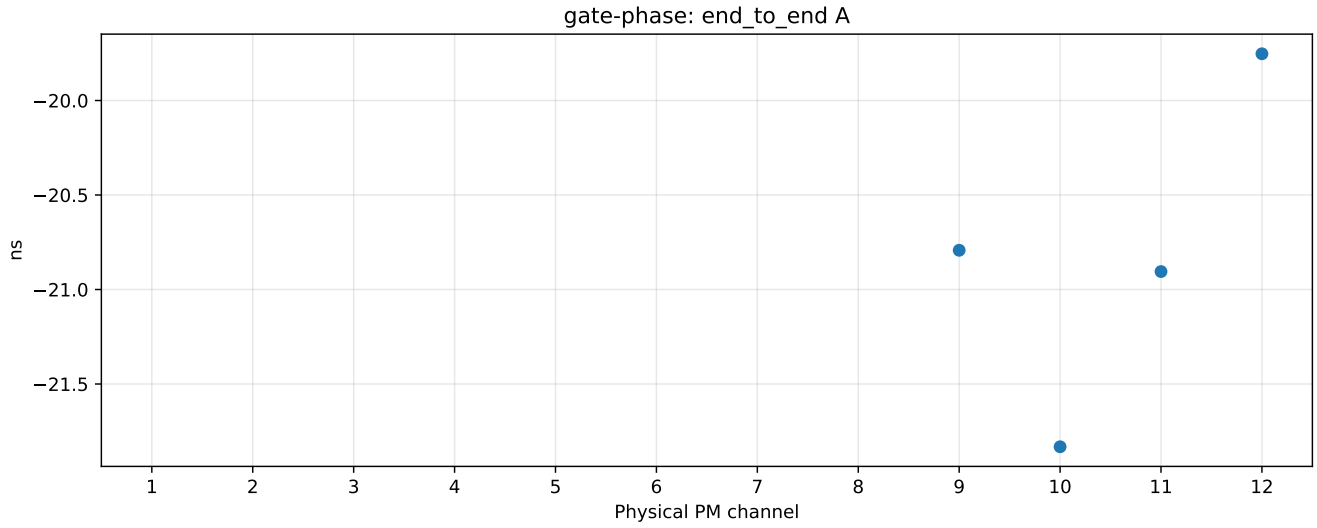

tdc: max_fit_residual
fine code
1.7
1.6
1.5
1.4
1.3
1.2
1.1
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

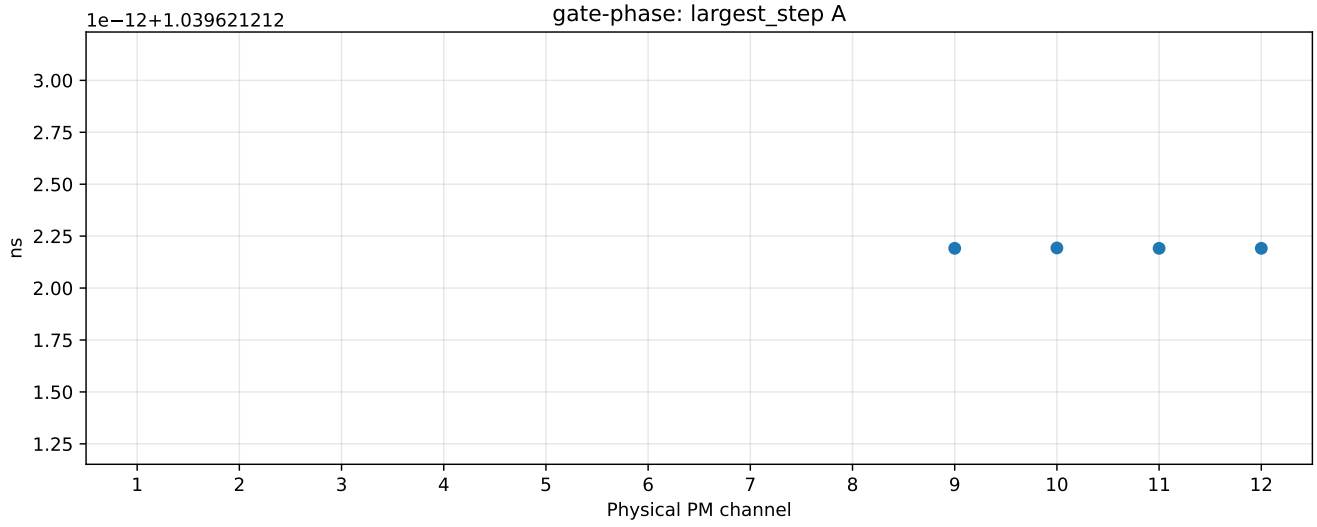

tdc: max_circular_spread
fine code
1.30
1.25
1.20
1.15
1.10
1.05
1 2 3 4 5 6 7 8 9 10 11 12
Physical PM channel

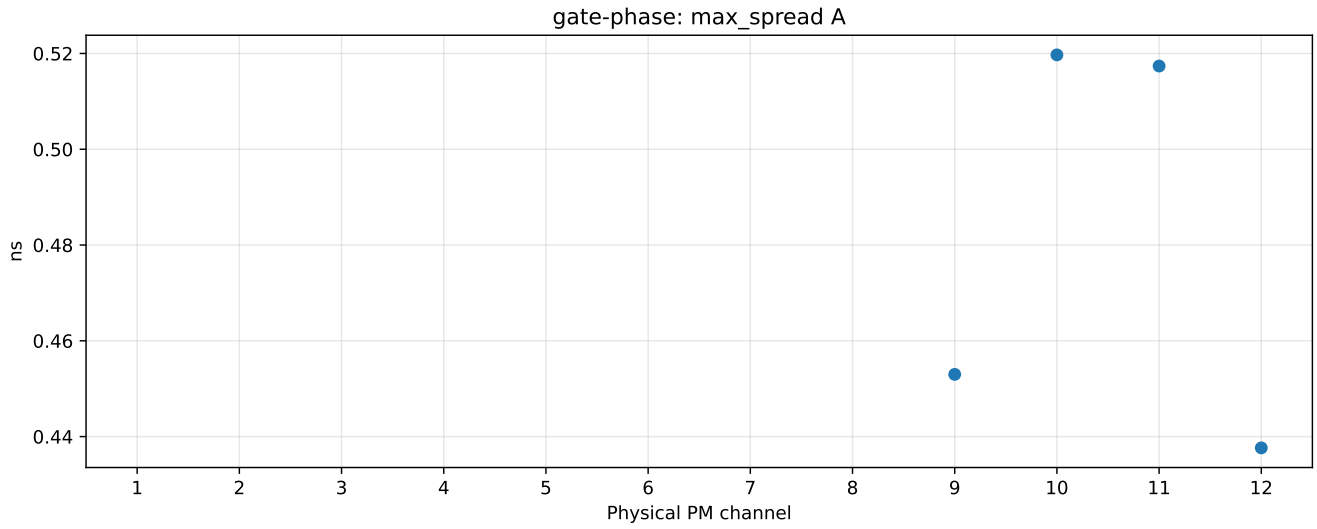

tdc: events
event
3350
3300
3250
3200
3150
3100
3050
1 2 3 4 5 6 7 8 9 10 11 12
Physical PM channel

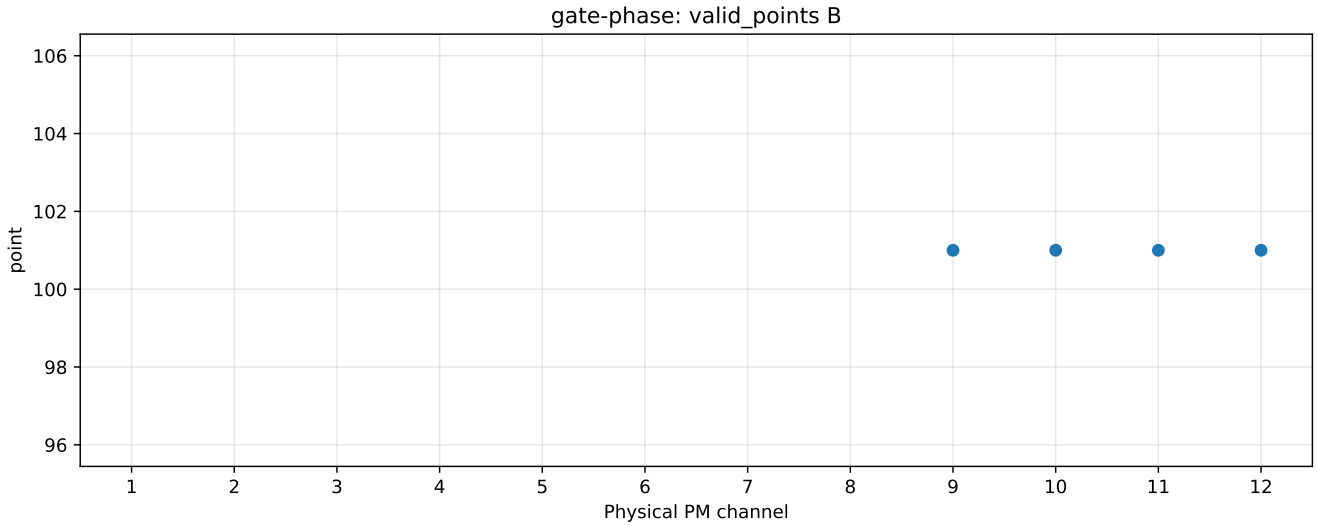

gate-phase: valid_points A
point
106
104
102
100
98
96
1 2 3 4 5 6 7 8 9 10 11 12
Physical PM channel

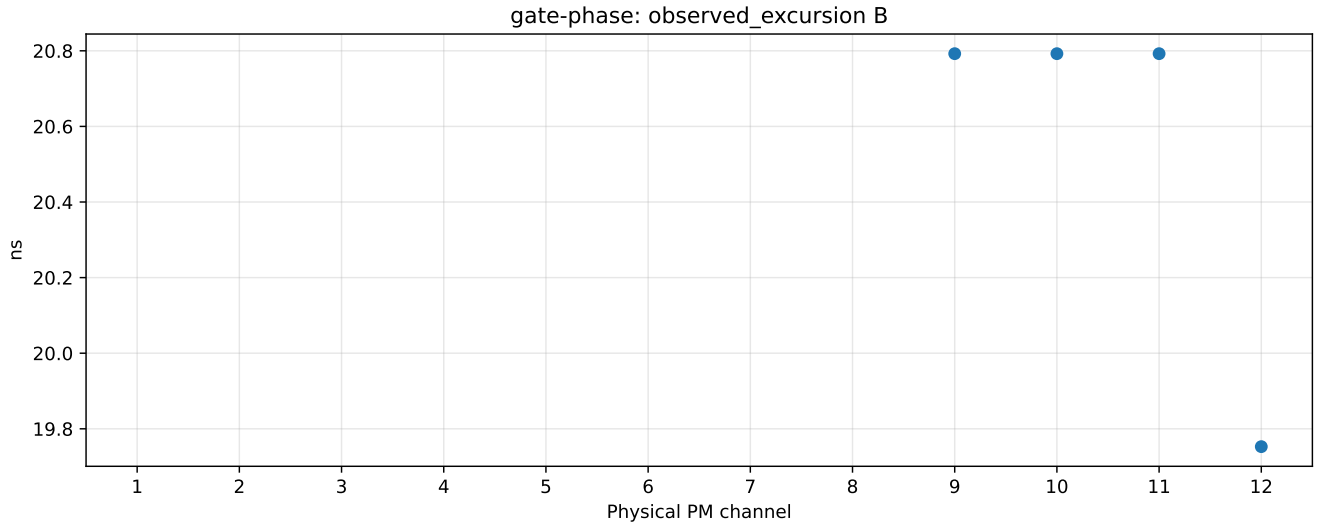

gate-phase: observed_excursion A
ns
21.5
21.0
20.5
20.0
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

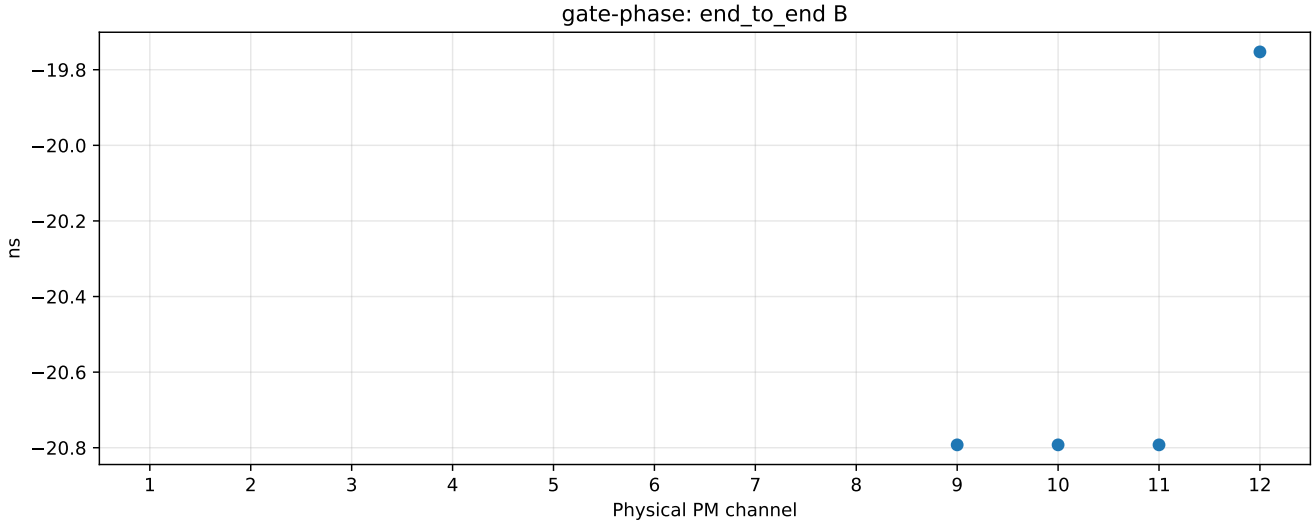

gate-phase: end_to_end A
ns
−20.0
−20.5
−21.0
−21.5
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

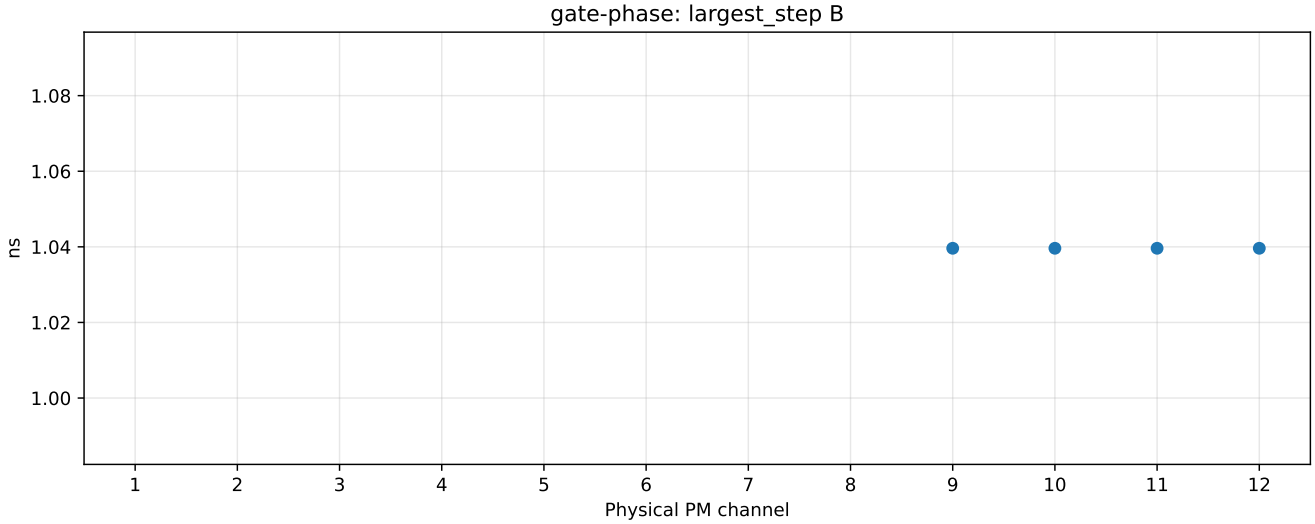

gate-phase: largest_step A
1e−12+1.039621212
ns
3.00
2.75
2.50
2.25
2.00
1.75
1.50
1.25
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

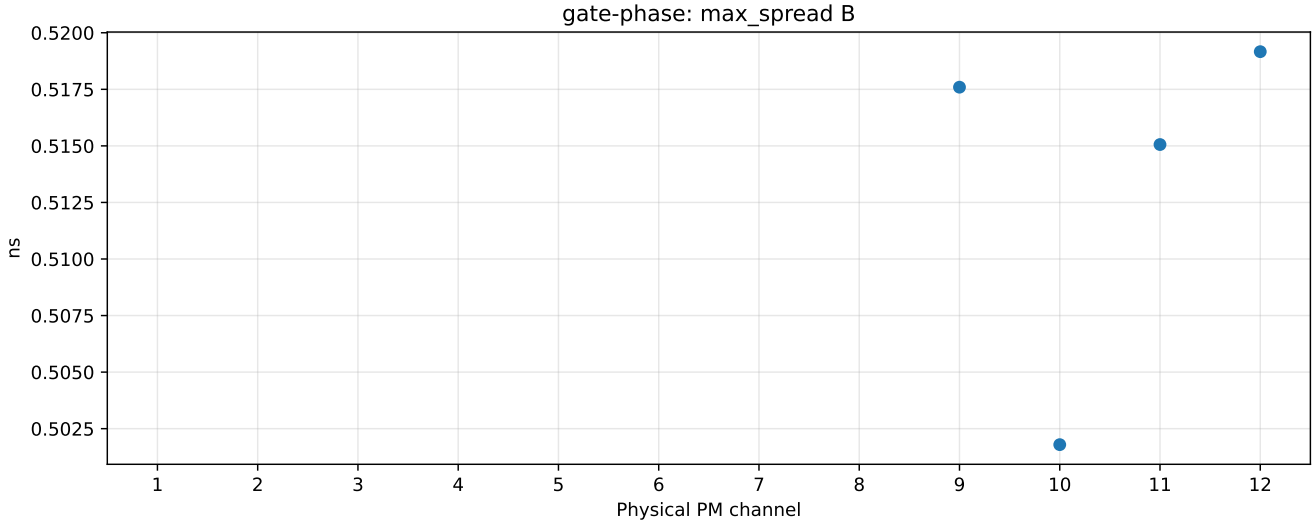

gate-phase: max_spread A
ns
0.52
0.50
0.48
0.46
0.44
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

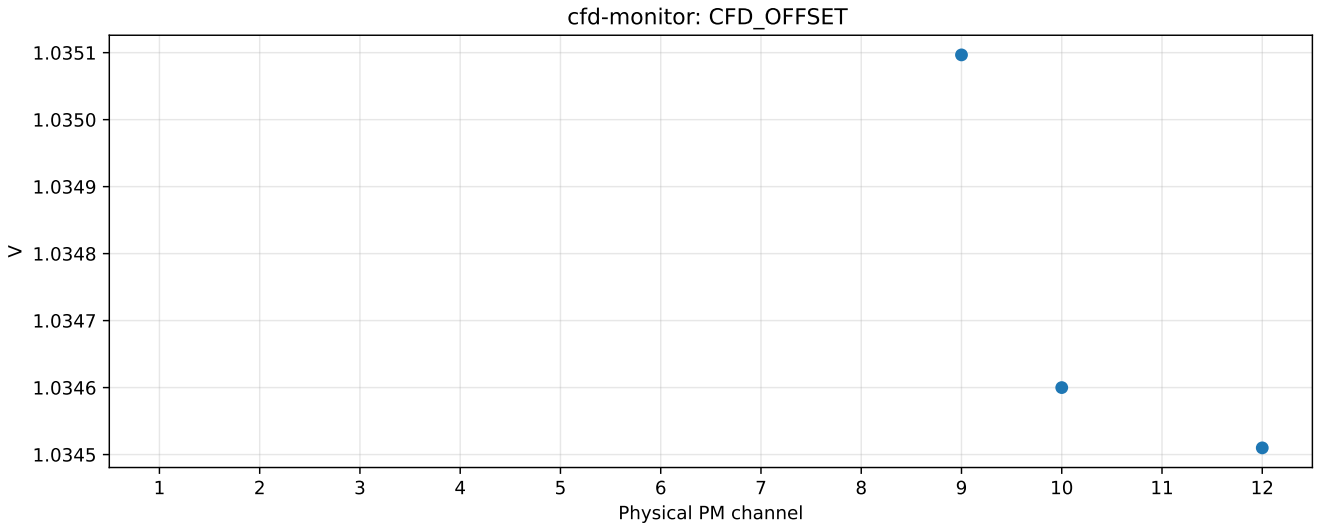

gate-phase: valid_points B
point
106
104
102
100
98
96
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

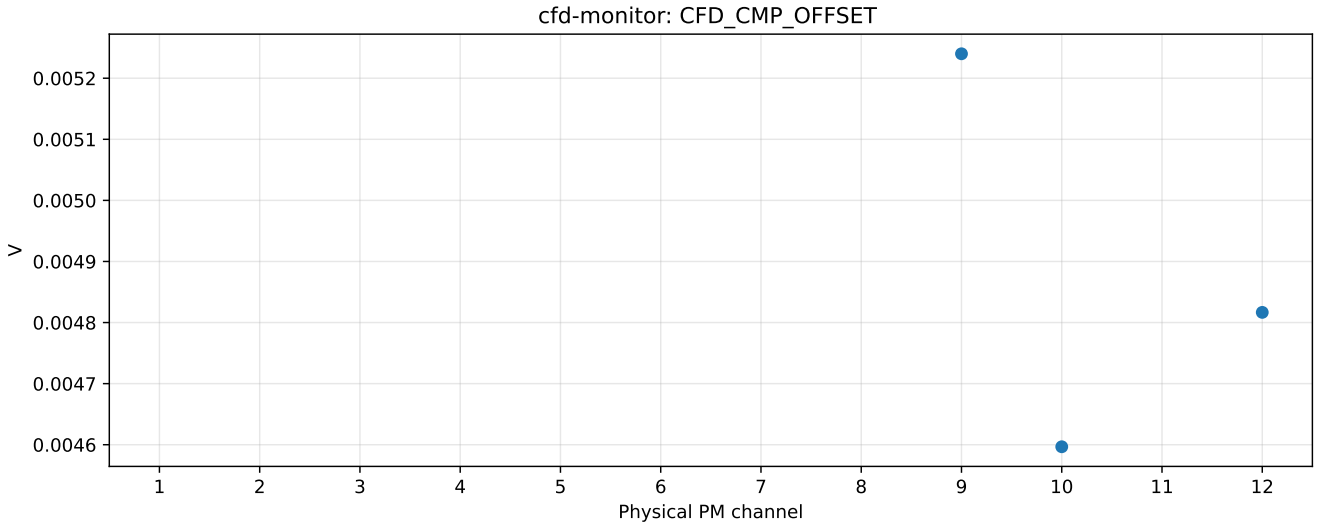

gate-phase: observed_excursion B
ns
20.8
20.6
20.4
20.2
20.0
19.8
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

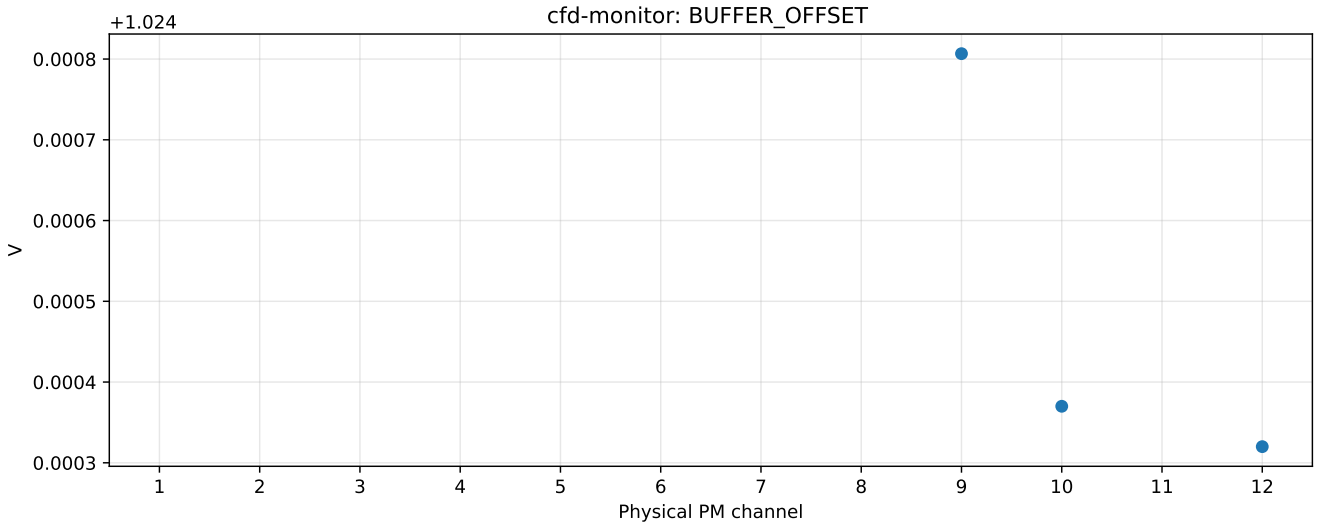

gate-phase: end_to_end B
ns
−19.8
−20.0
−20.2
−20.4
−20.6
−20.8
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

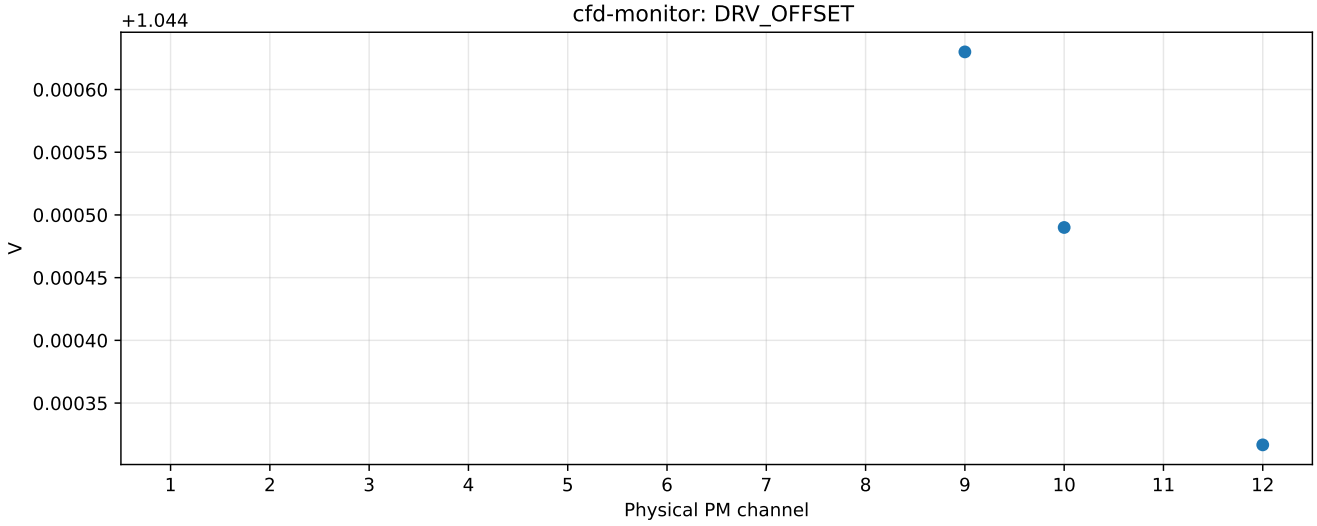

gate-phase: largest_step B
ns
1.08
1.06
1.04
1.02
1.00
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

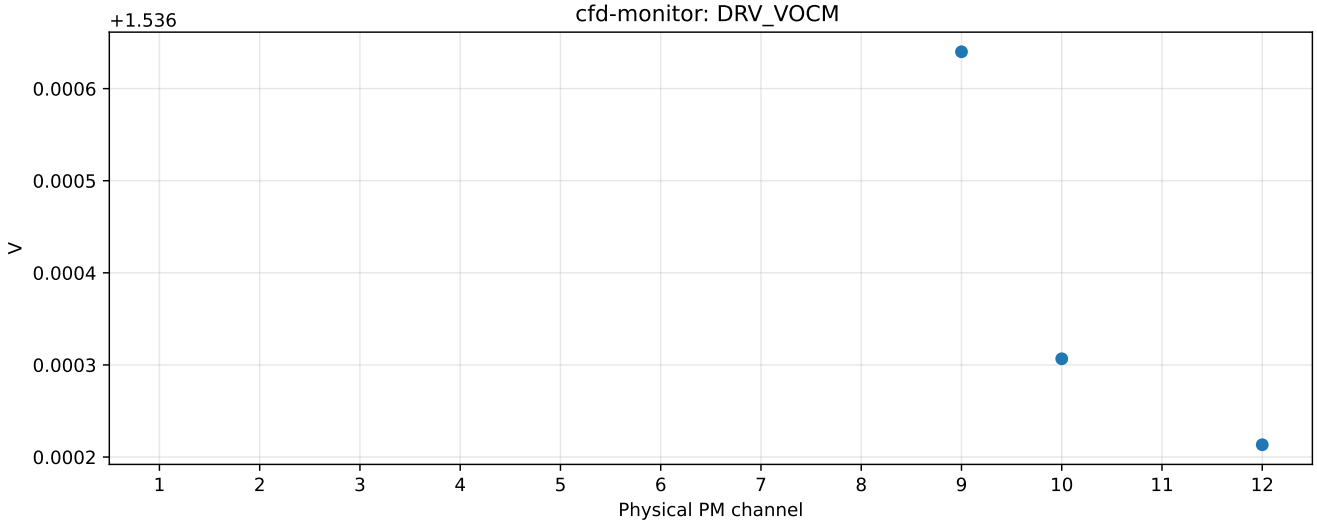

gate-phase: max_spread B
ns
0.5200
0.5175
0.5150
0.5125
0.5100
0.5075
0.5050
0.5025
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

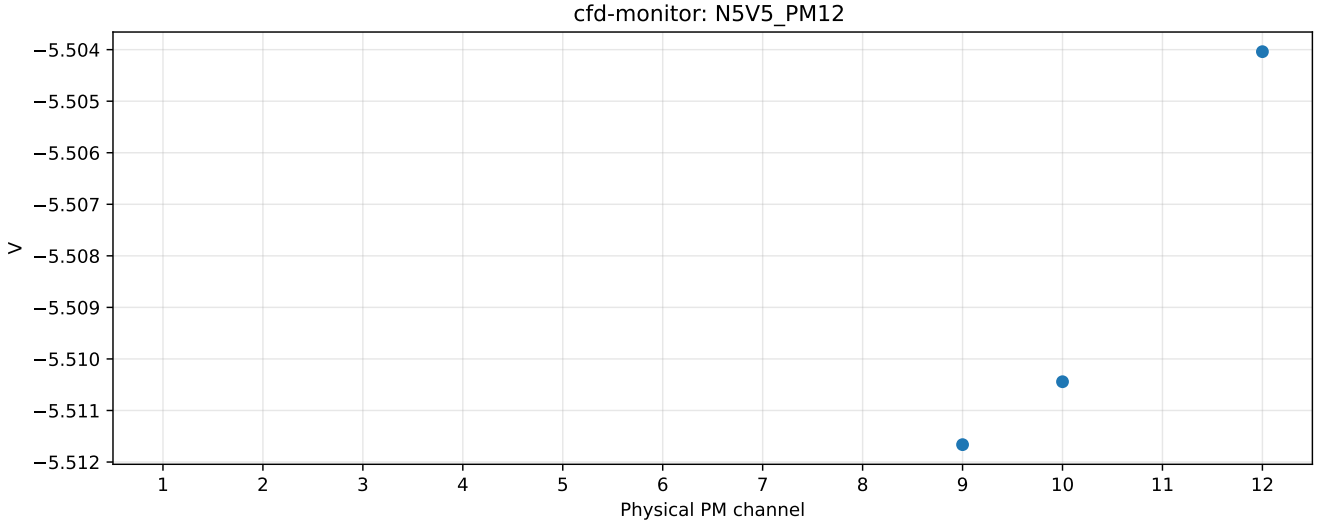

cfd-monitor: CFD_OFFSET
1.0351
1.0350
1.0349
1.0348
1.0347
1.0346
1.0345
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

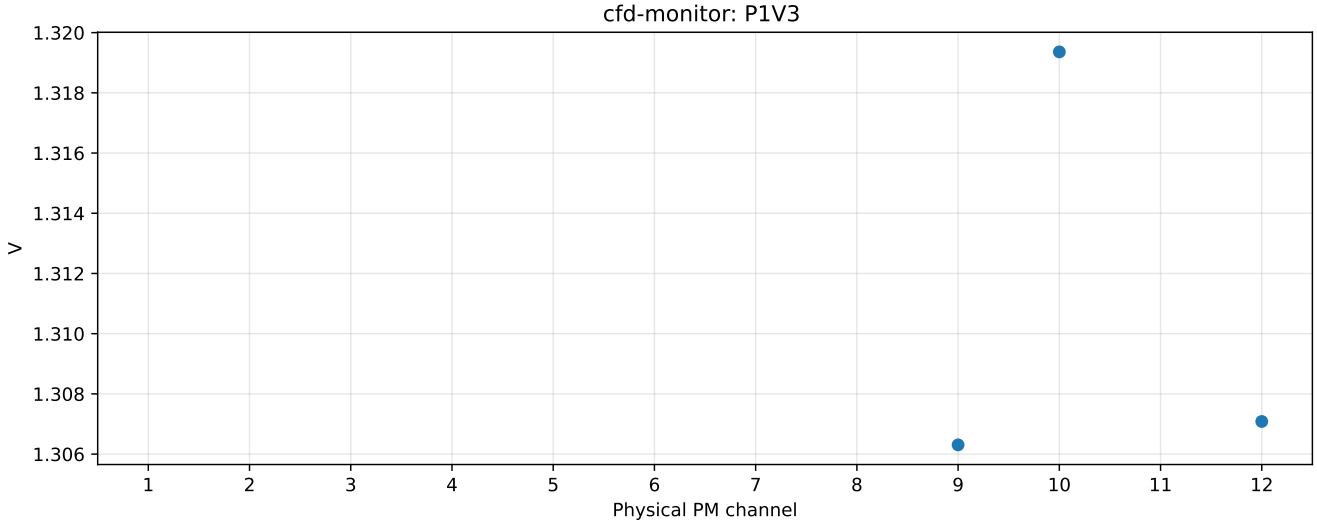

cfd-monitor: CFD_CMP_OFFSET
0.0052
0.0051
0.0050
0.0049
0.0048
0.0047
0.0046
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel

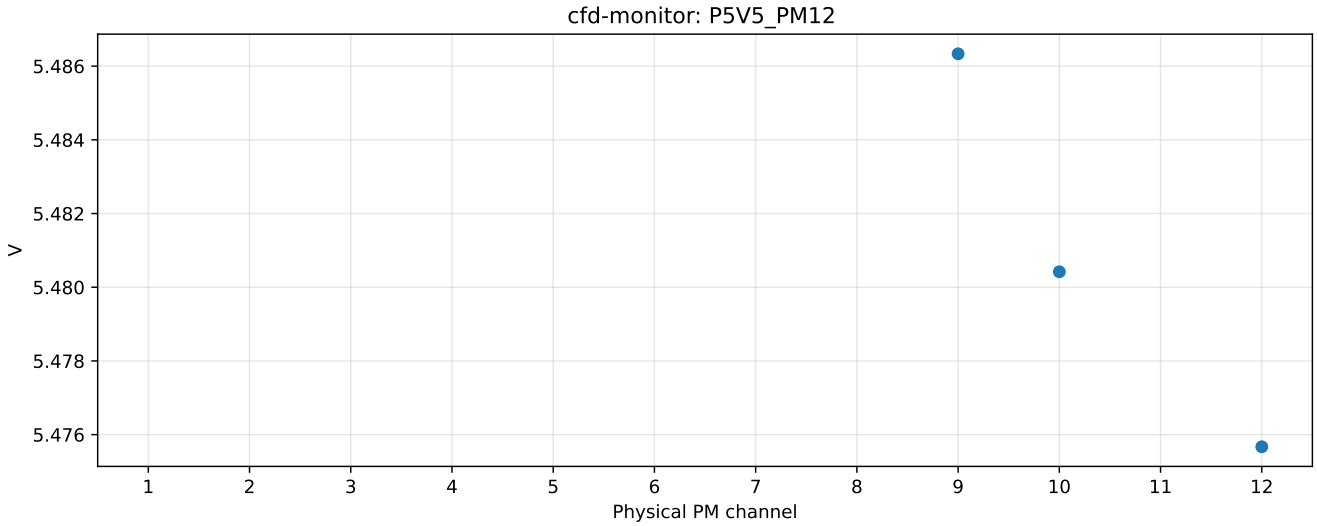

cfd-monitor: BUFFER_OFFSET
+1.024
0.0008
0.0007
0.0006
0.0005
0.0004
0.0003
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel


cfd-monitor: DRV_OFFSET
+1.044
0.00060
0.00055
0.00050
0.00045
0.00040
0.00035
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel


cfd-monitor: DRV_VOCM
+1.536
0.0006
0.0005
0.0004
0.0003
0.0002
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel


cfd-monitor: N5V5_PM12
−5.504
−5.505
−5.506
−5.507
−5.508
−5.509
−5.510
−5.511
−5.512
V
1
2
3
4
5
6
7
8
9
10
11
12
Physical PM channel





11 Static CPLD–FPGA connection discovery

Each path independently used338walking-one/zero vectors. All168expected sources match; one common table avoids repeated descriptions. This checks static connectivity, not at-speed margin or unconnected analog inputs.

PM channel	Signal	CPLD pin	FPGA ball	Gateware	Boundary
1	DI1[0]	99	T7	PASS	PASS
1	DI1[1]	97	U7	PASS	PASS
1	DI1[2]	96	U6	PASS	PASS
1	DI1[3]	95	U5	PASS	PASS
1	DI1[4]	94	U4	PASS	PASS
1	DI1[5]	93	U2	PASS	PASS
1	DI1[6]	92	U1	PASS	PASS
1	DI1[7]	91	V6	PASS	PASS
1	DI1[8]	90	V4	PASS	PASS
1	DI1[9]	89	W4	PASS	PASS
1	DI1[10]	87	V2	PASS	PASS
1	DI1[11]	86	V1	PASS	PASS
1	DI1[12]	85	W6	PASS	PASS
1	STR1	82	W5	PASS	PASS
2	DI2[0]	99	AF4	PASS	PASS
2	DI2[1]	97	AD5	PASS	PASS
2	DI2[2]	96	AE5	PASS	PASS

PM channel	Signal	CPLD pin	FPGA ball	Gateware	Boundary
2	DI2[3]	95	AF5	PASS	PASS
2	DI2[4]	94	AB6	PASS	PASS
2	DI2[5]	93	AC6	PASS	PASS
2	DI2[6]	92	AD6	PASS	PASS
2	DI2[7]	91	AE6	PASS	PASS
2	DI2[8]	90	U9	PASS	PASS
2	DI2[9]	89	V7	PASS	PASS
2	DI2[10]	87	V8	PASS	PASS
2	DI2[11]	86	Y7	PASS	PASS
2	DI2[12]	85	Y8	PASS	PASS
2	STR2	82	AA7	PASS	PASS
3	DI3[0]	99	AA8	PASS	PASS
3	DI3[1]	97	AD11	PASS	PASS
3	DI3[2]	96	AE11	PASS	PASS
3	DI3[3]	95	V12	PASS	PASS
3	DI3[4]	94	Y12	PASS	PASS
3	DI3[5]	93	Y13	PASS	PASS
3	DI3[6]	92	AA12	PASS	PASS
3	DI3[7]	91	AA13	PASS	PASS
3	DI3[8]	90	V13	PASS	PASS
3	DI3[9]	89	W13	PASS	PASS
3	DI3[10]	87	V14	PASS	PASS
3	DI3[11]	86	W14	PASS	PASS
3	DI3[12]	85	AA14	PASS	PASS
3	STR3	82	AA15	PASS	PASS
4	DI4[0]	99	V18	PASS	PASS
4	DI4[1]	97	W18	PASS	PASS
4	DI4[2]	96	AA19	PASS	PASS
4	DI4[3]	95	AC19	PASS	PASS
4	DI4[4]	94	AE18	PASS	PASS
4	DI4[5]	93	AF18	PASS	PASS
4	DI4[6]	92	AB20	PASS	PASS
4	DI4[7]	91	V19	PASS	PASS
4	DI4[8]	90	W19	PASS	PASS
4	DI4[9]	89	AB19	PASS	PASS
4	DI4[10]	87	AD19	PASS	PASS
4	DI4[11]	86	AF19	PASS	PASS
4	DI4[12]	85	AA20	PASS	PASS
4	STR4	82	AD20	PASS	PASS
5	DI5[0]	99	AE20	PASS	PASS
5	DI5[1]	97	AF20	PASS	PASS
5	DI5[2]	96	AE21	PASS	PASS
5	DI5[3]	95	AF22	PASS	PASS
5	DI5[4]	94	AE22	PASS	PASS
5	DI5[5]	93	AF23	PASS	PASS
5	DI5[6]	92	AE23	PASS	PASS
5	DI5[7]	91	AF24	PASS	PASS
5	DI5[8]	90	AF25	PASS	PASS
5	DI5[9]	89	AE25	PASS	PASS
5	DI5[10]	87	AE26	PASS	PASS
5	DI5[11]	86	AD21	PASS	PASS
5	DI5[12]	85	AD23	PASS	PASS
5	STR5	82	AB21	PASS	PASS
6	DI6[0]	99	AD25	PASS	PASS
6	DI6[1]	97	AD26	PASS	PASS
6	DI6[2]	96	AC21	PASS	PASS
6	DI6[3]	95	AC22	PASS	PASS
6	DI6[4]	94	AC23	PASS	PASS
6	DI6[5]	93	AD24	PASS	PASS
6	DI6[6]	92	AC26	PASS	PASS

PM channel	Signal	CPLD pin	FPGA ball	Gateware	Boundary
6	DI6[7]	91	AC24	PASS	PASS
6	DI6[8]	90	AB22	PASS	PASS
6	DI6[9]	89	AB24	PASS	PASS
6	DI6[10]	87	AB25	PASS	PASS
6	DI6[11]	86	AB26	PASS	PASS
6	DI6[12]	85	AA22	PASS	PASS
6	STR6	82	AA23	PASS	PASS
7	DI7[0]	99	AA24	PASS	PASS
7	DI7[1]	97	AA25	PASS	PASS
7	DI7[2]	96	Y20	PASS	PASS
7	DI7[3]	95	Y21	PASS	PASS
7	DI7[4]	94	Y22	PASS	PASS
7	DI7[5]	93	Y23	PASS	PASS
7	DI7[6]	92	Y25	PASS	PASS
7	DI7[7]	91	Y26	PASS	PASS
7	DI7[8]	90	W20	PASS	PASS
7	DI7[9]	89	W21	PASS	PASS
7	DI7[10]	87	W23	PASS	PASS
7	DI7[11]	86	W24	PASS	PASS
7	DI7[12]	85	W25	PASS	PASS
7	STR7	82	W26	PASS	PASS
8	DI8[0]	99	V21	PASS	PASS
8	DI8[1]	97	V22	PASS	PASS
8	DI8[2]	96	V23	PASS	PASS
8	DI8[3]	95	V24	PASS	PASS
8	DI8[4]	94	V26	PASS	PASS
8	DI8[5]	93	U21	PASS	PASS
8	DI8[6]	92	U22	PASS	PASS
8	DI8[7]	91	U24	PASS	PASS
8	DI8[8]	90	U25	PASS	PASS
8	DI8[9]	89	U26	PASS	PASS
8	DI8[10]	87	U16	PASS	PASS
8	DI8[11]	86	U17	PASS	PASS
8	DI8[12]	85	U19	PASS	PASS
8	STR8	82	U20	PASS	PASS
9	DI9[0]	99	T17	PASS	PASS
9	DI9[1]	97	T18	PASS	PASS
9	DI9[2]	96	T19	PASS	PASS
9	DI9[3]	95	T20	PASS	PASS
9	DI9[4]	94	T22	PASS	PASS
9	DI9[5]	93	T23	PASS	PASS
9	DI9[6]	92	T24	PASS	PASS
9	DI9[7]	91	T25	PASS	PASS
9	DI9[8]	90	R16	PASS	PASS
9	DI9[9]	89	R17	PASS	PASS
9	DI9[10]	87	R18	PASS	PASS
9	DI9[11]	86	R20	PASS	PASS
9	DI9[12]	85	R21	PASS	PASS
9	STR9	82	R22	PASS	PASS
10	DI10[0]	99	R23	PASS	PASS
10	DI10[1]	97	R25	PASS	PASS
10	DI10[2]	96	R26	PASS	PASS
10	DI10[3]	95	P16	PASS	PASS
10	DI10[4]	94	P18	PASS	PASS
10	DI10[5]	93	P19	PASS	PASS
10	DI10[6]	92	P20	PASS	PASS
10	DI10[7]	91	P21	PASS	PASS
10	DI10[8]	90	P23	PASS	PASS
10	DI10[9]	89	P24	PASS	PASS
10	DI10[10]	87	P25	PASS	PASS

PM channel	Signal	CPLD pin	FPGA ball	Gateware	Boundary
10	DI10[11]	86	P26	PASS	PASS
10	DI10[12]	85	N16	PASS	PASS
10	STR10	82	N17	PASS	PASS
11	DI11[0]	99	N18	PASS	PASS
11	DI11[1]	97	N19	PASS	PASS
11	DI11[2]	96	N21	PASS	PASS
11	DI11[3]	95	N22	PASS	PASS
11	DI11[4]	94	N23	PASS	PASS
11	DI11[5]	93	N24	PASS	PASS
11	DI11[6]	92	N26	PASS	PASS
11	DI11[7]	91	M19	PASS	PASS
11	DI11[8]	90	M20	PASS	PASS
11	DI11[9]	89	M21	PASS	PASS
11	DI11[10]	87	M22	PASS	PASS
11	DI11[11]	86	M24	PASS	PASS
11	DI11[12]	85	M25	PASS	PASS
11	STR11	82	M26	PASS	PASS
12	DI12[0]	99	L24	PASS	PASS
12	DI12[1]	97	L25	PASS	PASS
12	DI12[2]	96	K25	PASS	PASS
12	DI12[3]	95	K26	PASS	PASS
12	DI12[4]	94	M16	PASS	PASS
12	DI12[5]	93	M17	PASS	PASS
12	DI12[6]	92	L17	PASS	PASS
12	DI12[7]	91	L18	PASS	PASS
12	DI12[8]	90	L19	PASS	PASS
12	DI12[9]	89	L20	PASS	PASS
12	DI12[10]	87	K18	PASS	PASS
12	DI12[11]	86	K16	PASS	PASS
12	DI12[12]	85	K17	PASS	PASS
12	STR12	82	J19	PASS	PASS